

Unit 1 Study Guide

Biology · Chapters 1, 2, 3 and 21 · printable version

How to use this. Each card gives you the plain-words answer first, the exam wording underneath, and a memory trick. Read one, cover it, then say it back in your own words before you check. The dashed box at the bottom is the part that earns the depth points, so do not skip it.

Every section opens with a one-page picture sheet. If you only have ten minutes before the exam, look at those six pages and nothing else.

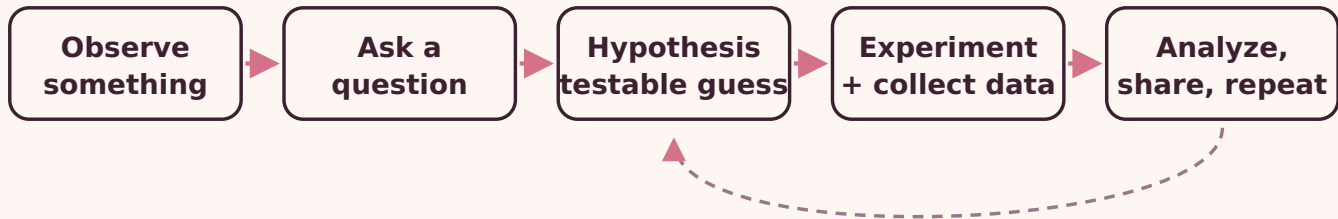
After the sections: diagrams to label from memory, a **Go deeper** appendix with a harder version of every question, then 68 multiple choice and 14 short answer questions, with the answer key at the very back.

Doing Science — picture sheet

How science actually works, and the vocabulary people mix up. Every diagram from this section, no text. Ten-minute review.

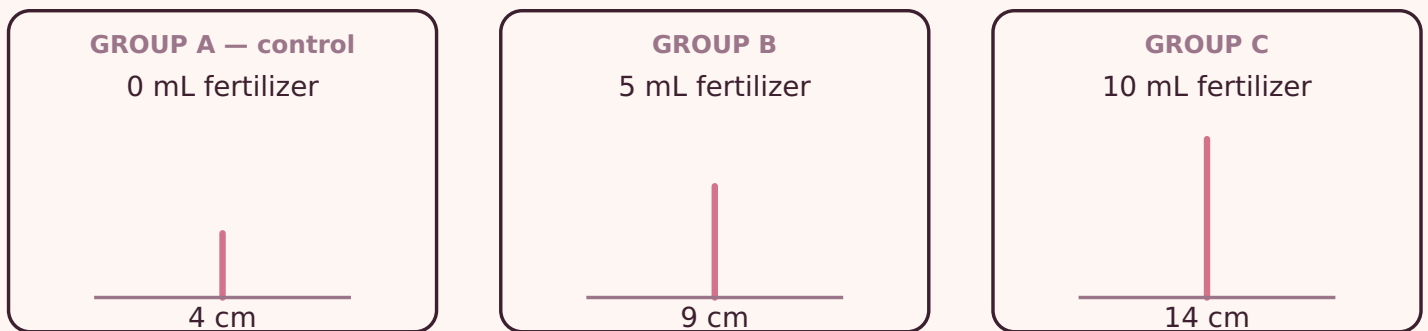
The scientific process

Peer review and repetition happen at every arrow, not just the last one



Results send you back to revise the hypothesis. Science loops, it does not end.

One experiment: does fertilizer make basil grow taller?



Independent = amount of fertilizer (what I change on purpose)

Dependent = height in cm (what I measure)

Controlled = same soil, light, water, pot, basil variety · **Control group** = Group A

Three words people mix up constantly

Hypothesis

Narrow, testable proposed explanation

"Ginkgo improves memory in Alzheimer's"

Scale: one question

Theory

Broad explanation backed by tons of evidence

"Evolution by natural selection"

Scale: whole field

Law

Describes WHAT happens, not why. Often math.

" $F = ma$ "
Gravity, thermodynamics

Scale: a pattern

A hypothesis never "grows up" into a law. They answer different kinds of questions.

A hypothesis has to be able to fail

YES Testable

“Plants given fertilizer will grow taller than plants without.”

a result could prove this wrong

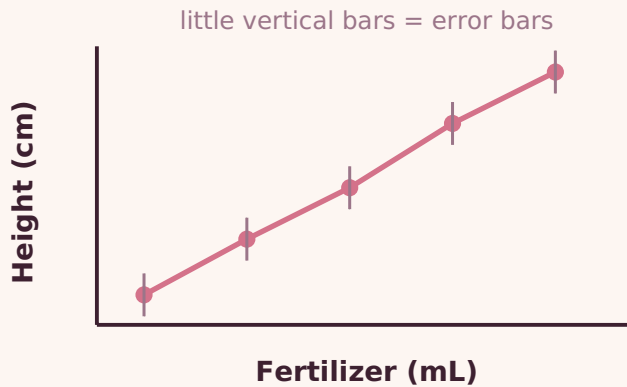
NO Not testable

“Plants grow because they want to.”

no result could ever prove this wrong

If nothing could disprove it, it is not a hypothesis. It is an opinion.

Building a graph, not just reading one



1. Independent on x, dependent on y

2. Label both axes WITH units

3. Pick the right type

bar = separate categories · line = continuous change
scatter = looking for a relationship

4. Scale evenly, start sensibly

uneven spacing is how graphs lie

5. Title it, and add error bars if you have them

Doing Science

Ch 1.1 · 11 cards

What is the broadest definition of biology?

IN PLAIN WORDS

Biology is the study of life. All of it, every level, from a single molecule inside a cell to an entire rainforest.

HOW TO SAY IT ON THE EXAM

Biology is the scientific study of life and living organisms, including their structure, function, growth, origin, evolution, and distribution, across every level of organization.

MEMORY TRICK

Bio = life, ology = study of. If it lives, biology claims it.

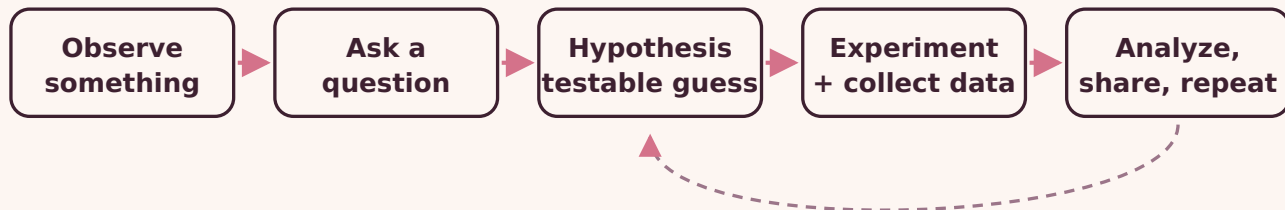
SAY IT OUT LOUD

Name three things biology covers that someone might not expect. (Viruses? Fossils? Your gut bacteria?)

Walk through the steps of the scientific process.

The scientific process

Peer review and repetition happen at every arrow, not just the last one



Results send you back to revise the hypothesis. Science loops, it does not end.

IN PLAIN WORDS

You notice something strange, ask why, make a guess you can actually test, run a test, then look at what happened and tell other people. Their questions send you back to the start.

HOW TO SAY IT ON THE EXAM

Observation leads to a question, which leads to a testable hypothesis with a prediction. An experiment generates data, data are analyzed, results are communicated through peer review and publication, and outcomes generate new questions. The process is iterative and non-linear, not a one-way staircase.

MEMORY TRICK

Ozzy Always Hides Extra Cash Regularly: Observe, Ask, Hypothesize, Experiment, Conclude, Repeat.

SAY IT OUT LOUD

Your professor loves this one: what do scientists actually DO at each step? Say the verb out loud for each box.

Control vs independent vs dependent variable. Define all three using one experiment.

One experiment: does fertilizer make basil grow taller?

GROUP A — control

0 mL fertilizer



GROUP B

5 mL fertilizer



GROUP C

10 mL fertilizer



Independent = amount of fertilizer (what I change on purpose)

Dependent = height in cm (what I measure)

Controlled = same soil, light, water, pot, basil variety · **Control group** = Group A

IN PLAIN WORDS

Independent is the one thing you change on purpose. Dependent is what you measure to see if your change did anything. Controlled variables are everything you deliberately keep identical. The control group is the version that gets no treatment, so you have something to compare against.

HOW TO SAY IT ON THE EXAM

The independent variable is manipulated by the researcher. The dependent variable is the measured response, and it depends on the independent variable. Controlled variables are held constant across all groups to eliminate confounding factors. The control group receives no experimental treatment and establishes the baseline.

MEMORY TRICK

I change the Independent. I Detect the Dependent. Control = everything constant, plus the group that gets nothing.

SAY IT OUT LOUD

Careful, this trips people up: a controlled VARIABLE and a control GROUP are not the same thing. Say the difference out loud.

Hypothesis vs scientific theory vs scientific law.

Three words people mix up constantly

Hypothesis

Narrow, testable
proposed explanation

"Ginkgo improves
memory in Alzheimer's"

Scale: one question

Theory

Broad explanation
backed by tons of evidence

"Evolution by
natural selection"

Scale: whole field

Law

Describes WHAT happens,
not why. Often math.

"F = ma"
Gravity, thermodynamics

Scale: a pattern

A hypothesis never “grows up” into a law. They answer different kinds of questions.

IN PLAIN WORDS

A hypothesis is a small testable guess about one specific thing. A theory is a huge explanation that survived decades of testing. A law describes a pattern that always happens but does not explain why it happens.

HOW TO SAY IT ON THE EXAM

A hypothesis is a proposed explanation for a narrow set of phenomena, based on prior observation and background knowledge, and it must be testable and falsifiable. A scientific theory is a broad, well-substantiated explanation supported by a large body of evidence from multiple independent lines of inquiry. A scientific law is a descriptive generalization, often mathematical, that states what consistently occurs under given conditions without explaining the mechanism.

MEMORY TRICK

Hypothesis = a Hunch. Theory = THE big explanation. Law = what happens, Like clockwork. A hypothesis does NOT get promoted into a theory and a theory does NOT become a law. They are different jobs, not ranks.

SAY IT OUT LOUD

The exam trap is 'just a theory.' Explain out loud why calling evolution 'just a theory' misuses the word.

What makes a hypothesis a good one?

A hypothesis has to be able to fail

YES Testable

“Plants given fertilizer will
grow taller than plants without.”

a result could prove this wrong

NO Not testable

“Plants grow because
they want to.”

no result could ever prove this wrong

If nothing could disprove it, it is not a hypothesis. It is an opinion.

IN PLAIN WORDS

You have to be able to prove it wrong. If no possible result would ever make you say 'okay, I was wrong,' it is not science.

HOW TO SAY IT ON THE EXAM

A scientific hypothesis must be testable and falsifiable. It generates a specific prediction in if-then form, and the experiment must be capable of producing a result that would disconfirm it.

MEMORY TRICK

If nothing could prove it wrong, it is not a hypothesis. It is an opinion.

SAY IT OUT LOUD

Give one testable hypothesis and one untestable claim, and say why the second one fails.

How do you read and interpret a scientific graph?

IN PLAIN WORDS

X axis is usually the thing the scientist changed. Y axis is what they measured. Read the title, read both axis labels with their units, then look for the trend before looking at any single point.

HOW TO SAY IT ON THE EXAM

Convention places the independent variable on the x-axis and the dependent variable on the y-axis. Interpretation requires identifying the variables and units, the trend or relationship (positive, negative, none), the presence of error bars indicating variability, and the sample size before drawing conclusions.

MEMORY TRICK

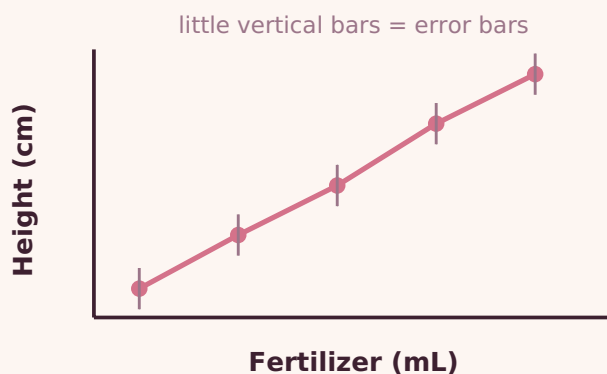
DRY MIX. Dependent, Responding, Y-axis. Manipulated, Independent, X-axis.

SAY IT OUT LOUD

Practice this sentence shape: 'As [independent] increased, [dependent] _____, which suggests _____.'

How do you construct a graph from your own data?

Building a graph, not just reading one



1. Independent on x, dependent on y

2. Label both axes WITH units

3. Pick the right type

bar = separate categories · line = continuous change

scatter = looking for a relationship

4. Scale evenly, start sensibly

uneven spacing is how graphs lie

5. Title it, and add error bars if you have them

IN PLAIN WORDS

Put what you changed on the bottom, what you measured on the side, label both with units, pick the right kind of graph, keep the spacing even, and give it a title. Bar graphs for separate categories, line graphs for something changing continuously, scatter plots when you are looking for a relationship.

HOW TO SAY IT ON THE EXAM

Graph construction follows convention: independent variable on the x-axis, dependent variable on the y-axis, both axes labeled with units, consistent interval scaling, and a descriptive title. Bar graphs display discrete categorical data, line graphs display continuous data across a range, and scatter plots display the relationship between two continuous variables. Error bars represent variability, typically standard deviation or standard error, and overlapping error bars suggest the difference between groups may not be meaningful.

MEMORY TRICK

DRY MIX again for the axes. For picking the type: Bars for Buckets, Lines for Long continuous change, Scatter for Searching for a relationship.

SAY IT OUT LOUD

Take the fertilizer experiment and describe the graph you would draw. Say the axis labels including the units, then say which type of graph and why.

How do you critique somebody else's experimental design?

IN PLAIN WORDS

Ask four things. Was there a control group? Did they only change one thing at a time? Did they test enough subjects? Could something else explain the result?

HOW TO SAY IT ON THE EXAM

Evaluate for: presence of an appropriate control group, isolation of a single independent variable, adequate sample size, random assignment, replication, blinding where relevant, and confounding variables that could offer an alternative explanation for the observed effect. Correlation in the data does not establish causation.

MEMORY TRICK

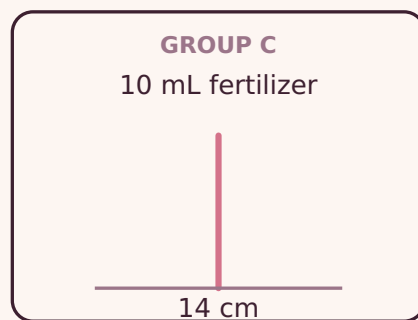
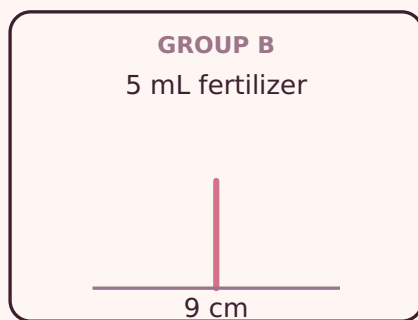
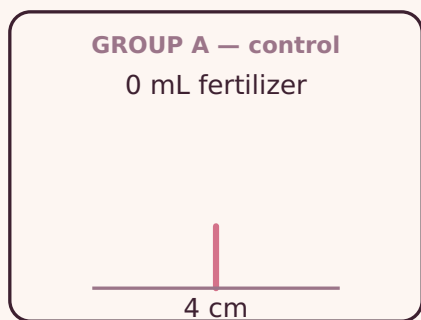
C-SOC: Control group, Sample size, One variable, Confounders.

SAY IT OUT LOUD

Design a deliberately bad experiment, then tear it apart out loud. Finding the flaw is easier than describing it, so make yourself describe it.

How would you design an experiment from scratch?

One experiment: does fertilizer make basil grow taller?



Independent = amount of fertilizer (what I change on purpose)

Dependent = height in cm (what I measure)

Controlled = same soil, light, water, pot, basil variety · **Control group** = Group A

IN PLAIN WORDS

Start with a question, turn it into a testable hypothesis, decide the one thing you will change and the one thing you will measure, list everything you will keep identical, include a group that gets no treatment, use enough subjects, repeat it, then decide in advance how you will analyze the results.

HOW TO SAY IT ON THE EXAM

Experimental design requires: a testable, falsifiable hypothesis with a specific prediction; a single manipulated independent variable; a defined and measurable dependent variable; identified controlled variables held constant across groups; an appropriate control group; adequate sample size with random assignment; replication; and a predetermined method of analysis. Blinding is used where expectation could bias measurement.

MEMORY TRICK

Same C-SOC checklist you use to tear an experiment apart, just run forwards instead of backwards. If you can critique one, you can build one.

SAY IT OUT LOUD

Design an experiment testing whether background music affects study performance. Say the hypothesis, the independent variable, the dependent variable, three controlled variables, and the control group.

Why do scientists publish and peer review?

IN PLAIN WORDS

Because one person can fool themselves. Other experts pick the work apart, and other labs try to get the same result. Anything that survives that is far more trustworthy.

HOW TO SAY IT ON THE EXAM

Peer review subjects methodology and conclusions to evaluation by independent experts before publication. Replication by independent groups tests reliability. Community scrutiny is a built-in error-correcting mechanism, and it is what distinguishes science from individual assertion.

MEMORY TRICK

Science is a group project where everybody is trying to catch your mistakes, and that is the feature, not the bug.

SAY IT OUT LOUD

Explain why a single unreplicated study is weak evidence, even if the scientist is famous.

Basic science vs applied science.

IN PLAIN WORDS

Basic science is finding out how something works because nobody knows yet. Applied science uses what we already know to solve a specific problem.

HOW TO SAY IT ON THE EXAM

Basic (pure) science expands fundamental knowledge without a predetermined practical application. Applied science uses existing knowledge to address a defined problem. Basic research frequently enables later applied breakthroughs that were not foreseeable at the time.

MEMORY TRICK

Basic asks 'how does this work?' Applied asks 'how do I fix this?' Basic today is applied in twenty years.

SAY IT OUT LOUD

Give a real example where basic research later made a medical treatment possible.

What Is Life — picture sheet

What counts as alive, and how biology stacks from atoms to the biosphere. Every diagram from this section, no text. Ten-minute review.

The checklist for being alive

Made of cells

one or more

Homeostasis

stays steady inside

Uses energy

metabolism

Grows

and develops

Responds

to its surroundings

Reproduces

makes more

Has DNA

heritable code

Evolves

as a population

A virus only ticks the last two boxes.

No cells, no metabolism, no homeostasis, no independent reproduction.

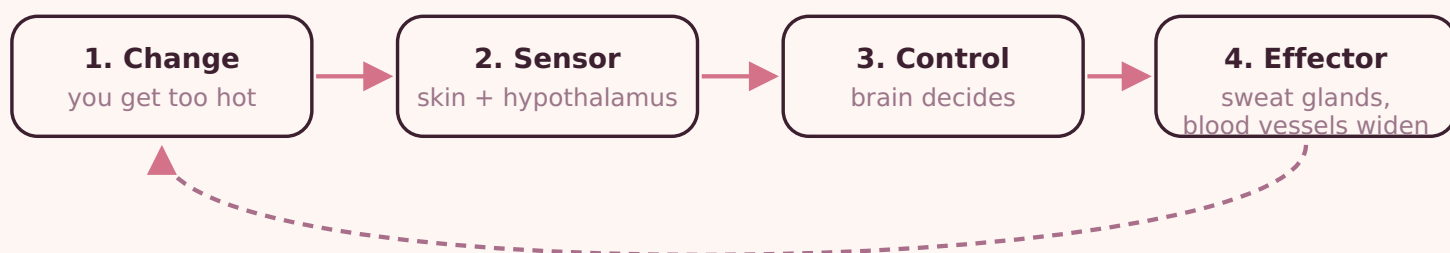
That is the whole “are viruses alive” debate in one picture.

Scoring a virus against the checklist

Made of cells	NO
Own metabolism / energy	NO
Homeostasis	NO
Grows and develops	NO
Reproduces on its own	NO
Has genetic material	YES
Evolves by natural selection	YES

Two out of seven. That is why the answer is “on the border of life,” not a plain yes or no.

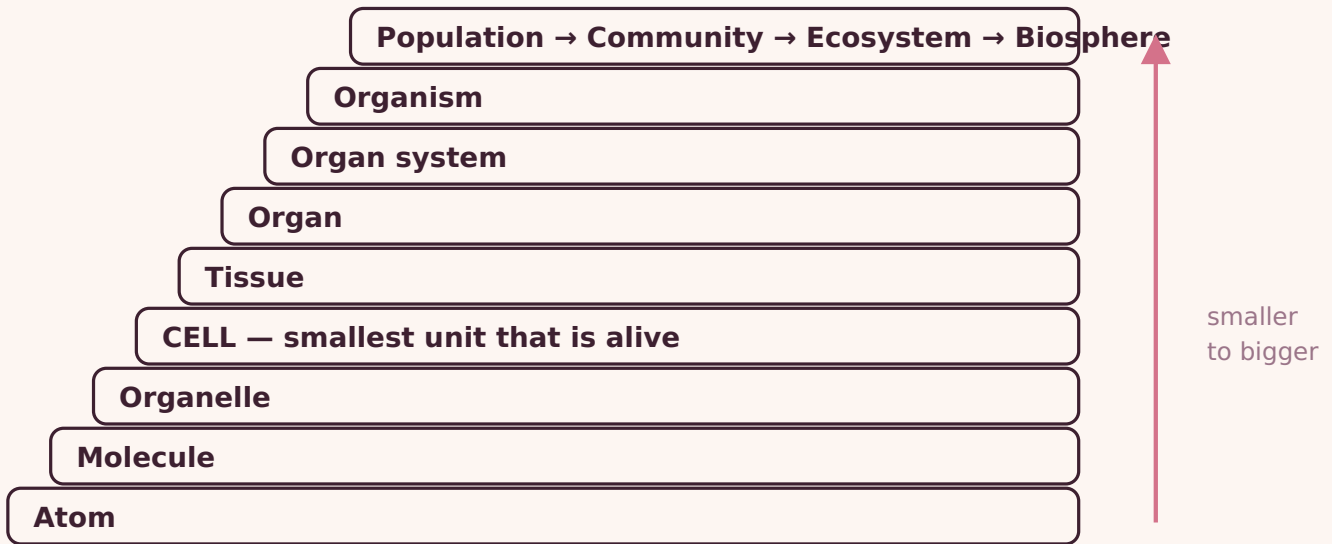
Negative feedback: how your body holds the line



5. Temperature drops back down, which switches the response off

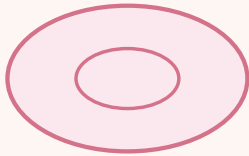
It is called NEGATIVE feedback because the response reverses the original change.

Levels of biological organization



Each step up gains properties the step below did not have. That is emergence.

Shape is the reason it can do its job



red blood cell

dented disc, no nucleus

More surface area

more room to pick up oxygen

Flexible and thin

squeezes single file through capillaries

No nucleus

more space inside for hemoglobin

The pattern to use

"It is shaped like ___,
which means it can ___,
which is why it ___."

Works for mitochondria folds,
root hairs, hair keratin, anything.

What Is Life

Ch 1.2 · 9 cards

What properties define something as alive?

The checklist for being alive

Made of cells

one or more

Homeostasis

stays steady inside

Uses energy

metabolism

Grows

and develops

Responds

to its surroundings

Reproduces

makes more

Has DNA

heritable code

Evolves

as a population

A virus only ticks the last two boxes.

No cells, no metabolism, no homeostasis, no independent reproduction.

That is the whole “are viruses alive” debate in one picture.

IN PLAIN WORDS

Living things are built from cells, they keep their insides stable, they use energy, they grow and develop, they react to their surroundings, they reproduce, they carry genetic instructions, and their populations evolve.

HOW TO SAY IT ON THE EXAM

The characteristics of life are: cellular organization, homeostasis, metabolism (energy processing), growth and development, response to stimuli, reproduction, heredity via genetic material, and evolutionary adaptation at the population level.

MEMORY TRICK

Mrs Gren, plus cells: Movement/metabolism, Reproduction, Sensitivity, Growth, Respiration, Excretion, Nutrition. Or remember it as: a living thing must EAT, REACT, GROW, COPY ITSELF, and STAY BALANCED.

SAY IT OUT LOUD

Pick your phone. Which properties does it have, which does it lack? That comparison is how you prove you understand the list instead of reciting it.

Are viruses alive? Argue both sides.

Scoring a virus against the checklist

Made of cells	NO
Own metabolism / energy	NO
Homeostasis	NO
Grows and develops	NO
Reproduces on its own	NO
Has genetic material	YES
Evolves by natural selection	YES

Two out of seven. That is why the answer is “on the border of life,” not a plain yes or no.

IN PLAIN WORDS

Not alive: a virus is not made of cells, it cannot make its own energy, and it cannot copy itself without hijacking someone else's machinery. Alive: it carries genetic material, it evolves fast, and it reproduces once it is inside a host.

HOW TO SAY IT ON THE EXAM

Arguments against life status: viruses are acellular, lack metabolic machinery and ribosomes, cannot generate ATP, and are obligate intracellular parasites incapable of autonomous replication. Arguments for: they possess nucleic acid genomes, undergo mutation and natural selection, replicate and produce progeny, and occupy ecological roles. Many biologists resolve this by describing viruses as being at the border of life, or as alive only within a host cell.

MEMORY TRICK

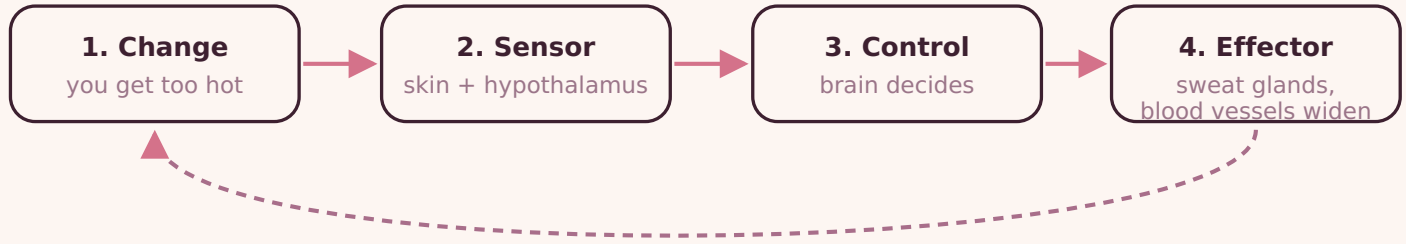
A virus is a USB stick. It holds real information, but plug it into nothing and it does nothing forever.

SAY IT OUT LOUD

The professor assigned a whole reading on this, so it is almost certainly a short answer. Practice a 4-sentence answer: one side, other side, the specific property that decides it, your position.

What is homeostasis? Give an example.

Negative feedback: how your body holds the line



5. Temperature drops back down, which switches the response off

It is called **NEGATIVE** feedback because the response reverses the original change.

IN PLAIN WORDS

Keeping your insides steady even when the outside changes. You stay around 98.6 F whether it is winter or summer.

HOW TO SAY IT ON THE EXAM

Homeostasis is the maintenance of a relatively stable internal environment despite external fluctuation, achieved through negative feedback loops involving a sensor, a control center, and an effector. Examples include thermoregulation by sweating and shivering, blood glucose regulation by insulin and glucagon, and blood pH held near 7.4 by buffer systems.

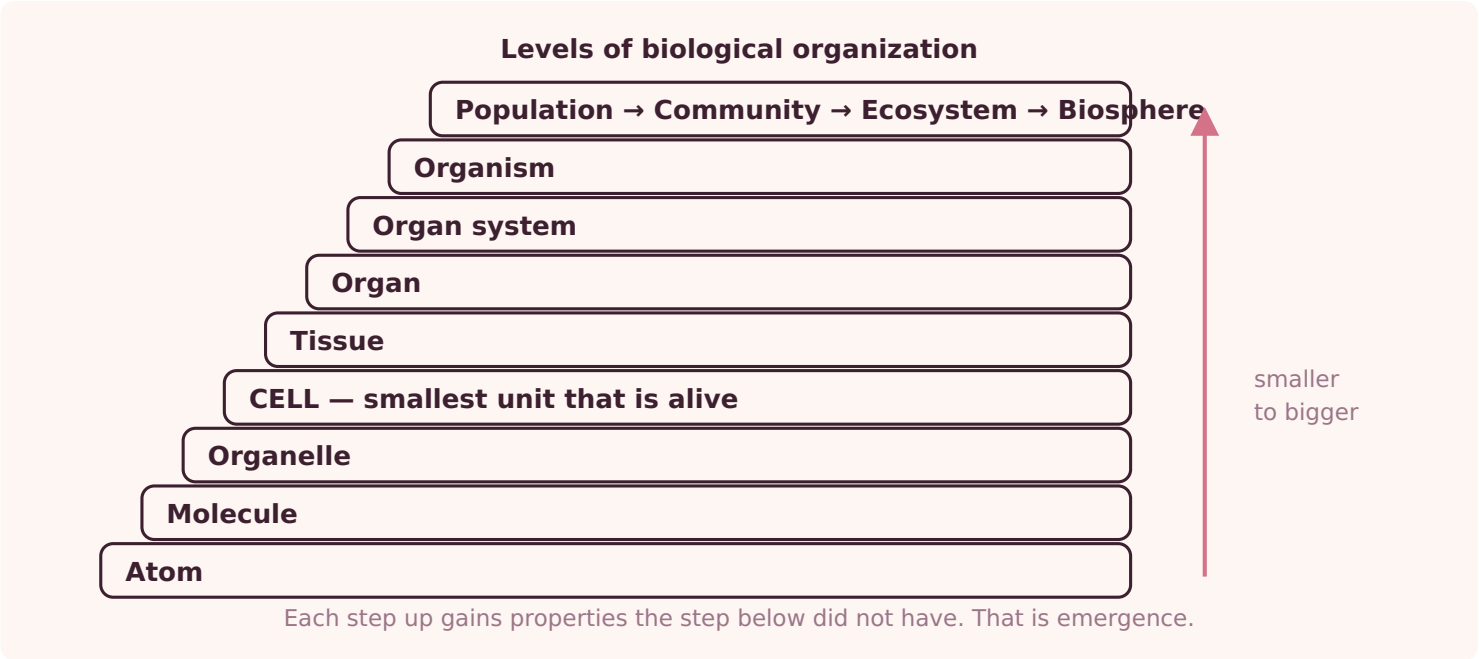
MEMORY TRICK

Home + stasis = keeping the house the same. A thermostat is literally a homeostasis machine.

SAY IT OUT LOUD

Trace one loop out loud: you get too hot, then what senses it, then what responds, then what happens to the temperature.

Rank the levels of biological organization.



IN PLAIN WORDS

Atom, molecule, organelle, cell, tissue, organ, organ system, organism, population, community, ecosystem, biosphere.

HOW TO SAY IT ON THE EXAM

The hierarchy proceeds from atom to molecule to macromolecule to organelle to cell to tissue to organ to organ system to organism to population to community to ecosystem to biosphere. Levels below the cell are subcellular; levels above the organism are ecological.

MEMORY TRICK

All My Organs Can Take Over Our Own Planet, Creating Excellent Biology: Atom, Molecule, Organelle, Cell, Tissue, Organ, Organ system, Organism, Population, Community, Ecosystem, Biosphere.

SAY IT OUT LOUD

Say where the line is: which level is the first one that is actually alive, and why does that line sit there?

What is emergence? Give an example using two levels.

IN PLAIN WORDS

When you combine parts, the combination can do something none of the parts could do alone. A single heart muscle cell twitches. Put millions together into a heart and you get a pump that moves blood through a whole body.

HOW TO SAY IT ON THE EXAM

Emergent properties are characteristics that arise from the specific arrangement and interaction of components at a lower level of organization, and that are not present in the isolated components. Example: individual neurons transmit electrical signals, but the organized network of neurons in a brain produces memory and consciousness, which no single neuron possesses.

MEMORY TRICK

The wetness of water. One H₂O molecule is not wet. A glass of them is. Nothing was added except arrangement.

SAY IT OUT LOUD

Make up your OWN emergence example using two levels from the ladder. Reciting the textbook one does not count unless you can explain the logic behind it.

What is reductionism? Give an example using two levels.

IN PLAIN WORDS

Breaking something complicated into its pieces to understand it. You study individual DNA molecules to understand how a whole organism inherits traits.

HOW TO SAY IT ON THE EXAM

Reductionism is the approach of explaining a complex system by analyzing its simpler constituent parts. Example: studying the molecular structure of DNA (molecule level) to explain heredity in organisms (organism level). Reductionism is powerful but incomplete, because it can miss emergent properties that only exist at higher levels.

MEMORY TRICK

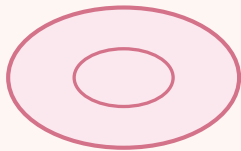
Reduce = zoom IN. Emerge = zoom OUT. They are opposite directions on the same ladder, and biology needs both.

SAY IT OUT LOUD

Explain why reductionism alone is not enough to understand a living thing. Link it back to emergence.

Give an example of structure relating to function.

Shape is the reason it can do its job



red blood cell

dented disc, no nucleus

More surface area

more room to pick up oxygen

Flexible and thin

squeezes single file through capillaries

No nucleus

more space inside for hemoglobin

The pattern to use

"It is shaped like ___,
which means it can ___,
which is why it ___."

Works for mitochondria folds,
root hairs, hair keratin, anything.

IN PLAIN WORDS

The shape of a thing is the reason it can do its job. Red blood cells are flat discs with a dent in the middle, which gives them more surface area to grab oxygen and lets them squeeze through narrow capillaries.

HOW TO SAY IT ON THE EXAM

Form determines function at every level of biology. Examples: the biconcave disc of an erythrocyte maximizes surface-area-to-volume ratio for gas exchange; the alpha helix of keratin provides tensile strength in hair; root hairs increase absorptive surface area; the folded cristae of mitochondria increase the membrane area available for ATP synthesis.

MEMORY TRICK

If you can describe the shape, you can usually guess the job. A funnel, a hook, a net, a sponge. Biology reuses the same shapes for the same reasons.

SAY IT OUT LOUD

Pick a structure and reason forward: 'it is shaped like ___, which means it can ___, which is why it ___.'

What is the most fundamental unit of life, and why?

IN PLAIN WORDS

The cell. It is the smallest thing that can do everything on the life checklist by itself. Anything smaller, like an organelle or a molecule, cannot survive alone.

HOW TO SAY IT ON THE EXAM

The cell is the basic unit of life, as stated in cell theory: all living organisms are composed of one or more cells, the cell is the fundamental unit of structure and function, and all cells arise from pre-existing cells. The cell is the smallest entity that independently exhibits all characteristics of life, including metabolism, homeostasis, and reproduction.

MEMORY TRICK

Cells are the smallest thing that can still tick every box on the life checklist.

SAY IT OUT LOUD

Explain why a mitochondrion is not the fundamental unit of life even though it has DNA.

Put natural selection in your own words.

IN PLAIN WORDS

Individuals in a population are all slightly different. Some of those differences make survival and reproduction easier in that particular environment. Those individuals have more offspring, and their traits become more common over generations. The environment does the choosing.

HOW TO SAY IT ON THE EXAM

Natural selection requires: heritable variation within a population, overproduction of offspring, differential survival and reproduction based on that variation, and change in allele frequency over generations. Individuals do not evolve; populations do. Selection acts on existing variation and does not create traits on demand.

MEMORY TRICK

VISTA: Variation, Inheritance, Selection, Time, Adaptation. Or shorter: nature does not create the winners, it just removes the losers.

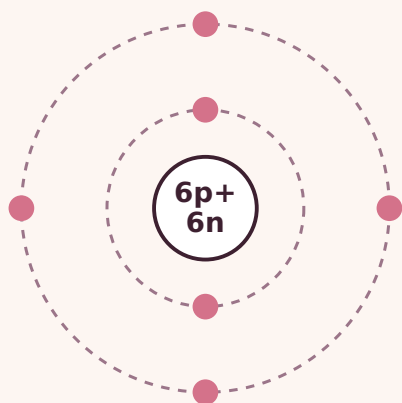
SAY IT OUT LOUD

The most common exam error is saying an organism 'adapts on purpose' or 'evolves because it needs to.' Say the correct version out loud until the wrong phrasing feels weird.

Chemistry of Life — picture sheet

The chemistry everything else in this course sits on top of. Every diagram from this section, no text. Ten-minute review.

Carbon atom



Atomic number = 6

protons only. This is the ID badge.

Mass number = 12

protons + neutrons

Inner shell holds 2

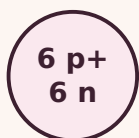
Next shell wants 8 (octet)

Carbon has 4 out there, needs 4 more, so it forms 4 bonds. Every time.

nucleus = protons (+) and neutrons (0) · electrons (–) orbit in shells

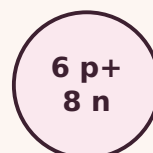
Isotopes: same element, extra neutrons

Carbon-12



stable, most common

Carbon-14



unstable, decays over time

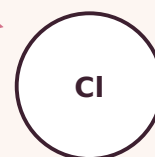
h are still carbon and behave identically in chemistry. Only the weight and stability differ, which is what makes carbon-14

How an ion is made

electron is handed over, not shared



1 lonely electron
in the outer shell

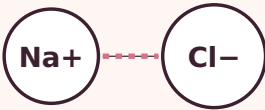


7 in the outer shell,
needs exactly 1 more

Opposite charges attract, and that attraction IS the ionic bond. In water it lets go and they separate again.

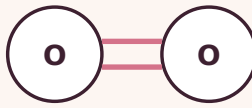
Three bonds, strongest to weakest

Ionic



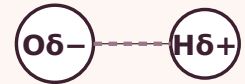
Electron is fully GIVEN away.
Opposite charges attract.
Falls apart in water → ions

Covalent



Electrons are SHARED.
Equal share = nonpolar
Unequal share = polar
Strongest of the three

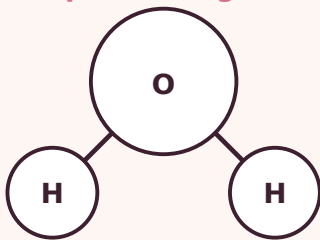
Hydrogen



No sharing at all. Just a weak tug between a $\delta+$ H and a nearby $\delta-$ O, N or F.
Weak alone, mighty in bulk

Why water is weird (and why you are alive because of it)

$\delta-$ partial negative



Oxygen hogs the shared electrons (higher electronegativity) → polar

1. Cohesion & adhesion

water sticks to water, and to other stuff → water climbs a tree trunk

2. High heat capacity

takes a lot of heat to warm up → your body temp stays steady

3. High heat of vaporization

escaping steam carries heat off → sweating cools you down

4. Universal solvent

dissolves anything polar or charged → blood, cytoplasm, ocean

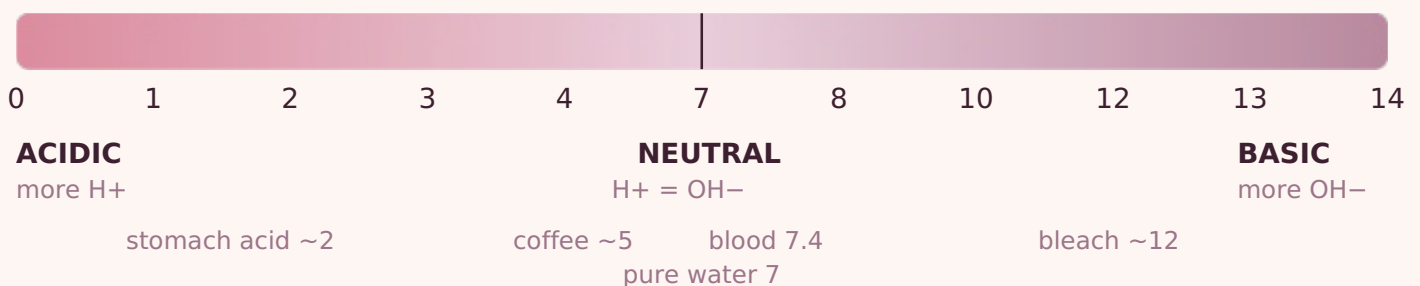
All four come from ONE thing: hydrogen bonds between polar water molec

Why water dissolves salt



Water surrounds each ion with its opposite partial charge. That jacket is a hydration shell, and it keeps the ions apart so they cannot snap back together. Nonpolar oil gets no such jacket.

The pH scale



step is 10×. pH 4 is TEN times more acidic than pH 5, not one unit more. Buffers soak up spare H^+ or OH^- to hold blood at

Strong vs weak: how much of it lets go of its H⁺

STRONG acid (HCl)



nearly every molecule releases its H⁺
dissociates completely, no going back

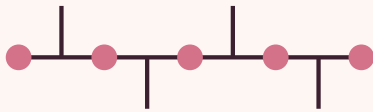
WEAK acid (carbonic)



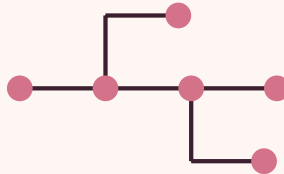
only a few let go, and they can take it back
a reversible reaction, both directions at once

That reversibility is exactly what makes a weak acid able to work as a buffer. A strong acid cannot buffer anything, because it has nothing left to give back.

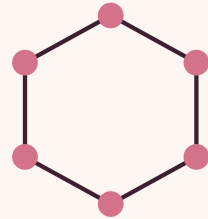
Four bonds is why carbon can build anything



Straight chain



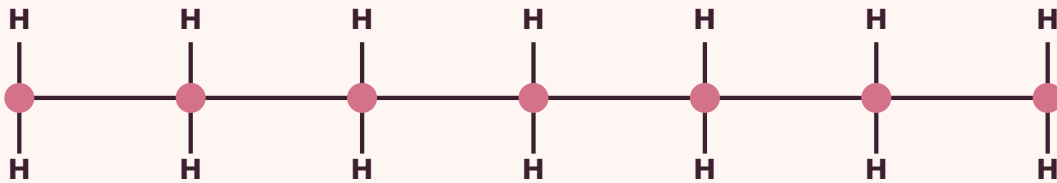
Branched



Ring

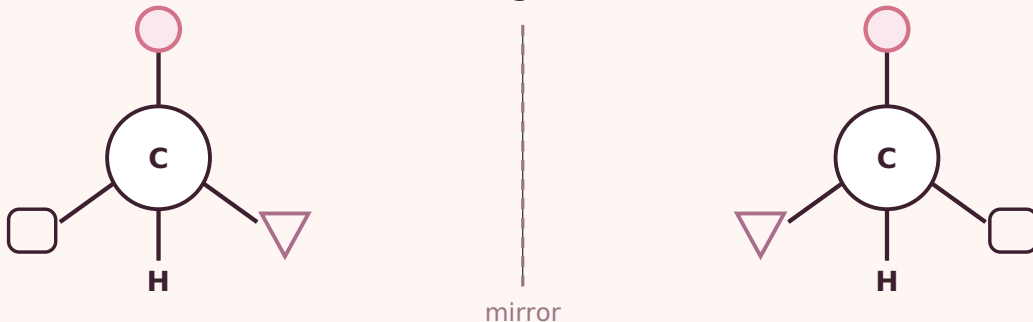
onds would only give you a line. Three gives you flat shapes. Four gives you three dimensions, and life needs three dimer

A hydrocarbon: only C and H, packed with energy



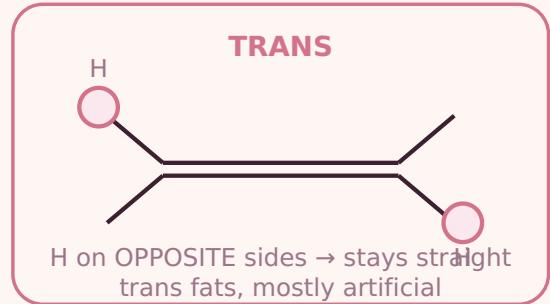
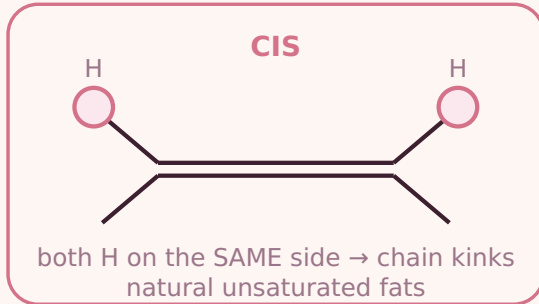
f those C-H bonds is a little packed battery, and none of them are polar, so the whole chain hides from water. That is a fa

Enantiomers: mirror images that are not the same



groups, same central carbon, but you cannot rotate one to land on the other. Your enzymes are shaped gloves, so only on

Cis vs trans: same bonds, different side of the double bond



A double bond cannot spin, which is the whole reason these are two different molecules instead of one.

Functional groups: the “charms” clipped onto a carbon skeleton

Hydroxyl —OH

polar · makes things water-loving
alcohols, sugars

Carbonyl —C=O

polar · ketone (middle) or
aldehyde (end of chain)

Carboxyl —COOH

ACIDIC · donates H⁺
amino acids, fatty acids

Amino —NH₂

BASIC · accepts H⁺
amino acids, the other end

Sulfhydryl —SH

forms disulfide bridges
locks protein shape (hair!)

Phosphate —PO₄²⁻

negative · energy transfer
ATP, DNA backbone, phospholipids

Methyl —CH₃

NONPOLAR · no reactivity
a tag, not a tool. Marks DNA.

Chemistry of Life

Ch 2 · 19 cards

What is matter, and what is an element?

IN PLAIN WORDS

Matter is anything that takes up space and has mass. An element is a pure substance that cannot be broken down any further by chemical means. There are 118 of them, and just four make up about 96 percent of your body.

HOW TO SAY IT ON THE EXAM

Matter is any substance possessing mass and occupying volume. An element is a substance composed of a single type of atom that cannot be decomposed into simpler substances by ordinary chemical reactions. Carbon, hydrogen, oxygen, and nitrogen constitute approximately 96 percent of living matter by mass.

MEMORY TRICK

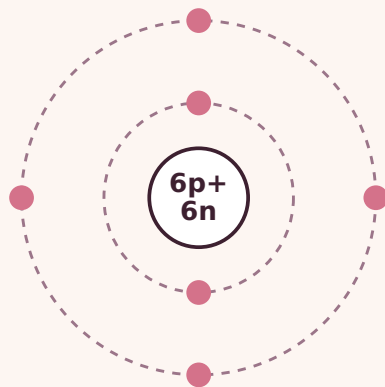
CHON is the four horsemen of biology: Carbon, Hydrogen, Oxygen, Nitrogen. Add phosphorus and sulfur and you have basically everything: C HOPKINS.

SAY IT OUT LOUD

What makes one element different from another? Answer with a number, not a word.

Describe the structure of an atom.

Carbon atom



Atomic number = 6

protons only. This is the ID badge.

Mass number = 12

protons + neutrons

Inner shell holds 2

Next shell wants 8 (octet)

Carbon has 4 out there, needs 4 more, so it forms 4 bonds. Every time.

nucleus = protons (+) and neutrons (0) · electrons (–) orbit in shells

IN PLAIN WORDS

A dense center made of protons (positive) and neutrons (no charge), surrounded by electrons (negative) that move around in layers called shells.

HOW TO SAY IT ON THE EXAM

The nucleus contains protons (charge +1, mass ~1 amu) and neutrons (charge 0, mass ~1 amu). Electrons (charge -1, negligible mass) occupy orbitals within electron shells surrounding the nucleus. In a neutral atom, the number of protons equals the number of electrons.

MEMORY TRICK

Protons are Positive. Neutrons are Neutral. That leaves the electron to be the negative one.

SAY IT OUT LOUD

Where is nearly all the mass, and where is nearly all the volume? They are not the same place.

How is an element's identity determined? Atomic number vs mass number.

IN PLAIN WORDS

The number of protons IS the element. Change the protons and it becomes a different element entirely. Atomic number = protons. Mass number = protons plus neutrons.

HOW TO SAY IT ON THE EXAM

Atomic number equals the number of protons and uniquely defines the element. Mass number equals the sum of protons and neutrons. Atomic mass on the periodic table is a weighted average across naturally occurring isotopes, which is why it is rarely a whole number.

MEMORY TRICK

Protons are the social security number of the atom. Everything else can change; that cannot.

SAY IT OUT LOUD

An atom has 6 protons, 8 neutrons, 6 electrons. Name the element, the atomic number, the mass number, and whether it is an ion or an isotope.

What are isotopes, and why do they matter in medicine?

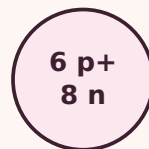
Isotopes: same element, extra neutrons

Carbon-12



stable, most common

Carbon-14



unstable, decays over time

Both are still carbon and behave identically in chemistry. Only the weight and stability differ, which is what makes carbon-14 useful in medicine.

IN PLAIN WORDS

Same element, different number of neutrons. Same chemistry, different weight. Some are unstable and give off radiation, and that radiation is something doctors can detect or aim.

HOW TO SAY IT ON THE EXAM

Isotopes are atoms of the same element with differing neutron numbers, therefore differing mass numbers but identical atomic numbers and chemical behavior. Radioisotopes decay at predictable rates. Applications include carbon-14 radiometric dating, radioactive tracers in PET imaging, iodine-131 in thyroid treatment, and targeted radiation therapy.

MEMORY TRICK

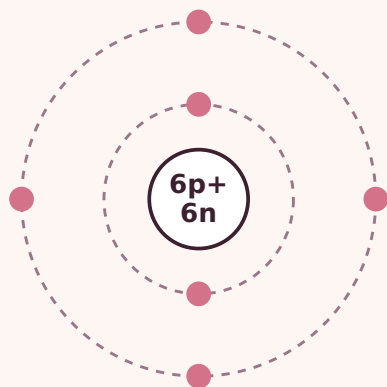
Iso = same, tope = place. Same place on the periodic table, different weight class. Isotope changes NEUTRONS; ion changes ELECTRONS. Different letters, different particles.

SAY IT OUT LOUD

Explain why an isotope behaves the same chemically but differently physically.

Explain electron shells and the octet rule.

Carbon atom



nucleus = protons (+) and neutrons (0) · electrons (–) orbit in shells

Atomic number = 6

protons only. This is the ID badge.

Mass number = 12

protons + neutrons

Inner shell holds 2

Next shell wants 8 (octet)

Carbon has 4 out there, needs 4 more, so it forms 4 bonds. Every time.

IN PLAIN WORDS

Electrons sit in layers around the nucleus. The first layer holds 2, the next ones want 8. An atom with a full outer layer is content and does not react. An atom with a partly filled outer layer will bond with something to fill it.

HOW TO SAY IT ON THE EXAM

Electrons occupy shells of increasing energy. The first shell holds 2 electrons, the second and third hold 8 each in biologically relevant elements. The octet rule states that atoms tend to gain, lose, or share valence electrons to achieve a full outer shell. Valence electron count therefore determines reactivity and bonding capacity. Noble gases have full valence shells and are inert.

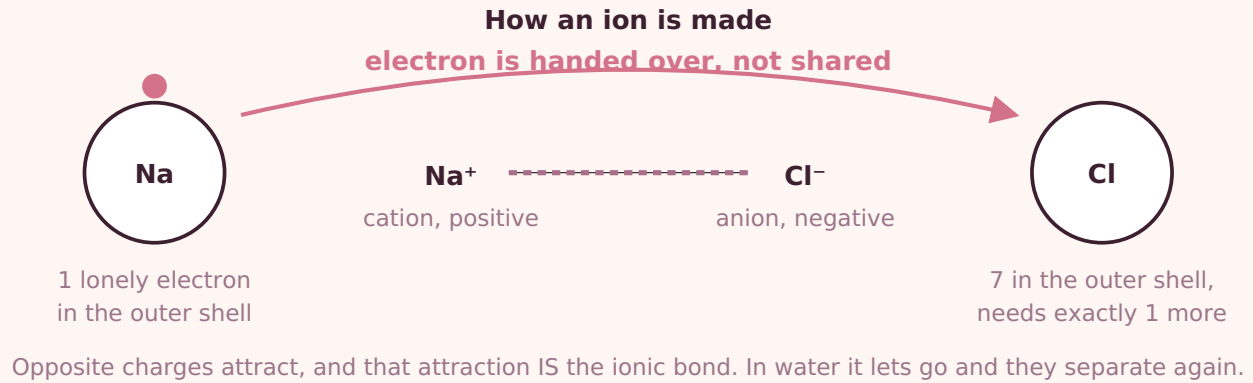
MEMORY TRICK

The outer shell is the atom's dating profile. Full = not looking. Missing one or two = extremely available.

SAY IT OUT LOUD

Carbon has 4 valence electrons. Predict how many bonds it forms and explain why, without saying 'because the book said four.'

What is an ion? Cation vs anion, and how does an ionic bond form?



IN PLAIN WORDS

An ion is an atom that gained or lost electrons, so it now has a charge. Lose electrons and you go positive (cation). Gain electrons and you go negative (anion). Two opposite ions stick together, and that attraction is the ionic bond. Living systems run on ions: sodium and potassium moving across membranes are what fire a nerve signal, calcium ions trigger muscle contraction and build bone, and chloride balances charge in blood.

HOW TO SAY IT ON THE EXAM

An ion is an atom or molecule with unequal proton and electron counts. A cation carries a net positive charge from electron loss; an anion carries a net negative charge from electron gain. An ionic bond is the electrostatic attraction between oppositely charged ions, formed by complete transfer of one or more electrons. Sodium chloride is the standard example; in aqueous solution it dissociates into Na^+ and Cl^- . Biologically important ionic species include Na^+ and K^+ , whose gradients across the plasma membrane generate the resting membrane potential and action potentials; Ca^{2+} , which triggers muscle contraction, neurotransmitter release, and forms hydroxyapatite in bone; Cl^- , which maintains charge balance and osmotic pressure; and Mg^{2+} , a cofactor for ATP-dependent enzymes.

MEMORY TRICK

CAT-ion is PAWS-itive. An ANION is A Negative ION. And ionic bonds are a breakup, not a partnership: one atom straight up takes the electron.

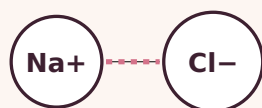
SAY IT OUT LOUD

Name three ions your body depends on and say what each one does. Then explain why ionic compounds fall apart in water but covalent ones usually do not.

Covalent bonds: polar vs nonpolar, and what is electronegativity?

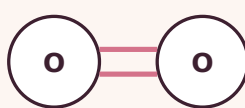
Three bonds, strongest to weakest

Ionic



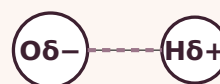
Electron is fully GIVEN away.
Opposite charges attract.
Falls apart in water → ions

Covalent



Electrons are SHARED.
Equal share = nonpolar
Unequal share = polar
Strongest of the three

Hydrogen



No sharing at all. Just a weak tug between a $\delta+$ H and a nearby $\delta-$ O, N or F.
Weak alone, mighty in bulk

IN PLAIN WORDS

A covalent bond is two atoms sharing electrons. If they pull equally, the sharing is fair and the bond is nonpolar. If one pulls harder, it hogs the electrons and gets a slightly negative end while the other gets a slightly positive end. That is polar. Electronegativity is just how hard an atom pulls.

HOW TO SAY IT ON THE EXAM

A covalent bond forms when two atoms share one or more electron pairs. Electronegativity is an atom's tendency to attract shared electrons. Equal or near-equal electronegativity produces a nonpolar covalent bond (O_2 , CH_4 , hydrocarbons). Unequal electronegativity produces a polar covalent bond with partial charges denoted delta-plus and delta-minus (H_2O , NH_3). Oxygen, nitrogen, and fluorine are the most electronegative elements in biology.

MEMORY TRICK

Electronegativity ranking to remember: FONCl, pronounced 'fonkle.' $\text{F} > \text{O} > \text{N} > \text{Cl}$. Oxygen is the electron thief of biology.

SAY IT OUT LOUD

Given any two atoms, explain the reasoning chain: electronegativity difference, then electron pull, then partial charges, then polarity.

What is a hydrogen bond, and why does something so weak matter so much?

IN PLAIN WORDS

When a hydrogen is attached to an oxygen or nitrogen, it becomes slightly positive and gets attracted to a slightly negative atom nearby. One of these is easy to break. But your body has trillions of them, and together they hold water, DNA, and proteins in shape.

HOW TO SAY IT ON THE EXAM

A hydrogen bond is a weak electrostatic attraction between a hydrogen atom covalently bonded to a highly electronegative atom (O, N, or F) and another electronegative atom bearing a partial negative charge. Individual hydrogen bonds are roughly 1/20 the strength of a covalent bond, but their collective effect is substantial. They produce the emergent properties of water, hold the DNA double helix together (A-T with 2, G-C with 3), and stabilize protein secondary structure.

MEMORY TRICK

Hydrogen bonds are velcro. One hook is nothing. Ten thousand hooks hold your jacket shut. And their weakness is the point: DNA has to be able to unzip.

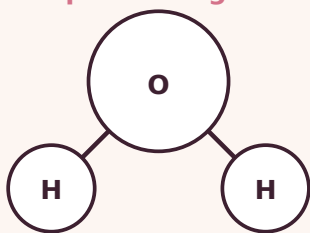
SAY IT OUT LOUD

Explain why it would actually be bad if hydrogen bonds were strong.

Why is water polar, and what does that cause?

Why water is weird (and why you are alive because of it)

δ^- partial negative



δ^+

δ^+

Oxygen hogs the shared electrons (higher electronegativity) → polar

1. Cohesion & adhesion

water sticks to water, and to other stuff → water climbs a tree trunk

2. High heat capacity

takes a lot of heat to warm up → your body temp stays steady

3. High heat of vaporization

escaping steam carries heat off → sweating cools you down

4. Universal solvent

dissolves anything polar or charged → blood, cytoplasm, ocean

All four come from ONE thing: hydrogen bonds between polar water molecules

IN PLAIN WORDS

Oxygen pulls the shared electrons harder than hydrogen does, so the oxygen side of a water molecule is slightly negative and the hydrogen side is slightly positive. Opposite ends attract each other, so water molecules cling together.

HOW TO SAY IT ON THE EXAM

Water's bent molecular geometry combined with oxygen's high electronegativity produces an asymmetric charge distribution: partial negative at oxygen, partial positive at each hydrogen. This permanent dipole allows each water molecule to form up to four hydrogen bonds with neighbors, and this hydrogen bonding network is the single cause of all of water's emergent properties.

MEMORY TRICK

Water is a tiny magnet with a Mickey Mouse shape. If the molecule were straight instead of bent, the charges would cancel and water would be boring.

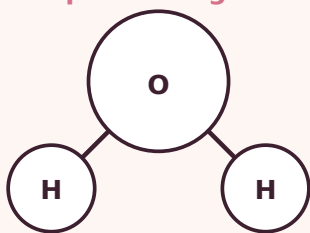
SAY IT OUT LOUD

Trace it: electronegativity, then partial charges, then hydrogen bonds, then ONE specific property. That chain is the answer to most water questions.

The four emergent properties of water, each with a biological example.

Why water is weird (and why you are alive because of it)

δ^- – partial negative



δ^+

δ^+

Oxygen hogs the shared electrons
(higher electronegativity) → polar

1. Cohesion & adhesion

water sticks to water, and to other stuff → water climbs a tree trunk

2. High heat capacity

takes a lot of heat to warm up → your body temp stays steady

3. High heat of vaporization

escaping steam carries heat off → sweating cools you down

4. Universal solvent

dissolves anything polar or charged → blood, cytoplasm, ocean

All four come from ONE thing: hydrogen bonds between polar water molecules

IN PLAIN WORDS

1. It sticks to itself and to other things, so water climbs up trees. 2. It takes a lot of heat to warm up, so your body temperature stays steady. 3. It takes a lot of heat to evaporate, so sweating cools you off. 4. It dissolves anything with a charge, so it can carry nutrients through blood.

HOW TO SAY IT ON THE EXAM

Cohesion (water to water) and adhesion (water to other surfaces) produce surface tension and capillary action, which drive transpirational water transport in xylem. High specific heat capacity means water absorbs substantial heat with minimal temperature change, buffering organismal and global temperature. High heat of vaporization means evaporation removes large amounts of heat, enabling evaporative cooling through perspiration. Polarity makes water a solvent for polar and ionic solutes, forming hydration shells, which is why cytoplasm, blood plasma, and interstitial fluid are aqueous.

MEMORY TRICK

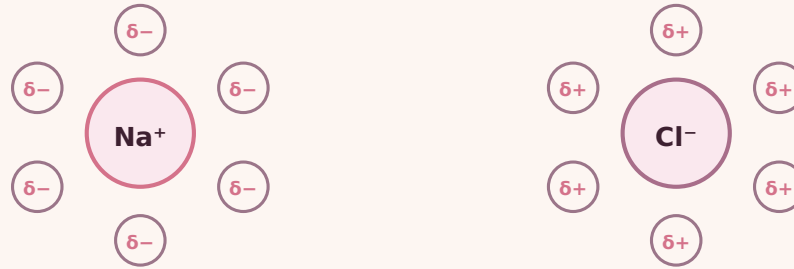
CHHS, say it as 'see H-H-S': Cohesion/adhesion, High heat capacity, High heat of vaporization, Solvent. Every one of them traces back to hydrogen bonding.

SAY IT OUT LOUD

For each property, name the biological consequence. Property alone gets partial credit; property plus consequence gets full credit.

Hydrophilic vs hydrophobic, and why water is called the universal solvent.

Why water dissolves salt



Water surrounds each ion with its opposite partial charge. That jacket is a hydration shell, and it keeps the ions apart so they cannot snap back together. Nonpolar oil gets no such jacket.

IN PLAIN WORDS

Hydrophilic means water-loving: anything polar or charged mixes with water. Hydrophobic means water-fearing: anything nonpolar, like oil, refuses to mix and clumps together instead.

HOW TO SAY IT ON THE EXAM

Hydrophilic substances are polar or ionic and form favorable interactions with water, becoming solvated by hydration shells. Hydrophobic substances are nonpolar and cannot form hydrogen bonds with water, so they aggregate to minimize contact with it. Water is described as a universal solvent because it dissolves a wider range of polar and ionic substances than any other liquid, not because it dissolves literally everything. The hydrophobic effect drives membrane formation and protein folding.

MEMORY TRICK

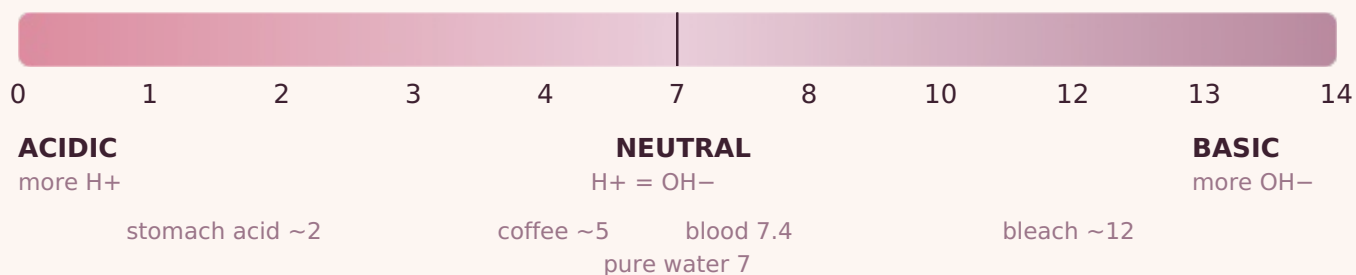
Philic = friendly. Phobic = phobia, afraid. And remember: LIKE DISSOLVES LIKE. Polar dissolves polar, nonpolar dissolves nonpolar.

SAY IT OUT LOUD

Explain why oil and water separate. The real answer is about what water does, not what oil does.

Acids, bases, and the pH scale.

The pH scale



1 step is 10 $\times$. pH 4 is TEN times more acidic than pH 5, not one unit more. Buffers soak up spare H^+ or OH^- to hold blood at

IN PLAIN WORDS

An acid dumps extra H^+ into a solution. A base soaks H^+ up, or adds OH^- . pH measures how much H^+ is floating around: low pH means lots of H^+ and acidic, high pH means little H^+ and basic, 7 is neutral.

HOW TO SAY IT ON THE EXAM

An acid is a proton donor, increasing hydrogen ion concentration. A base is a proton acceptor, decreasing hydrogen ion concentration or increasing hydroxide ion concentration. pH is the negative base-10 logarithm of hydrogen ion concentration, so each unit represents a tenfold change. At pH 7, hydrogen and hydroxide concentrations are equal. Human blood is maintained near 7.4.

MEMORY TRICK

Acid Adds H^+ . Base Bags H^+ . And the scale is backwards on purpose: MORE H^+ means LOWER number, because of the negative in the log.

SAY IT OUT LOUD

How many times more acidic is pH 3 than pH 6? Say the reasoning, not just the number.

Strong vs weak acids and bases.

Strong vs weak: how much of it lets go of its H^+

STRONG acid (HCl)



nearly every molecule releases its H^+ dissociates completely, no going back

WEAK acid (carbonic)



only a few let go, and they can take it back a reversible reaction, both directions at once

That reversibility is exactly what makes a weak acid able to work as a buffer. A strong acid cannot buffer anything, because it has nothing left to give back.

IN PLAIN WORDS

A strong acid gives up almost all of its H^+ the moment it hits water, and it does not take them back. A weak acid only lets a few go, and it can grab them again. That ability to go both directions is what makes weak acids useful in the body.

HOW TO SAY IT ON THE EXAM

Strong acids and bases dissociate essentially completely in aqueous solution, so hydrochloric acid becomes H^+ and Cl^- with effectively no reverse reaction. Weak acids and bases dissociate only partially and exist in equilibrium with their undissociated form, meaning the reaction proceeds in both directions simultaneously. Because a weak acid and its conjugate base coexist, the pair can absorb added H^+ or release H^+ as needed, which is the mechanism behind every biological buffer system, including bicarbonate in blood.

MEMORY TRICK

Strong acids are a one-way door. Weak acids are a revolving door, and only a revolving door can let traffic go both ways, which is exactly what a buffer has to do.

SAY IT OUT LOUD

Explain why a buffer has to be made from a WEAK acid. Then say what would happen to blood if the bicarbonate system used a strong acid instead.

What is a buffer, and why is it crucial?

IN PLAIN WORDS

A buffer is a chemical sponge for acidity. When acid shows up it absorbs H^+ , and when things get too basic it releases H^+ , so the pH barely moves.

HOW TO SAY IT ON THE EXAM

A buffer is a system, typically a weak acid and its conjugate base, that resists changes in pH by accepting or donating hydrogen ions. The bicarbonate buffer system (carbonic acid and bicarbonate) maintains blood pH within roughly 7.35 to 7.45. Deviation outside this narrow range denatures enzymes and is rapidly fatal, which is why buffering is non-optional for life.

MEMORY TRICK

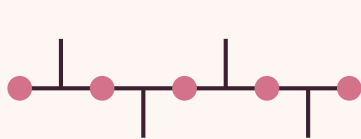
A buffer is a shock absorber for pH. Enzymes are the reason it matters: wrong pH, wrong protein shape, no reaction, dead.

SAY IT OUT LOUD

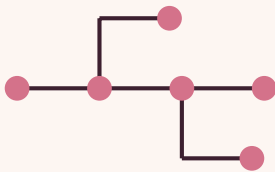
Connect two chapters out loud: what does a pH change physically DO to a protein? That link between chapters is exactly the kind of depth question that earns full credit.

Why is carbon the backbone of life?

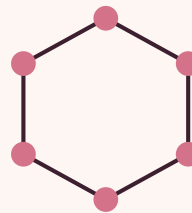
Four bonds is why carbon can build anything



Straight chain



Branched



Ring

bonds would only give you a line. Three gives you flat shapes. Four gives you three dimensions, and life needs three dimensions.

IN PLAIN WORDS

Carbon has four open bonding spots, so it can link to four different things at once, including other carbons. That lets it build chains, branches, and rings of basically unlimited complexity. No other common element is that versatile.

HOW TO SAY IT ON THE EXAM

Carbon has four valence electrons and forms four stable covalent bonds. It bonds readily with itself (catenation) to form chains, branched structures, and rings, and it bonds with hydrogen, oxygen, nitrogen, phosphorus, and sulfur. This tetravalence combined with moderate bond strength generates the structural diversity required for biological macromolecules.

MEMORY TRICK

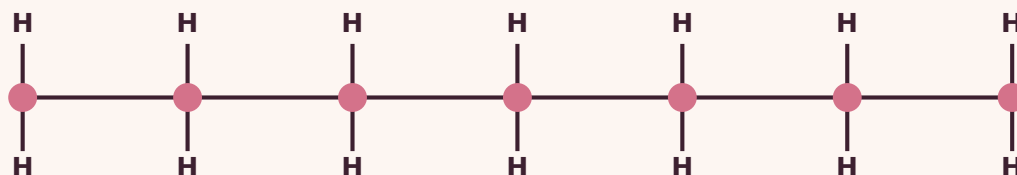
Carbon is LEGO. Four connection points means you can build anything. Silicon has four too, but its bonds are clumsier and it cannot make the same variety.

SAY IT OUT LOUD

Explain why four bonds specifically creates diversity. Two would give you a chain, three gives flat shapes, four gives 3D branching.

What are hydrocarbons, and why do they store energy well?

A hydrocarbon: only C and H, packed with energy



of those C-H bonds is a little packed battery, and none of them are polar, so the whole chain hides from water. That is a fatt

IN PLAIN WORDS

Molecules made of only carbon and hydrogen. They form because carbon links to itself in chains and fills every leftover bonding spot with hydrogen. Their C-H bonds hold a lot of energy, and because they are nonpolar they do not mix with water, so the body can pack them together tightly.

HOW TO SAY IT ON THE EXAM

Hydrocarbons consist exclusively of carbon and hydrogen joined by nonpolar covalent bonds. They form by catenation, carbon bonding to carbon to build a skeleton, with hydrogen saturating the remaining valence positions. The many energy-rich C-H bonds release substantial energy on oxidation, making hydrocarbon chains efficient energy storage. Their nonpolarity makes them hydrophobic, so they aggregate rather than dissolve. Fatty acid tails are hydrocarbon chains, which is why fat stores more than twice the energy per gram of carbohydrate.

MEMORY TRICK

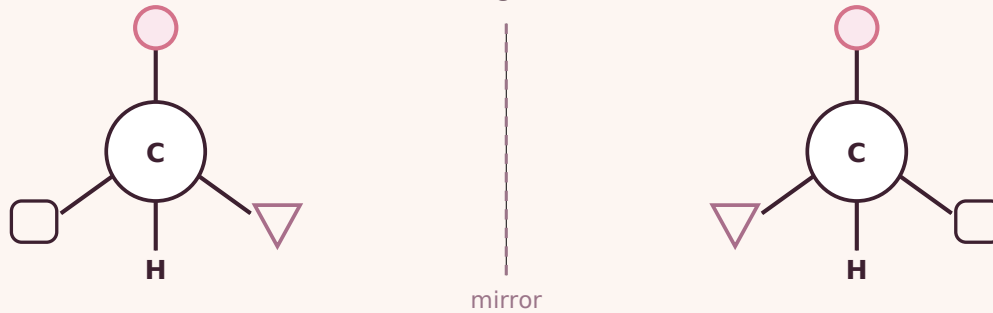
Hydrocarbon = just H and C, nothing else invited. All those C-H bonds are packed batteries.

SAY IT OUT LOUD

Explain why fat stores more energy per gram than sugar, using the words C-H bonds and water.

Isomers: structural, cis-trans, and enantiomers.

Enantiomers: mirror images that are not the same

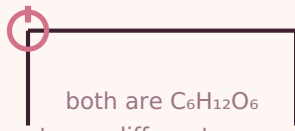


...r groups, same central carbon, but you cannot rotate one to land on the other. Your enzymes are shaped gloves, so only one

Aldose vs ketose: where is the C=O?

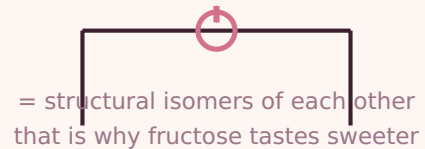
ALDOSE — glucose

C=O sits at the END of the chain



KETOSE — fructose

C=O sits in the MIDDLE of the chain



IN PLAIN WORDS

Isomers are molecules with the exact same atoms arranged differently. Structural isomers have the atoms hooked up in a different order. Cis-trans isomers have the same order but different 3D positions around a double bond. Enantiomers are mirror images, like your left and right hand.

HOW TO SAY IT ON THE EXAM

Isomers share a molecular formula but differ in structure. Structural isomers differ in the covalent arrangement of atoms (glucose vs fructose). Cis-trans (geometric) isomers differ in spatial arrangement around a carbon-carbon double bond, which cannot rotate freely. Enantiomers are non-superimposable mirror images arising from an asymmetric carbon. Biological receptors and enzymes are stereospecific, so enantiomers often have radically different effects. The thalidomide tragedy is the standard example: one enantiomer treated nausea, the other caused severe birth defects.

MEMORY TRICK

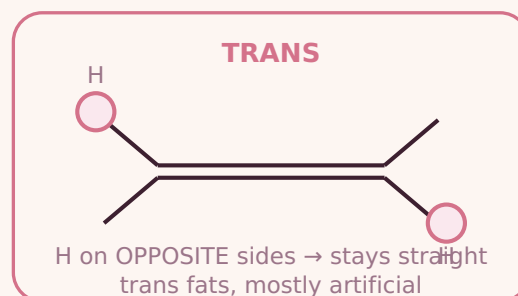
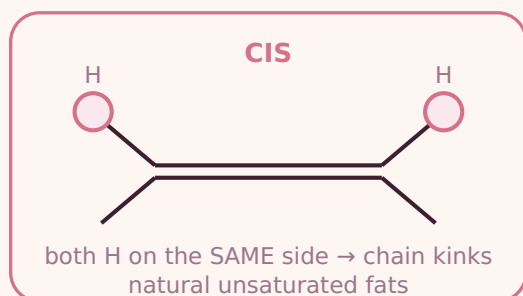
Enantiomer = your hands. Same fingers, same order, but a right glove will never fit your left hand. Your enzymes are the glove.

SAY IT OUT LOUD

Explain why a cell can tell two enantiomers apart. The answer is about shape and receptors, not about chemistry.

Cis-trans isomers: how are they different from structural isomers?

Cis vs trans: same bonds, different side of the double bond



A double bond cannot spin, which is the whole reason these are two different molecules instead of one.

IN PLAIN WORDS

Structural isomers have their atoms connected in a different order. Cis-trans isomers have the exact same connections, but groups sit on different sides of a double bond. Double bonds cannot spin, so the two arrangements are locked in and become genuinely different molecules.

HOW TO SAY IT ON THE EXAM

Structural isomers differ in the covalent connectivity of their atoms. Geometric (cis-trans) isomers have identical connectivity but differ in spatial arrangement around a carbon-carbon double bond, which cannot rotate freely. In the cis configuration the like groups sit on the same side, producing a bend in the chain; in the trans configuration they sit on opposite sides, leaving the chain straight. Naturally occurring unsaturated fatty acids are almost entirely cis, which is why they kink; industrial hydrogenation produces trans fats that pack like saturated fats and raise LDL while lowering HDL.

MEMORY TRICK

CIS = same Side (both start with S). TRANS = across, like transatlantic. And the reason it matters at all: a double bond is a locked hinge, not a swivel.

SAY IT OUT LOUD

Explain why cis and trans are two different molecules but two ends of a single bond are not. The answer is about rotation.

Name all seven functional groups and what each one does.

Functional groups: the “charms” clipped onto a carbon skeleton

Hydroxyl —OH

polar · makes things water-loving
alcohols, sugars

Carbonyl —C=O

polar · ketone (middle) or
aldehyde (end of chain)

Carboxyl —COOH

ACIDIC · donates H⁺
amino acids, fatty acids

Amino —NH₂

BASIC · accepts H⁺
amino acids, the other end

Sulfhydryl —SH

forms disulfide bridges
locks protein shape (hair!)

Phosphate —PO₄²⁻

negative · energy transfer
ATP, DNA backbone, phospholipids

Methyl —CH₃

NONPOLAR · no reactivity
a tag, not a tool. Marks DNA.

IN PLAIN WORDS

They are the little attachments on a carbon skeleton that decide how a molecule actually behaves. Hydroxyl makes it water-friendly. Carbonyl makes sugars. Carboxyl is acidic. Amino is basic. Sulfhydryl makes bridges that lock shape. Phosphate carries energy and charge. Methyl is an inert tag.

HOW TO SAY IT ON THE EXAM

Hydroxyl (-OH): polar, increases solubility, found in alcohols and sugars. Carbonyl (C=O): polar, aldehyde at a terminal carbon, ketone at an internal carbon. Carboxyl (-COOH): acidic, donates H⁺, present in amino acids and fatty acids. Amino (-NH₂): basic, accepts H⁺, present in amino acids. Sulfhydryl (-SH): forms disulfide bridges that stabilize tertiary protein structure. Phosphate (-OPO₃²⁻): negatively charged, central to ATP energy transfer and the nucleic acid backbone. Methyl (-CH₃): nonpolar and unreactive, functions as a tag that regulates gene expression.

MEMORY TRICK

Hey Cool Cats, All Sharks Phear Mermaids: Hydroxyl, Carbonyl, Carboxyl, Amino, Sulfhydryl, Phosphate, Methyl. Remember the acid/base pair: carbOXYL gives H⁺ away, aMINO takes it. They are the two ends of every amino acid.

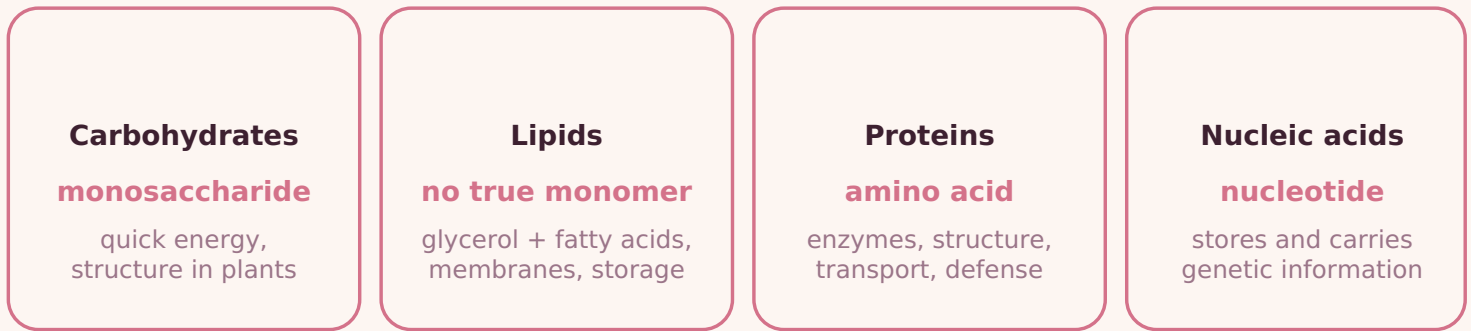
SAY IT OUT LOUD

Pick any three groups and say what they do to a molecule's solubility, charge, and reactivity. That is the depth version of this question.

Macromolecules — picture sheet

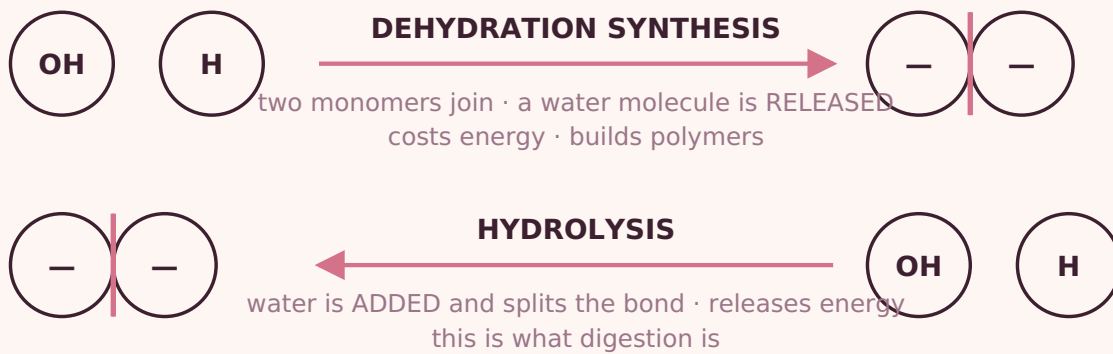
The four building blocks of every living thing. Every diagram from this section, no text. Ten-minute review.

Four classes, four building blocks

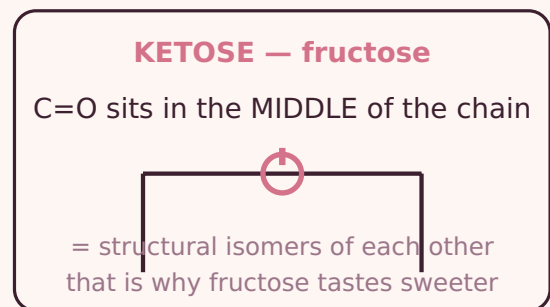
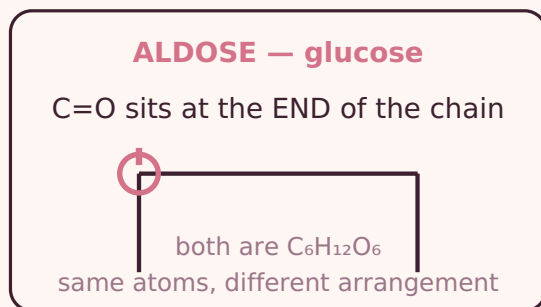


Lipids are the odd one out: they are grouped by a property (hating water), not by a shared building block.

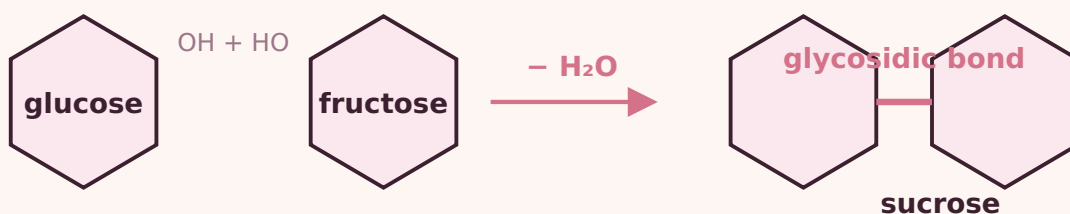
Building up vs breaking down



Aldose vs ketose: where is the C=O?



Making a disaccharide



lactose = glucose + galactose. Maltose = glucose + glucose. Lactose intolerance is simply missing the enzyme that cuts this bond

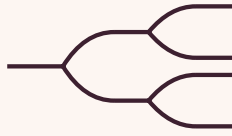
Same glucose monomer, three totally different builds

Starch



lightly branched
PLANT energy storage
alpha bonds
potatoes, rice, bread

Glycogen



heavily branched
ANIMAL energy storage
alpha bonds · liver & muscle
more ends = faster release

Cellulose



straight, stacked, zero branching
PLANT structure (cell wall)
BETA bonds
we lack the enzyme → fiber

Structural polysaccharides: sugar used as scaffolding

Cellulose

plant cell walls
pure glucose, beta bonds

Chitin

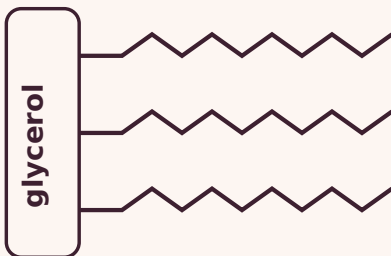
insect and crab shells
glucose with nitrogen added

Chitin

fungal cell walls
same molecule as the shells

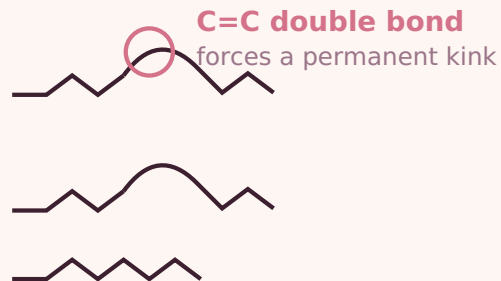
Polysaccharides share the same trick: straight chains, beta linkages, stacked and hydrogen bonded into something rigid. Storage

Triglyceride: 1 glycerol + 3 fatty acid tails



SATURATED

no double bonds · straight · packs tight
SOLID at room temp · butter, lard



UNSATURATED

kinks stop tight packing
LIQUID at room temp · olive oil, fish

The molecule that makes cells possible



Head: phosphate group

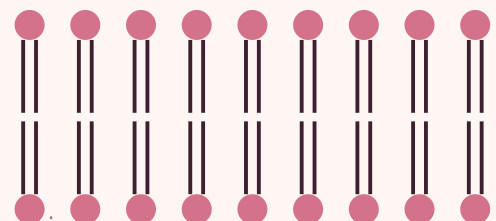
charged → HYDROPHILIC, loves water

Tails: 2 fatty acids

nonpolar → HYDROPHOBIC, hides from water

One molecule, two opposite personalities = amphipathic

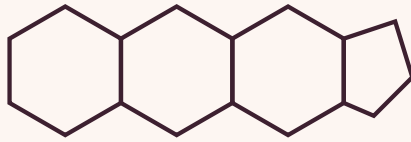
water outside the cell



water inside the cell

Heads face the water, tails hide in the middle.
Nobody built this. It assembles itself.

Steroids: four fused rings, no tails at all



three six-sided rings + one five-sided ring

Cholesterol

keeps membranes fluid, parent of the rest

Testosterone & estrogen

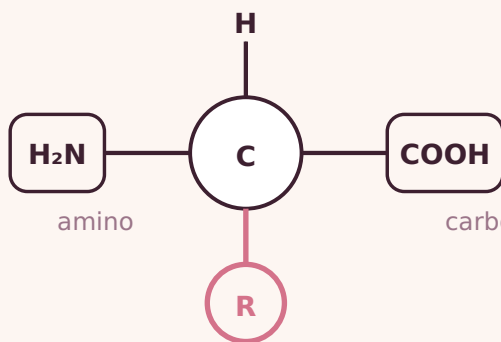
sex hormones

Cortisol

stress and metabolism

nothing like a fat, but they are still nonpolar, so they are still lipids. Being nonpolar lets them walk straight through a membrane

Amino acid: four things on one central carbon



The R group is the ONLY difference

Peptide bond



carboxyl of one meets amino of the next
H₂O comes out → it is dehydration synthesis again
Chain of these = polypeptide

Four levels of protein structure

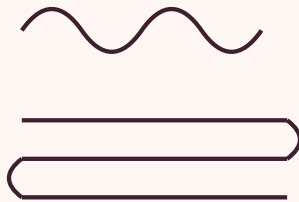
1° Primary



the sequence of amino acids

held by peptide bonds
DNA writes this

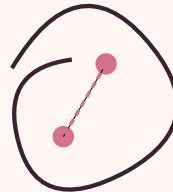
2° Secondary



alpha helix (coil) or beta pleated sheet

hydrogen bonds in backbone

3° Tertiary



whole chain folds into a 3D blob

R-group interactions, disulfide

4° Quaternary



2+ finished chains assemble together
hemoglobin = 4 chains

Denaturation: the shape dies, the sequence lives



folded and working

Peptide bonds are untouched, so primary structure survives. Sometimes it can refold. A fried egg cannot.

heat, wrong pH, salt

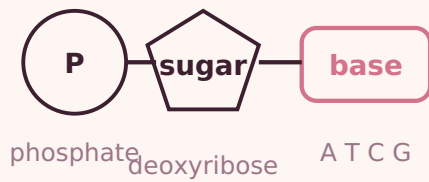


weak interactions break

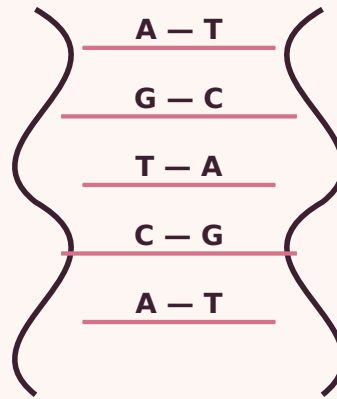


unfolded and useless

One nucleotide, then the double helix



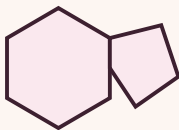
Three parts. Every single time.
Sugar + phosphate link up to form
the backbone of the whole strand.



Backbone of sugar and phosphate runs up the outside. Bases point inward.
A pairs with T using 2 hydrogen bonds, G pairs with C using 3.
Each strand is a template for rebuilding the other one.

Purines and pyrimidines: count the rings

PURINES — two rings



Adenine · Guanine
bigger

PYRIMIDINES — one ring



Cytosine · Thymine · Uracil
smaller

A purine always pairs with a pyrimidine. Big plus small keeps the helix exactly the same width all the way down.

Pure As Gold = PURines are A and G. CUT the PY = pyrimidines are C, U, T.

DNA vs RNA: four differences

DNA

double stranded helix
sugar = deoxyribose
bases A T C G
stays in the nucleus
the master blueprint, kept safe

RNA

single stranded
sugar = ribose (extra OH)
bases A U C G (U replaces T)
travels out to the ribosome
mRNA carries · tRNA delivers · rRNA builds

Macromolecules

Ch 3 · 20 cards

What are the four classes of biological macromolecules, and what are their monomers?

Four classes, four building blocks

Carbohydrates

monosaccharide

quick energy,
structure in plants

Lipids

no true monomer

glycerol + fatty acids,
membranes, storage

Proteins

amino acid

enzymes, structure,
transport, defense

Nucleic acids

nucleotide

stores and carries
genetic information

Lipids are the odd one out: they are grouped by a property (hating water), not by a shared building block.

IN PLAIN WORDS

Carbohydrates, built from monosaccharides. Lipids, built from glycerol and fatty acids (the odd one out). Proteins, built from amino acids. Nucleic acids, built from nucleotides.

HOW TO SAY IT ON THE EXAM

The four classes are carbohydrates (monomer: monosaccharide, polymer: polysaccharide), lipids (not true polymers; assembled from glycerol and fatty acid subunits), proteins (monomer: amino acid, polymer: polypeptide), and nucleic acids (monomer: nucleotide, polymer: DNA or RNA). Three of the four are true polymers; lipids are the exception because their subunits are not repeating identical units joined into a chain.

MEMORY TRICK

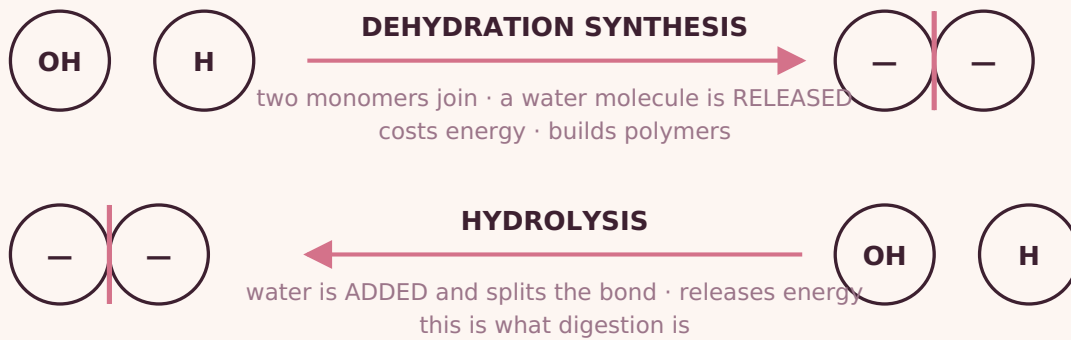
CLiPN, like 'clip-n': Carbs, Lipids, Proteins, Nucleic acids. Lipids are the rule-breaker in the group, remember that separately.

SAY IT OUT LOUD

Why are lipids not technically polymers? If you can answer that, you understand what a polymer actually is.

Dehydration synthesis vs hydrolysis.

Building up vs breaking down



IN PLAIN WORDS

To join two building blocks, the cell pulls out a water molecule and uses the leftover ends to link them. To break them apart, the cell adds a water molecule back in and splits the bond. Building removes water, breaking adds water.

HOW TO SAY IT ON THE EXAM

Dehydration synthesis (condensation) joins two monomers with covalent bond formation and releases one molecule of water; it is endergonic and requires enzymes. Hydrolysis cleaves a covalent bond by adding one molecule of water, with the hydroxyl going to one fragment and the hydrogen to the other; it is exergonic and is the chemical basis of digestion.

MEMORY TRICK

DE-hydration removes water. HYDRO-lysis uses water to LYSE (cut). 'Lysis' always means splitting.

SAY IT OUT LOUD

How many water molecules are released when 10 amino acids form one chain? Say why it is 9, not 10.

What are carbohydrates, and what is the general formula?

IN PLAIN WORDS

Sugars and starches. Quick energy plus some structural jobs. The formula is roughly one carbon, two hydrogens, one oxygen, repeated.

HOW TO SAY IT ON THE EXAM

Carbohydrates have the general empirical formula $(CH_2O)_n$, meaning a 1:2:1 ratio of carbon, hydrogen, and oxygen. The name means hydrated carbon. Functions include short-term energy supply, energy storage, structural support, and cell-surface recognition markers.

MEMORY TRICK

Carbo-HYDRATE literally means watered carbon. C with H₂O stuck to it.

SAY IT OUT LOUD

Glucose is C₆H₁₂O₆. Check it against the ratio out loud.

Monosaccharides: name three and why they matter.

IN PLAIN WORDS

Glucose, fructose, and galactose. All three have the exact same atoms, $C_6H_{12}O_6$, but arranged differently. Glucose is the fuel every cell runs on.

HOW TO SAY IT ON THE EXAM

Monosaccharides are single sugar units and the simplest carbohydrates. Glucose is the primary cellular fuel and the input to glycolysis and cellular respiration. Fructose is found in fruit and is the sweetest natural sugar. Galactose is a component of lactose. All three share the formula $C_6H_{12}O_6$ and are structural isomers of each other. Ribose and deoxyribose are five-carbon monosaccharides found in nucleic acids.

MEMORY TRICK

Mono = one. All three end in -ose, which is the sugar suffix. If it ends in -ose, it is a sugar.

SAY IT OUT LOUD

Same formula, different molecule. Explain how that is possible, and name what we call that relationship.

Aldose vs ketose.

Aldose vs ketose: where is the $C=O$?

ALDOSE — glucose

$C=O$ sits at the END of the chain



both are $C_6H_{12}O_6$
same atoms, different arrangement

KETOSE — fructose

$C=O$ sits in the MIDDLE of the chain



= structural isomers of each other
that is why fructose tastes sweeter

IN PLAIN WORDS

Both have a $C=O$ group. If it sits at the end of the chain, it is an aldose, like glucose. If it sits in the middle, it is a ketose, like fructose.

HOW TO SAY IT ON THE EXAM

Aldose sugars carry a carbonyl group at a terminal carbon, forming an aldehyde; glucose and galactose are aldoses. Ketose sugars carry the carbonyl at an internal carbon, forming a ketone; fructose is the standard ketose. The distinction affects ring structure and reactivity.

MEMORY TRICK

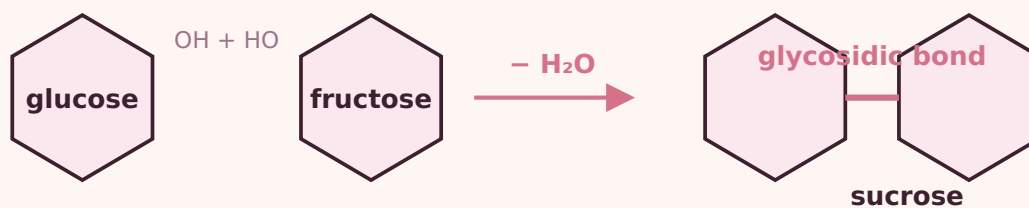
Aldose = At the end (A for the top of the chain). KETOse = Kept in the middle. Glucose is an aldose, fructose is a ketose: 'G is at the end of the alphabet stretch, F is in the middle' if that helps.

SAY IT OUT LOUD

Given any sugar structure, describe how to classify it in one sentence.

How do disaccharides form? Name three.

Making a disaccharide



tose = glucose + galactose. Maltose = glucose + glucose. Lactose intolerance is simply missing the enzyme that cuts this bc

IN PLAIN WORDS

Two single sugars join by dehydration synthesis, and the link between them is called a glycosidic bond. Sucrose is glucose plus fructose (table sugar). Lactose is glucose plus galactose (milk sugar). Maltose is glucose plus glucose (malt sugar).

HOW TO SAY IT ON THE EXAM

A disaccharide forms when two monosaccharides are joined by a glycosidic linkage via dehydration synthesis. Sucrose = glucose + fructose. Lactose = glucose + galactose. Maltose = glucose + glucose. Lactose intolerance results from insufficient lactase, the enzyme that hydrolyzes the glycosidic bond in lactose.

MEMORY TRICK

GF, GG, GG-lactose. Sucrose = Sweet Fructose. Lactose = Lactation, so milk. Maltose = Malt, so two glucoses from grain.

SAY IT OUT LOUD

Explain lactose intolerance at the molecular level: which bond, which enzyme, what happens when the enzyme is missing.

Starch, glycogen, and cellulose: compare structure and function.

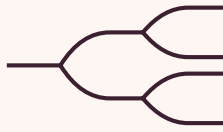
Same glucose monomer, three totally different builds

Starch



lightly branched
PLANT energy storage
alpha bonds
potatoes, rice, bread

Glycogen



heavily branched
ANIMAL energy storage
alpha bonds · liver & muscle
more ends = faster release

Cellulose



straight, stacked, zero branching
PLANT structure (cell wall)
BETA bonds
we lack the enzyme → fiber

IN PLAIN WORDS

All three are long chains of glucose, but built differently. Starch is plant storage, lightly branched. Glycogen is animal storage, heavily branched so energy can be released fast. Cellulose is plant structure, perfectly straight with a different kind of bond, which is why we cannot digest it.

HOW TO SAY IT ON THE EXAM

All three are glucose polymers. Starch (amylose and amylopectin) is the plant storage polysaccharide, with alpha glycosidic linkages and limited branching. Glycogen is the animal storage polysaccharide, stored in liver and muscle, with alpha linkages and extensive branching that provides many terminal ends for rapid enzymatic hydrolysis. Cellulose is the structural polysaccharide of plant cell walls, with beta glycosidic linkages that produce straight chains capable of hydrogen bonding into rigid microfibrils. Humans lack cellulase and cannot hydrolyze beta linkages, so cellulose passes through as dietary fiber. Chitin, the structural polysaccharide of arthropod exoskeletons and fungal cell walls, is a modified glucose polymer containing nitrogen.

MEMORY TRICK

Alpha = A for Available energy. Beta = B for Building material. Glycogen is branchier because animals move and need fuel NOW; plants can wait.

SAY IT OUT LOUD

Why can a cow digest cellulose and you cannot? The answer is not about the cow's stomach shape.

Chitin and the other structural polysaccharides.

Structural polysaccharides: sugar used as scaffolding

Cellulose

plant cell walls
pure glucose, beta bonds

Chitin

insect and crab shells
glucose with nitrogen added

Chitin

fungus cell walls
same molecule as the shells

accharides share the same trick: straight chains, beta linkages, stacked and hydrogen bonded into something rigid. Storage

IN PLAIN WORDS

Chitin is what crab shells, insect exoskeletons, and mushroom cell walls are made of. It is glucose with a nitrogen group attached, built in straight beta chains just like cellulose. Same idea as cellulose: structure, not fuel.

HOW TO SAY IT ON THE EXAM

Chitin is a structural polysaccharide of N-acetylglucosamine, a glucose derivative containing a nitrogen-bearing group, joined by beta glycosidic linkages. It forms the exoskeleton of arthropods and the cell walls of fungi. Like cellulose, its beta linkages permit straight chains that hydrogen bond into rigid sheets, and most animals lack the enzyme to digest it. Structural polysaccharides consistently use beta linkages and straight chains, while storage polysaccharides use alpha linkages and coil.

MEMORY TRICK

Alpha coils for storage, Beta is a Beam for building. Chitin is cellulose that went to the beach: same construction, nitrogen added, shells instead of stems.

SAY IT OUT LOUD

Say the general rule out loud: which linkage type builds structure, which stores energy, and what shape each one produces.

What defines a lipid, and what do lipids do?

IN PLAIN WORDS

Lipids are the greasy ones. What unites them is not a shared building block, it is that they are nonpolar and will not dissolve in water. They store energy, build membranes, insulate, cushion organs, and act as hormones.

HOW TO SAY IT ON THE EXAM

Lipids are a chemically diverse group unified by hydrophobicity rather than by a common monomer. They are predominantly nonpolar and insoluble in water. Functions include long-term energy storage, membrane structure, thermal insulation, physical protection of organs, and chemical signaling as steroid hormones. Fat-soluble vitamins A, D, E, and K are also lipids.

MEMORY TRICK

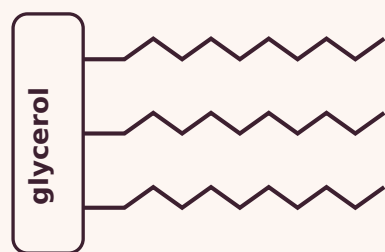
Lipids are defined by what they refuse to do: mix with water. It is the only macromolecule class defined by a physical property instead of a building block.

SAY IT OUT LOUD

Name three structurally different lipids and explain what makes all three count as lipids anyway.

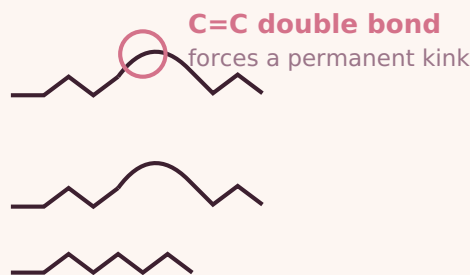
Structure of a triglyceride, and saturated vs unsaturated.

Triglyceride: 1 glycerol + 3 fatty acid tails



SATURATED

no double bonds · straight · packs tight
SOLID at room temp · butter, lard



UNSATURATED

kinks stop tight packing
LIQUID at room temp · olive oil, fish

IN PLAIN WORDS

One glycerol with three fatty acid tails attached. If the tails have no double bonds they are straight, pack tightly, and the fat is solid, like butter. If they have double bonds the tails kink, cannot pack tightly, and the fat is liquid, like olive oil.

HOW TO SAY IT ON THE EXAM

A triglyceride consists of a glycerol backbone joined to three fatty acids by ester linkages formed through dehydration synthesis. Saturated fatty acids contain no carbon-carbon double bonds, are saturated with hydrogen, pack densely, and are solid at room temperature; they are predominantly of animal origin and are associated with raised LDL cholesterol. Unsaturated fatty acids contain one or more double bonds that introduce cis kinks preventing dense packing, and are liquid at room temperature; they are predominantly plant and fish oils. Trans fats are artificially hydrogenated unsaturated fats whose trans double bonds do not kink, so they behave like saturated fats while being more harmful, raising LDL and lowering HDL.

MEMORY TRICK

SATURATED = the carbon is SAT with hydrogen, no room for more, straight and stiff and SOLID. Unsaturated has room, so it bends. 'Fats are solid, oils are oily' and now you know why.

SAY IT OUT LOUD

Explain the physical reason butter is solid using the words double bond, kink, and packing. That causal chain is what earns full credit.

Structure of a phospholipid, and why it forms membranes.

The molecule that makes cells possible

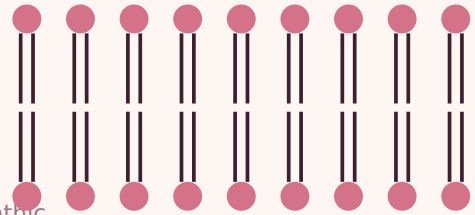
**Head: phosphate group**

charged → HYDROPHILIC, loves water

Tails: 2 fatty acids

nonpolar → HYDROPHOBIC, hides from water

One molecule, two opposite personalities = amphipathic

water outside the cell**water inside the cell**

Heads face the water, tails hide in the middle.
Nobody built this. It assembles itself.

IN PLAIN WORDS

A phosphate head that loves water, attached to two fatty acid tails that hate water. Drop a bunch in water and they automatically arrange into a double layer with all the heads facing out toward water and all the tails hiding inward. That double layer is your cell membrane.

HOW TO SAY IT ON THE EXAM

A phospholipid has a glycerol backbone with two fatty acid chains and one phosphate-containing head group. The phosphate head is hydrophilic and the fatty acid tails are hydrophobic, making the molecule amphipathic. In aqueous environments, phospholipids spontaneously self-assemble into a bilayer with heads oriented toward water on both surfaces and tails sequestered in the interior. This arrangement is driven by the hydrophobic effect and creates a selectively permeable barrier, which is the physical basis of the cell as a distinct compartment.

MEMORY TRICK

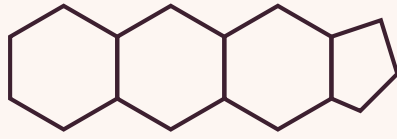
A phospholipid is a lollipop with two legs. The head loves water, the legs hate it. Nobody assembles the membrane; it builds itself because that arrangement is the most stable.

SAY IT OUT LOUD

This is a top-tier depth question: explain why the bilayer forms WITHOUT saying the cell builds it. The driver is the water, not the cell.

What makes steroids structurally different, and why are they still lipids?

Steroids: four fused rings, no tails at all



three six-sided rings + one five-sided ring

Cholesterol

keeps membranes fluid, parent of the rest

Testosterone & estrogen

sex hormones

Cortisol

stress and metabolism

ok nothing like a fat, but they are still nonpolar, so they are still lipids. Being nonpolar lets them walk straight through a men

IN PLAIN WORDS

Steroids are four rings fused together, not chains. They look nothing like fats. They count as lipids because they are still nonpolar and still refuse to mix with water.

HOW TO SAY IT ON THE EXAM

Steroids consist of four fused carbon rings, three six-membered and one five-membered, with varying functional groups attached. They lack fatty acid chains entirely, yet are classified as lipids on the basis of hydrophobicity. Cholesterol is the precursor for all other steroids and regulates membrane fluidity. Testosterone and estrogen are steroid hormones; cortisol regulates stress response and metabolism. Being nonpolar, steroid hormones pass directly through the plasma membrane and bind intracellular receptors.

MEMORY TRICK

Four rings fused = steroid. Cholesterol gets a bad reputation but it is the parent molecule of your sex hormones and it keeps membranes from freezing solid.

SAY IT OUT LOUD

Explain why steroid hormones can walk straight through the cell membrane while protein hormones cannot.

List the functions of proteins.

IN PLAIN WORDS

Proteins do almost every job in the body. They speed up reactions, build structure, move things, carry messages, defend against infection, transport molecules, and store materials.

HOW TO SAY IT ON THE EXAM

Protein functions include: enzymatic catalysis (amylase, DNA polymerase), structural support (collagen, keratin), transport (hemoglobin, membrane channel proteins), movement (actin and myosin), defense (antibodies), signaling (insulin and other peptide hormones), storage (casein, ferritin), and receptors for cell communication.

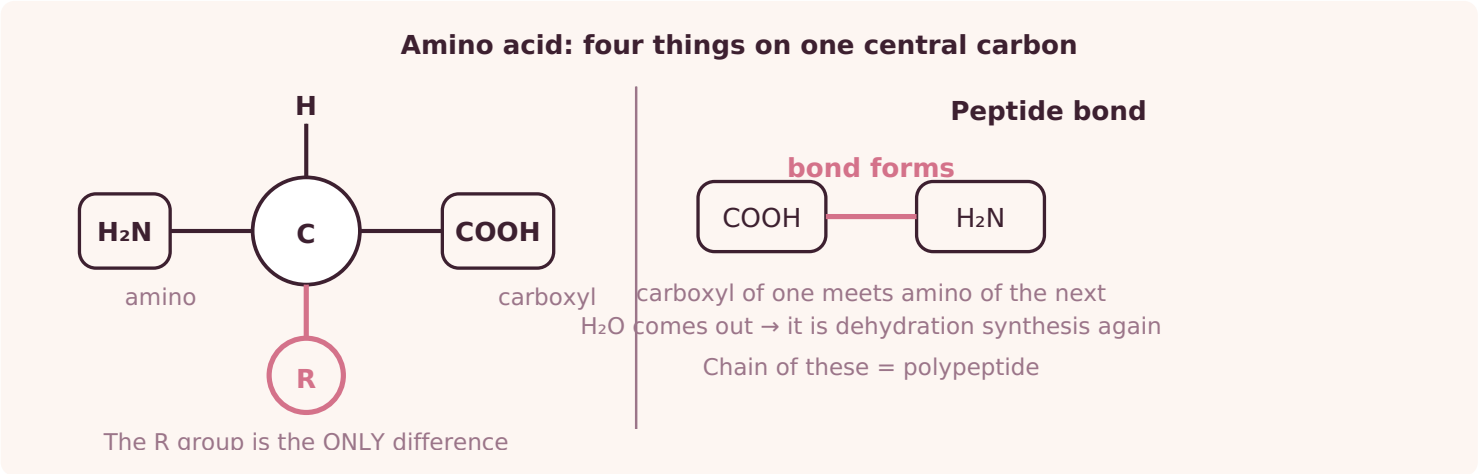
MEMORY TRICK

Proteins are the workforce. DNA is the manual, proteins are the workers. Every job you can name, some protein is doing it.

SAY IT OUT LOUD

Name one specific protein for each function. Specific examples are what separate a B answer from an A answer.

Structure of an amino acid, and how peptide bonds form.



IN PLAIN WORDS

A central carbon holding four things: an amino group, a carboxyl group, a hydrogen, and an R group. The R group is the only part that changes between the 20 amino acids, and it decides that amino acid's personality. Two amino acids join when the carboxyl of one bonds to the amino of the next, releasing water.

HOW TO SAY IT ON THE EXAM

An amino acid consists of a central alpha carbon bonded to an amino group (-NH₂), a carboxyl group (-COOH), a hydrogen atom, and a variable R group (side chain). The R group determines whether the amino acid is nonpolar, polar, acidic, or basic, and therefore determines how that residue behaves during folding. A peptide bond is a covalent bond formed by dehydration synthesis between the carboxyl carbon of one amino acid and the amino nitrogen of the next. Chains of amino acids are polypeptides; one or more folded polypeptides constitute a protein.

MEMORY TRICK

Every amino acid is the same except the R group. R for 'the Rest of it' and R for 'the Reason they differ.' Twenty amino acids, twenty different R groups, that is the whole story.

SAY IT OUT LOUD

Why do R groups matter so much for folding? Link it: an R group's chemistry determines where it wants to sit relative to water.

The four levels of protein structure.

Four levels of protein structure

1° Primary

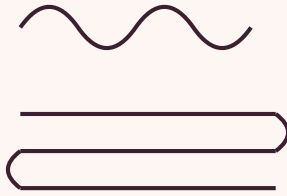


the sequence of
amino acids

held by peptide bonds

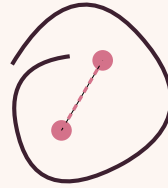
DNA writes this

2° Secondary



alpha helix (coil) or
beta pleated sheet
hydrogen bonds in backbone

3° Tertiary



whole chain folds into
a 3D blob

R-group interactions, disulfide

4° Quaternary



2+ finished chains
assemble together
hemoglobin = 4 chains

IN PLAIN WORDS

Primary is the order of the amino acids, like letters in a word. Secondary is local twisting into coils or pleats, held by hydrogen bonds along the backbone. Tertiary is the whole chain folding into one 3D shape, driven by the R groups. Quaternary is two or more finished chains joining into one working unit.

HOW TO SAY IT ON THE EXAM

Primary structure is the linear amino acid sequence, determined by the gene and joined by peptide bonds. Secondary structure comprises localized alpha helices and beta pleated sheets, stabilized by hydrogen bonds between backbone atoms, not R groups. Tertiary structure is the overall three-dimensional conformation of a single polypeptide, stabilized by R-group interactions including hydrophobic clustering, hydrogen bonds, ionic bonds, and covalent disulfide bridges. Quaternary structure is the assembly of multiple polypeptide subunits into a functional complex; hemoglobin with four subunits is the standard example. Not all proteins have quaternary structure.

MEMORY TRICK

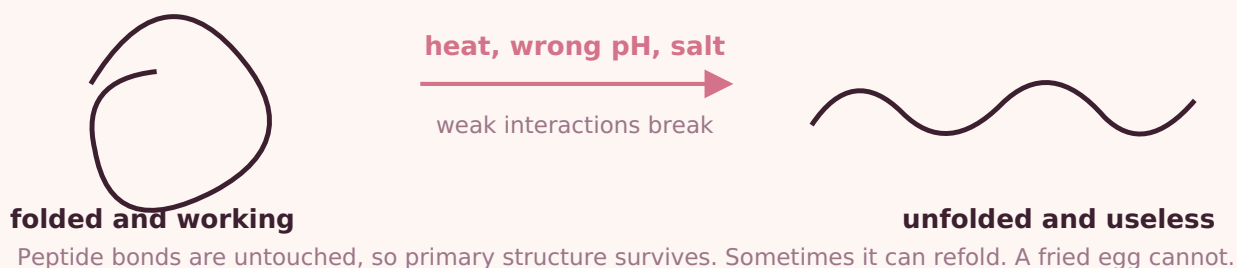
Think of a phone cord: the plastic and wire is primary, the coil is secondary, the tangled mess in your bag is tertiary, and two cords knotted together is quaternary. Key exam detail: SECONDARY uses backbone hydrogen bonds, TERTIARY uses R-group interactions. Professors love that distinction.

SAY IT OUT LOUD

Alpha helix and beta pleated sheet are which level? Say it, then say what holds them together.

What is denaturation, and is it reversible?

Denaturation: the shape dies, the sequence lives



IN PLAIN WORDS

Denaturation is when a protein loses its 3D shape and stops working. Heat, wrong pH, salt, or harsh chemicals can do it. The order of amino acids is untouched, only the folding is wrecked. Sometimes it can refold, often it cannot.

HOW TO SAY IT ON THE EXAM

Denaturation is the disruption of a protein's secondary, tertiary, and quaternary structure without cleaving peptide bonds, so primary structure is retained. Causes include elevated temperature, extremes of pH, high salt concentration, and organic solvents, all of which disrupt the weak interactions maintaining the fold. Because function depends on conformation, denatured proteins lose activity. Some proteins renature spontaneously when normal conditions return, but many aggregate irreversibly; a fried egg is the standard irreversible example, where denatured albumin cannot revert to its soluble form.

MEMORY TRICK

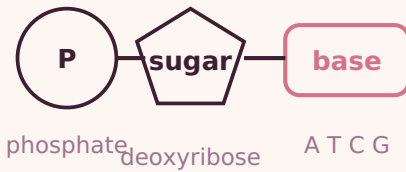
A fried egg never un-fries. Denaturation destroys the SHAPE, not the SEQUENCE, which is why primary structure survives every time.

SAY IT OUT LOUD

Tie chapters together: why does a fever above about 104 F become dangerous? Answer in terms of enzymes and shape.

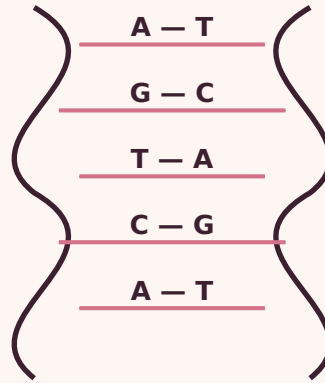
What are nucleic acids and nucleotides?

One nucleotide, then the double helix



Three parts. Every single time.

Sugar + phosphate link up to form the backbone of the whole strand.



Backbone of sugar and phosphate runs up the outside. Bases point inward.
A pairs with T using 2 hydrogen bonds, G pairs with C using 3.
Each strand is a template for rebuilding the other one.

IN PLAIN WORDS

Nucleic acids are DNA and RNA, the molecules that store and carry genetic instructions. Their building block is the nucleotide, which is always three parts: a phosphate, a five-carbon sugar, and a nitrogenous base.

HOW TO SAY IT ON THE EXAM

Nucleic acids are polymers of nucleotides that store, transmit, and express genetic information. Each nucleotide comprises a phosphate group, a pentose sugar (deoxyribose in DNA, ribose in RNA), and one nitrogenous base. Purines (adenine, guanine) have two rings; pyrimidines (cytosine, thymine, uracil) have one. Nucleotides are joined by phosphodiester bonds between the phosphate of one and the sugar of the next, forming the sugar-phosphate backbone.

MEMORY TRICK

Three parts: Phosphate, Sugar, Base. PSB, like 'peanut butter sandwich' out of order. Pure As Gold = PURines are Adenine and Guanine (two rings). CUT the PY = pyrimidines are Cytosine, Uracil, Thymine (one ring).

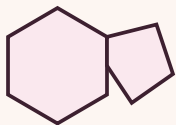
SAY IT OUT LOUD

Name the three parts without looking, then say which part differs between DNA and RNA.

Purines vs pyrimidines.

Purines and pyrimidines: count the rings

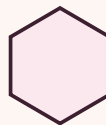
PURINES — two rings



Adenine • Guanine

bigger

PYRIMIDINES — one ring



Cytosine • Thymine • Uracil

smaller

A purine always pairs with a pyrimidine. Big plus small keeps the helix exactly the same width all the way down.

Pure As Gold = PURines are A and G. CUT the PY = pyrimidines are C, U, T.

IN PLAIN WORDS

Two kinds of bases. Purines have two rings and are bigger: adenine and guanine. Pyrimidines have one ring and are smaller: cytosine, thymine, and uracil. A big one always pairs with a small one, which keeps the DNA ladder exactly the same width the whole way down.

HOW TO SAY IT ON THE EXAM

Purines are double-ring nitrogenous bases: adenine and guanine. Pyrimidines are single-ring bases: cytosine, thymine (DNA only), and uracil (RNA only). Complementary base pairing always joins a purine to a pyrimidine, maintaining a constant helix diameter of about 2 nanometers. Adenine pairs with thymine via two hydrogen bonds; guanine pairs with cytosine via three, making G-C pairs more thermally stable.

MEMORY TRICK

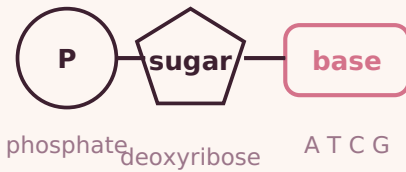
Pure As Gold: PURines are Adenine and Guanine. CUT the PY: PYrimidines are Cytosine, Uracil, Thymine. And PYrimidines have a Y in the name and one ring, like one Y-shape.

SAY IT OUT LOUD

Explain why the helix would not work if two purines ever paired together. Say it in terms of width.

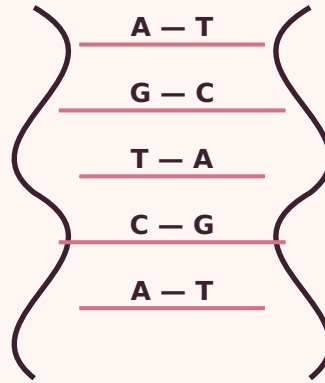
Describe DNA's double helix and complementary base pairing.

One nucleotide, then the double helix



Three parts. Every single time.

Sugar + phosphate link up to form the backbone of the whole strand.



Backbone of sugar and phosphate runs up the outside. Bases point inward.
A pairs with T using 2 hydrogen bonds, G pairs with C using 3.
Each strand is a template for rebuilding the other one.

IN PLAIN WORDS

Two strands twisted around each other like a spiral ladder. The sides are sugar and phosphate. The rungs are pairs of bases: A always pairs with T, G always pairs with C, held by hydrogen bonds. Because the pairing is fixed, one strand tells you exactly what the other says.

HOW TO SAY IT ON THE EXAM

DNA is an antiparallel double helix with a sugar-phosphate backbone on the exterior and nitrogenous bases oriented inward. Complementary base pairing joins adenine to thymine by two hydrogen bonds and guanine to cytosine by three hydrogen bonds. A purine always pairs with a pyrimidine, keeping helix width constant. Because each strand specifies the sequence of the other, each serves as a template during semiconservative replication, which is the structural basis for accurate inheritance.

MEMORY TRICK

Apples in the Tree, Cars in the Garage. A-T, C-G. And G-C has 3 hydrogen bonds while A-T has 2, so GC-rich DNA is harder to pull apart.

SAY IT OUT LOUD

The depth question: explain how the STRUCTURE makes accurate copying possible. Structure to function, again.

DNA vs RNA, and the three types of RNA.

DNA vs RNA: four differences

DNA

double stranded helix
sugar = deoxyribose
bases A T C G
stays in the nucleus
the master blueprint, kept safe

RNA

single stranded
sugar = ribose (extra OH)
bases A U C G (U replaces T)
travels out to the ribosome
mRNA carries · tRNA delivers · rRNA builds

IN PLAIN WORDS

DNA is double stranded, uses deoxyribose, uses the base T, and stays in the nucleus. RNA is single stranded, uses ribose, uses U instead of T, and travels out to do the work. mRNA carries the message, tRNA delivers amino acids, rRNA builds the ribosome itself.

HOW TO SAY IT ON THE EXAM

DNA is double-stranded, contains deoxyribose, uses thymine, and remains in the nucleus in eukaryotes as long-term genetic storage. RNA is typically single-stranded, contains ribose (with an additional hydroxyl group making it less chemically stable), uses uracil in place of thymine, and functions in the cytoplasm. Messenger RNA carries the transcribed coding sequence from nucleus to ribosome. Transfer RNA carries specific amino acids to the ribosome and reads codons via its anticodon. Ribosomal RNA is a structural and catalytic component of the ribosome.

MEMORY TRICK

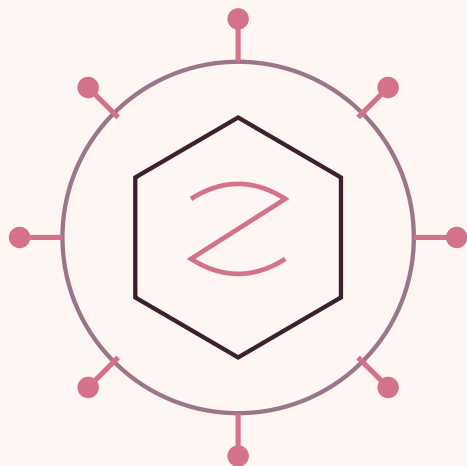
DNA is the library book that never leaves the library. mRNA is the photocopy you carry out. U replaces T in RNA: 'U R the one that leaves.' Types: M for Message, T for Transfer/Taxi, R for Ribosome.

SAY IT OUT LOUD

Why does the cell bother making a copy instead of sending the DNA itself? Answer in terms of protecting the original.

Viruses — picture sheet

Small section, high chance of a short answer question. Every diagram from this section, no text. Ten-minute review.



Parts of a virus

Spikes (glycoproteins)

the key that fits a host cell's lock. No match, no infection.

Envelope

stolen host membrane. Not all viruses have one. Soap destroys it.

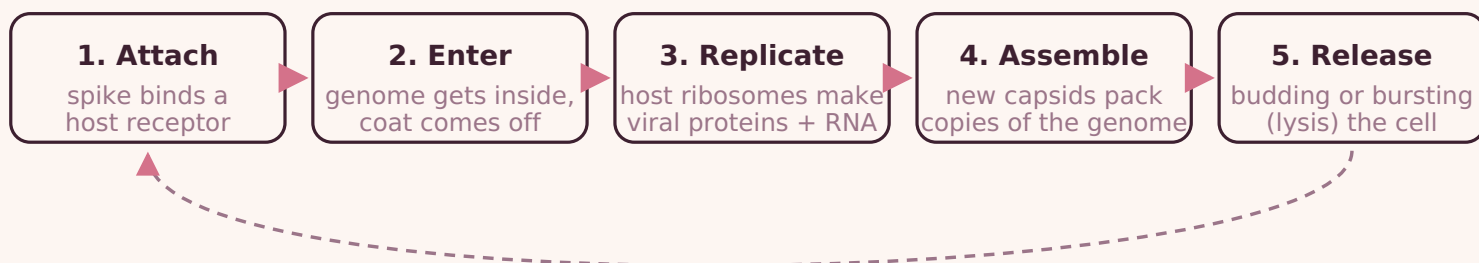
Capsid

protein shell built from repeating subunits. Protects the cargo.

Nucleic acid

DNA or RNA, single or double stranded. The instructions. That is

How an RNA virus hijacks a cell



New viruses go find new cells. One infected cell can release thousands.

The whole point: a virus owns no ribosomes, no enzymes, no ATP. Step 3 is impossible without a host.

Why the flu shot changes every year

DNA polymerase

ATGCATG
ATGCATG

proofreads and fixes its mistakes

RNA polymerase

AUGCAUG
AUGAAUG

no proofreading, the error stays in

Every uncorrected error is a mutation. Multiply by thousands of copies per infected cell and new variants appear fast.

This is natural selection happening in real time, inside a body.

Scoring a virus against the checklist

Made of cells	NO
Own metabolism / energy	NO
Homeostasis	NO
Grows and develops	NO
Reproduces on its own	NO
Has genetic material	YES
Evolves by natural selection	YES

Two out of seven. That is why the answer is “on the border of life,” not a plain yes or no.

The checklist for being alive

Made of cells

one or more

Homeostasis

stays steady inside

Uses energy

metabolism

Grows

and develops

Responds

to its surroundings

Reproduces

makes more

Has DNA

heritable code

Evolves

as a population

A virus only ticks the last two boxes.

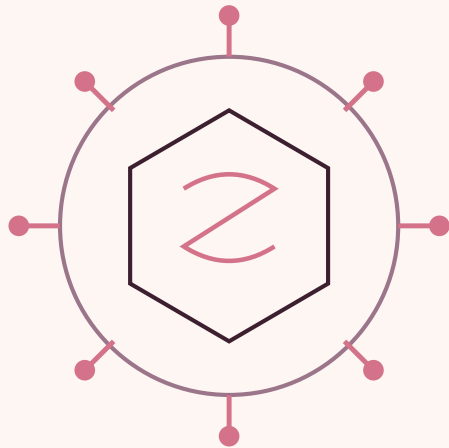
No cells, no metabolism, no homeostasis, no independent reproduction.

That is the whole “are viruses alive” debate in one picture.

Viruses

Ch 21 · 6 cards

Describe the basic structure of a virus.



Parts of a virus

Spikes (glycoproteins)

the key that fits a host cell's lock. No match, no infection.

Envelope

stolen host membrane. Not all viruses have one. Soap destroys it.

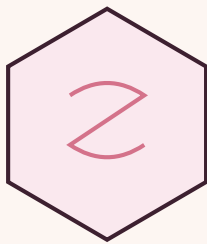
Capsid

protein shell built from repeating subunits. Protects the cargo.

Nucleic acid

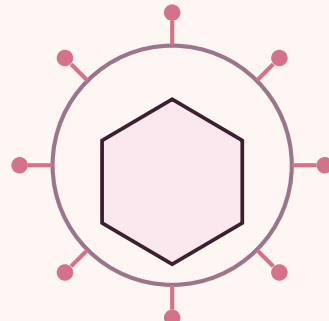
DNA or RNA, single or double stranded. The instructions. That is all

Naked vs enveloped



Naked

capsid only - tough, survives on surfaces



Enveloped

stolen host membrane, and it is made of lipid. Soap pulls lipids apart, which is why hand sanitizer wrecks enveloped viruses like

IN PLAIN WORDS

At minimum: genetic material wrapped in a protein shell. Some also have a stolen piece of membrane around the outside with protein spikes sticking out that let them latch onto host cells.

HOW TO SAY IT ON THE EXAM

All viruses possess a nucleic acid genome (DNA or RNA, single- or double-stranded) enclosed in a capsid built from repeating protein subunits called capsomeres. Enveloped viruses additionally possess a lipid bilayer envelope derived from host cell membrane during budding, studded with virally encoded glycoprotein spikes that mediate attachment to specific host cell receptors. Naked (non-enveloped) viruses lack this envelope and are generally more resistant to environmental conditions and disinfectants.

MEMORY TRICK

Capsid = the CAPsule holding the cargo. Spikes = the keys. Envelope = the stolen jacket, and it is made of lipid, which is exactly why soap wrecks enveloped viruses.

SAY IT OUT LOUD

Why does soap work against enveloped viruses? Connect it to the lipid card from Chapter 3.

Why is a virus so specific about which cells it infects?

IN PLAIN WORDS

The spikes on the virus have to physically fit a receptor protein on the host cell, like a key in a lock. No matching receptor, no entry, no infection.

HOW TO SAY IT ON THE EXAM

Host range and tissue tropism are determined by the molecular complementarity between viral attachment proteins and specific host cell surface receptors. Absence of the required receptor confers resistance. This is why most viruses infect only one species, or only particular tissues within a species, and it also explains how mutations in attachment proteins can enable cross-species transmission.

MEMORY TRICK

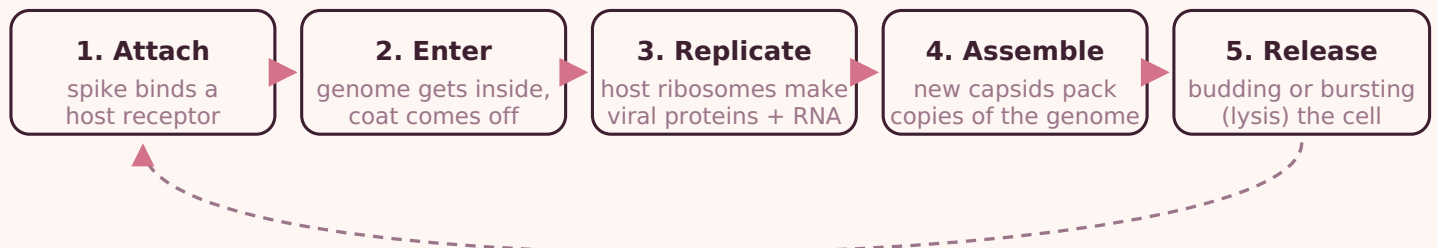
Key and lock. Change the key shape (mutation) and suddenly a new species' door opens. That is how a virus jumps species.

SAY IT OUT LOUD

Explain why you cannot catch a plant virus.

Walk through RNA virus replication inside a host cell.

How an RNA virus hijacks a cell



New viruses go find new cells. One infected cell can release thousands.

The whole point: a virus owns no ribosomes, no enzymes, no ATP. Step 3 is impossible without a host.

IN PLAIN WORDS

Attach to the cell, get inside and shed the coat, hijack the cell's machinery to copy the genome and make viral proteins, assemble new viruses, then leave by budding out or bursting the cell open.

HOW TO SAY IT ON THE EXAM

The general cycle is attachment (spike binds host receptor), entry and uncoating (genome enters the cytoplasm), replication and biosynthesis (host ribosomes translate viral proteins; viral RNA-dependent RNA polymerase replicates the genome), assembly (new genomes packaged into newly synthesized capsids), and release by budding, which preserves the cell temporarily, or by lysis, which destroys it. RNA viruses mutate rapidly because RNA-dependent RNA polymerase lacks proofreading activity, which is why influenza vaccines are reformulated annually.

MEMORY TRICK

AURAR, say it as 'a roar': Attach, Uncoat, Replicate, Assemble, Release. The virus brings instructions and nothing else. Every tool belongs to the host.

SAY IT OUT LOUD

Which step is literally impossible without a host, and why? That is the answer that decides the 'are viruses alive' question.

Why do RNA viruses mutate so fast?

Why the flu shot changes every year

DNA polymerase

ATGCATG
ATGCATG

proofreads and fixes its mistakes

RNA polymerase

AUGCAUG
AUGAUG

no proofreading, the error stays in

Every uncorrected error is a mutation. Multiply by thousands of copies per infected cell and new variants appear fast.
This is natural selection happening in real time, inside a body.

IN PLAIN WORDS

The enzyme copying their genome makes mistakes and has no spell-check. Every mistake is a mutation, and with millions of copies made per infection, useful mutations show up fast.

HOW TO SAY IT ON THE EXAM

RNA-dependent RNA polymerase lacks the 3' to 5' exonuclease proofreading activity found in DNA polymerases, resulting in error rates orders of magnitude higher than DNA replication. Combined with enormous burst sizes and short generation times, this produces high genetic variability, rapid evolution of new variants, immune escape, and antiviral resistance.

MEMORY TRICK

No spell-check. DNA polymerase proofreads, RNA polymerase does not. That single missing feature explains flu shots every year.

SAY IT OUT LOUD

Connect this to Chapter 1: how is viral variant emergence a real-time demonstration of natural selection?

Build a full answer to 'are viruses alive?'

Scoring a virus against the checklist

Made of cells	NO
Own metabolism / energy	NO
Homeostasis	NO
Grows and develops	NO
Reproduces on its own	NO
Has genetic material	YES
Evolves by natural selection	YES

Two out of seven. That is why the answer is “on the border of life,” not a plain yes or no.

IN PLAIN WORDS

Say what a virus can do (carry genes, reproduce inside a host, evolve), say what it cannot do (make energy, make proteins, reproduce alone, it is not even made of cells), then pick the property you think decides it, and commit to a position.

HOW TO SAY IT ON THE EXAM

Case against: viruses are acellular, violating cell theory; they possess no metabolism, no ribosomes, and no capacity to generate ATP; they cannot reproduce independently and are obligate intracellular parasites; outside a host they are chemically inert particles. Case for: they possess genetic material, they replicate and produce progeny, they undergo mutation and natural selection, and they exert ecological and evolutionary pressure on all cellular life. Many biologists describe viruses as replicating entities at the boundary of life, or as alive only in the context of an infected cell.

MEMORY TRICK

Structure the answer: cannot, can, decide. The professor is grading the reasoning and the vocabulary, not which side you pick.

SAY IT OUT LOUD

Say a full 5-sentence answer out loud, timed. This is the single most likely short-answer question on the whole exam.

How do viruses fit into the characteristics-of-life checklist?

The checklist for being alive

Made of cells one or more	Homeostasis stays steady inside	Uses energy metabolism	Grows and develops
Responds to its surroundings	Reproduces makes more	Has DNA heritable code	Evolves as a population

A virus only ticks the last two boxes.

No cells, no metabolism, no homeostasis, no independent reproduction.
That is the whole “are viruses alive” debate in one picture.

IN PLAIN WORDS

Go down the list. Cells? No. Metabolism? No. Homeostasis? No. Response to stimuli? Arguably no. Growth and development? No. Reproduction? Only with help. Heredity? Yes. Evolution? Yes, and fast.

HOW TO SAY IT ON THE EXAM

Applying the standard criteria: viruses fail cellular organization, independent metabolism, homeostasis, and autonomous growth. Reproduction occurs only within a host cell, so it is not independent. Viruses clearly satisfy heredity through nucleic acid genomes and evolutionary adaptation through mutation and selection. Meeting two of roughly eight criteria is why classification remains contested.

MEMORY TRICK

Two out of eight. She should be able to walk the checklist item by item and score the virus, because that is exactly what a good answer looks like.

SAY IT OUT LOUD

Walk the whole checklist aloud, giving a yes or no plus one reason for each. Do it twice.

Label it from memory

Fill these in with a pen, then check them against the picture sheets

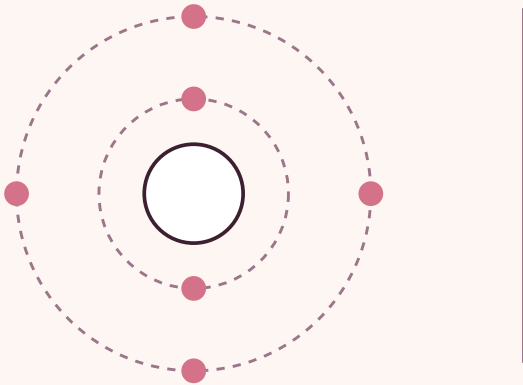
The ladder of biological organization

Name every part, and for each one write what it does.



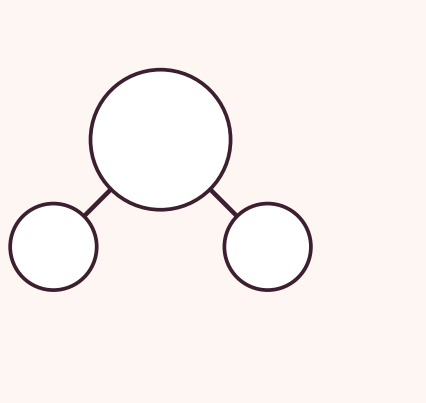
The atom

Name every part, and for each one write what it does.



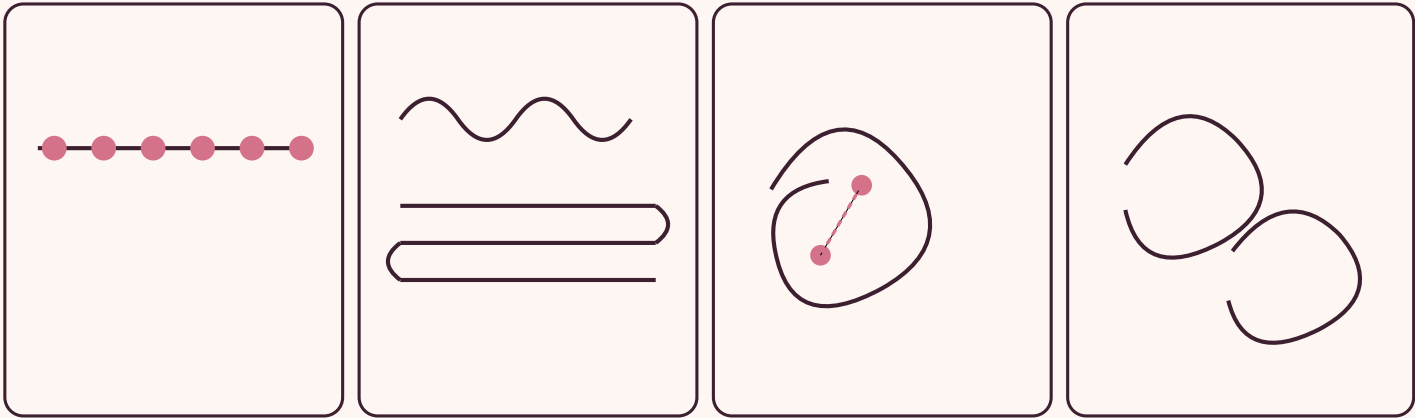
Water's polarity and its four properties

Name every part, and for each one write what it does.



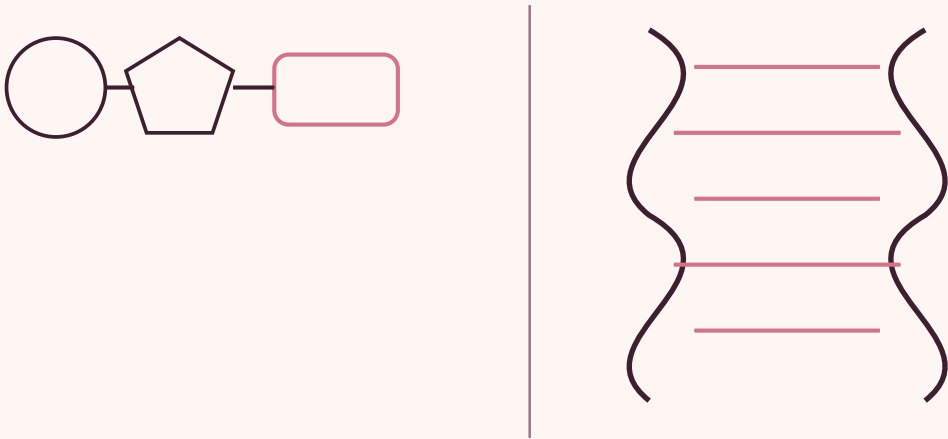
The four levels of protein structure

Name every part, and for each one write what it does.



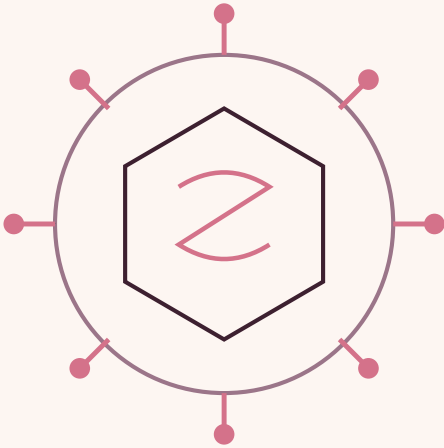
The nucleotide and the double helix

Name every part, and for each one write what it does.



The parts of a virus

Name every part, and for each one write what it does.



The seven functional groups

Name every part, and for each one write what it does.

The viral replication cycle

Name every part, and for each one write what it does.

Go deeper

Optional. One entry per card, in the same order. A harder version of each question, why it matters in a pathology lab, and where it comes back later.

Skip this entirely if you are cramming. It is here for the day after the exam, and for the tutoring session you could not make.

Doing Science

What is the broadest definition of biology?

A HARDER VERSION OF THIS QUESTION

Biology overlaps almost everything. Where does it stop being biology and start being chemistry? Try answering with the levels of organization ladder: biology owns the levels from molecules upward once those molecules are part of something living. A chemist studying water is doing chemistry. A biologist studying water inside a cell is doing biology, because the question is about function in a living system.

WHAT IT SETS UP LATER

Every course you take from here narrows this definition. Microbiology takes the cell level, physiology takes organs and systems, pathology takes what happens when any of it fails.

Walk through the steps of the scientific process.

A HARDER VERSION OF THIS QUESTION

The textbook draws this as a clean loop, but real research almost never runs in order. Name two ways a real study departs from it. One: observation and hypothesis often swap, because a researcher starts with a hypothesis and goes looking for observations to test it. Two: results routinely send you back to fix the method rather than the hypothesis, which is not a step on any diagram.

WHY IT MATTERS IN PATHOLOGY

A pathology lab lives in the analysis and communication steps. A pathologist takes data (a slide, a stain, a lab value), interprets it against known patterns, and issues a report that another physician acts on. The report is the communication step, and it has consequences.

WHAT IT SETS UP LATER

Every lab report you write in Chem 1A and A&B has this structure baked into it: purpose, hypothesis, method, results, discussion.

Control vs independent vs dependent variable. Define all three using one experiment.

A HARDER VERSION OF THIS QUESTION

A study finds that people who drink more coffee have lower rates of a disease. Name the independent variable, the dependent variable, and then say why this study cannot claim coffee prevents the disease. The answer: it is observational, not experimental. Nobody assigned the coffee, so coffee drinkers may differ in a dozen other ways, and any of those could be the real cause. Correlation is not causation, and the reason is uncontrolled variables.

WHY IT MATTERS IN PATHOLOGY

Clinical trials are this exact structure with lives attached. The treatment is the independent variable, the outcome is the dependent variable, and the control group gets a placebo. Blinding exists because knowing which group you are in changes the result.

WHAT IT SETS UP LATER

This is the backbone of the MCAT's research design questions, which appear in every section, not just biology.

Hypothesis vs scientific theory vs scientific law.

A HARDER VERSION OF THIS QUESTION

If a theory is so well supported, can it ever be wrong? Yes, and it has happened. The theory of spontaneous generation was mainstream until Pasteur's experiments. The point is not that theories are fragile, it is that they are the strongest thing science offers while still being revisable. Certainty is not on the menu.

WHAT IT SETS UP LATER

You will hear people misuse this word for the rest of your life. Being able to explain the difference cleanly is a small but real credibility marker in any science setting.

What makes a hypothesis a good one?

A HARDER VERSION OF THIS QUESTION

Rewrite this into a testable hypothesis: 'Stress is bad for health.' A workable version specifies a measurable exposure and a measurable outcome, for example: 'Adults reporting high perceived stress on a validated scale will show higher resting cortisol than adults reporting low stress.' Now both sides are measurable, and a result could contradict it.

WHY IT MATTERS IN PATHOLOGY

Diagnostic reasoning works the same way. A differential diagnosis is a set of competing hypotheses, and each test ordered is designed to rule one out. A diagnosis you cannot test is not a diagnosis.

WHAT IT SETS UP LATER

Writing testable hypotheses is the first graded skill in any research methods course and in any lab you join.

How do you read and interpret a scientific graph?

A HARDER VERSION OF THIS QUESTION

A graph shows a treatment group averaging 20 and a control averaging 18, with error bars that overlap heavily. What can you conclude? Very little. Overlapping error bars suggest the difference may be within normal variation rather than a real effect. Always check the y-axis too, because starting it at 17 instead of 0 makes that small gap look enormous.

WHAT IT SETS UP LATER

Reading figures critically is most of what you do with a scientific paper, and it is exactly what the MCAT tests in its passage sections.

How do you construct a graph from your own data?

A HARDER VERSION OF THIS QUESTION

You measured heart rate at rest, after 1 minute, 3 minutes, and 5 minutes of exercise, in 20 people. What graph, and why? A line graph, because time is continuous and you want to show change across it, with error bars showing variation between the 20 people. A bar graph would work only if you were comparing separate categories, like three different exercise types.

WHAT IT SETS UP LATER

Every lab report in Chem 1A will ask you to graph data and justify the format you picked.

How do you critique somebody else's experimental design?

A HARDER VERSION OF THIS QUESTION

A supplement company reports that 8 out of 10 people felt more energetic after taking their product for a week. List every flaw. No control group, no blinding, self-reported and unmeasurable outcome, tiny sample, no randomization, no time control for placebo effect, and nothing ruling out that people who choose to take a supplement expect to feel better.

WHY IT MATTERS IN PATHOLOGY

Journal clubs are this exercise done professionally, every week, for the rest of a medical career. The skill is not finding flaws, it is deciding whether a flaw is fatal to the conclusion or merely a limitation.

WHAT IT SETS UP LATER

This is the highest-value transferable skill in this entire chapter.

How would you design an experiment from scratch?

A HARDER VERSION OF THIS QUESTION

Your design will have a flaw. Name yours before someone else does. In the music-and-studying example: how do you keep the difficulty of the study material equal across groups, and how do you stop people who already study with music from having an advantage? Random assignment handles the second. The first needs matched materials, and saying so out loud is what separates a designed experiment from a guess.

WHAT IT SETS UP LATER

When you approach a professor about research, showing up with a designed experiment rather than an idea is the difference between a polite no and a yes.

Why do scientists publish and peer review?

A HARDER VERSION OF THIS QUESTION

Peer review catches a lot, but it does not catch everything. What does it not catch? Fabricated data, usually. Reviewers evaluate the reasoning and methods described, not the raw numbers, and they rarely replicate anything. That is why independent replication, not peer review, is the real test.

WHY IT MATTERS IN PATHOLOGY

Pathology reports get a version of this too. Difficult cases go to a second pathologist, and unusual findings go to a consensus conference. Same principle, faster timeline.

WHAT IT SETS UP LATER

Understanding what peer review does and does not guarantee will matter the first time you read a preprint.

Basic science vs applied science.

A HARDER VERSION OF THIS QUESTION

CRISPR gene editing came from researchers studying how bacteria defend themselves against viruses. Nobody was trying to build a gene editing tool. Explain what that says about funding basic research. The useful answer: you cannot know in advance which basic question turns into a technology, so a system that only funds applications will systematically miss the biggest breakthroughs.

WHAT IT SETS UP LATER

If you end up in a research lab, this distinction will decide which grants you can apply for.

What Is Life

What properties define something as alive?

A HARDER VERSION OF THIS QUESTION

Fire grows, consumes fuel, produces waste, responds to its environment, and spreads. Why is it not alive? It fails cellular organization, it has no genetic material, and it does not maintain homeostasis or evolve. This is the classic trap question, and the answer is that life requires the whole list, not a majority of it.

WHAT IT SETS UP LATER

Every definition problem in biology, from viruses to prions to synthetic cells, comes back to this list.

Are viruses alive? Argue both sides.

A HARDER VERSION OF THIS QUESTION

Prions are worse. They are misfolded proteins with no genetic material at all, and they reproduce by forcing normal proteins to misfold into the same shape. Are they alive? By every criterion, no. They do not even have a genome. Yet they replicate and cause fatal disease. If you can explain why a prion is not alive but still dangerous, you have understood that being alive and being biologically active are different questions.

WHY IT MATTERS IN PATHOLOGY

Prion diseases like Creutzfeldt-Jakob are diagnosed on autopsy by a pathologist, and the tissue shows a distinctive spongy pattern. It is one of the few diagnoses that is definitively pathological rather than clinical.

WHAT IT SETS UP LATER

The virus question reappears in microbiology and in every discussion of what counts as an infectious agent.

What is homeostasis? Give an example.

A HARDER VERSION OF THIS QUESTION

Distinguish negative from positive feedback with one example each. Negative feedback reverses a change and is the basis of nearly all homeostasis, like sweating when hot. Positive feedback amplifies a change and is rare because it is unstable, but childbirth uses it: contractions release oxytocin, which causes stronger contractions, until delivery ends the loop.

WHY IT MATTERS IN PATHOLOGY

Disease is very often homeostasis failing. Diabetes is failed blood glucose regulation, hypertension is failed blood pressure regulation, and fever is homeostasis working with the set point deliberately moved.

WHAT IT SETS UP LATER

A&P spends entire units on this. Nearly every organ system is explained as a homeostatic mechanism.

Rank the levels of biological organization.

A HARDER VERSION OF THIS QUESTION

Which level would a pathologist work at most of the time? Tissue, primarily, which is what histology is. But the answer is really that pathology crosses levels constantly: a tissue abnormality gets explained by a cellular change, which gets explained by a molecular one. Being able to move up and down this ladder is the actual skill.

WHY IT MATTERS IN PATHOLOGY

Histopathology is literally the study of the tissue level. A biopsy is a tissue sample, read for cellular abnormalities that predict organism-level disease.

WHAT IT SETS UP LATER

This ladder is how every later biology course is organized.

What is emergence? Give an example using two levels.

A HARDER VERSION OF THIS QUESTION

Is consciousness an emergent property, and can it be explained by studying neurons alone? This is a real open argument, not a settled answer, and that is the point. A strong answer says: emergent properties can be studied at the lower level, but they cannot be predicted from it, which is why reductionism alone runs out.

WHAT IT SETS UP LATER

Systems biology as a field exists because of this limitation.

What is reductionism? Give an example using two levels.

A HARDER VERSION OF THIS QUESTION

Give one thing reductionism explains beautifully and one thing it fails at. It explains sickle cell disease perfectly: one DNA base change, one amino acid change, one misshapen protein, one disease. It fails at something like depression, where no single molecule accounts for the condition and the explanation has to sit at several levels at once.

WHY IT MATTERS IN PATHOLOGY

Molecular pathology is reductionism applied: identify the exact genetic lesion causing a cancer, then target it with a drug built for that lesion.

WHAT IT SETS UP LATER

This tension shows up in every genetics and neuroscience course you take.

Give an example of structure relating to function.

A HARDER VERSION OF THIS QUESTION

Run the logic backwards. You are shown an unfamiliar cell under a microscope that is packed with mitochondria and has a hugely folded membrane. What does it probably do? Something enormously energy-expensive with a lot of transport, like a kidney tubule cell or cardiac muscle. Predicting function from structure is exactly what a pathologist does with an unfamiliar slide.

WHY IT MATTERS IN PATHOLOGY

This is the core reasoning skill of histology. You identify tissue and spot abnormality by reading shape, arrangement, and staining pattern, then inferring what the cell was doing or failing to do.

WHAT IT SETS UP LATER

Histology and A&P are essentially two years of this one skill.

What is the most fundamental unit of life, and why?

A HARDER VERSION OF THIS QUESTION

Mitochondria have their own DNA, their own ribosomes, and divide independently. Why are they still not alive? Because they cannot survive or reproduce outside a host cell, and they depend on nuclear genes for most of their proteins. The endosymbiotic theory says they used to be free-living bacteria, which makes this a case of something that was alive and is no longer independently so.

WHY IT MATTERS IN PATHOLOGY

Mitochondrial DNA is inherited only from the mother and mutates faster than nuclear DNA, which makes it useful in both genetic disease diagnosis and forensic identification.

WHAT IT SETS UP LATER

Endosymbiotic theory gets a full treatment in cell biology and evolution.

Put natural selection in your own words.

A HARDER VERSION OF THIS QUESTION

Antibiotic resistance is natural selection you can watch in a week. Walk through it: a bacterial population already contains rare variants with resistance; the antibiotic kills the susceptible ones; the resistant ones survive and reproduce; the population is now mostly resistant. Note what did NOT happen: the bacteria did not develop resistance because of the antibiotic. The variation was already there.

WHY IT MATTERS IN PATHOLOGY

Antibiotic resistance patterns are tracked by clinical microbiology labs, and hospital prescribing guidelines are set from that data. It is one of the most direct ways evolution touches daily medicine.

WHAT IT SETS UP LATER

Evolution is a full unit of the MCAT biology section and underlies all of genetics.

Chemistry of Life

What is matter, and what is an element?

A HARDER VERSION OF THIS QUESTION

Why is it CHON specifically, and not any other four elements? Because they are small, they form strong covalent bonds, and they can form multiple bonds, which lets them build stable, complex, varied molecules. Bigger atoms form weaker, longer bonds that break too easily to hold a genome together.

WHAT IT SETS UP LATER

Chem 1A opens with atomic structure and periodicity, which is the formal version of this.

Describe the structure of an atom.

A HARDER VERSION OF THIS QUESTION

Almost all of an atom's mass is in a nucleus that occupies almost none of its volume. If an atom were the size of a football stadium, the nucleus would be a marble at the center. So why does matter feel solid? Because you never touch nuclei, you feel electron shells repelling each other.

WHAT IT SETS UP LATER

Chem 1A will push this into orbitals and quantum numbers. The shell model you are learning now is a simplification you will later replace.

How is an element's identity determined? Atomic number vs mass number.

A HARDER VERSION OF THIS QUESTION

Why is the atomic mass on a periodic table a decimal, like 12.011 for carbon, when protons and neutrons are whole numbers? Because it is a weighted average of all naturally occurring isotopes. Carbon is mostly carbon-12 with a small fraction of carbon-13, and averaging them lands just above 12.

WHAT IT SETS UP LATER

You will use atomic masses constantly in Chem 1A stoichiometry, so knowing why they are decimals helps.

What are isotopes, and why do they matter in medicine?

A HARDER VERSION OF THIS QUESTION

Carbon-14 dating has a limit of roughly 50,000 years. Why? Because half-life is exponential: after enough half-lives, so little carbon-14 remains that measurement error swamps the signal. Older samples need isotopes with longer half-lives, like potassium-40.

WHY IT MATTERS IN PATHOLOGY

PET imaging uses a radioactive glucose analog. Tumors consume glucose faster than normal tissue, so they light up. That is an isotope being used as a functional diagnostic, not just a structural image.

WHAT IT SETS UP LATER

Half-life calculations show up in Chem 1A and on the MCAT.

Explain electron shells and the octet rule.

A HARDER VERSION OF THIS QUESTION

Predict the bonding behavior of oxygen (6 valence electrons), nitrogen (5), and hydrogen (1) without memorizing. Oxygen needs 2 more, so 2 bonds. Nitrogen needs 3, so 3 bonds. Hydrogen needs 1, so 1 bond. Now look at a water molecule, an ammonia molecule, and methane, and notice you could have predicted all three structures.

WHAT IT SETS UP LATER

Lewis structures in Chem 1A are this rule applied systematically.

What is an ion? Cation vs anion, and how does an ionic bond form?

A HARDER VERSION OF THIS QUESTION

Why is the inside of a resting nerve cell negative compared to the outside? Because a pump moves three sodium ions out for every two potassium ions in, and the membrane leaks potassium back out. The result is a charge difference stored across the membrane, ready to be released as a nerve signal. Every thought you have is ions moving.

WHY IT MATTERS IN PATHOLOGY

Electrolyte panels are among the most ordered lab tests in medicine. Abnormal potassium can stop a heart, and abnormal sodium can cause seizures, both because of this same membrane physics.

WHAT IT SETS UP LATER

Membrane potential is a major topic in A&P and in the MCAT biology section.

Covalent bonds: polar vs nonpolar, and what is electronegativity?

A HARDER VERSION OF THIS QUESTION

Carbon dioxide has two polar C=O bonds, yet the molecule as a whole is nonpolar. Why? Because it is linear and the two bond dipoles point in exactly opposite directions and cancel. Water has two polar bonds too, but it is bent, so they do not cancel. Shape decides molecular polarity, not just bond polarity.

WHAT IT SETS UP LATER

Molecular geometry and VSEPR theory in Chem 1A formalize exactly this.

What is a hydrogen bond, and why does something so weak matter so much?

A HARDER VERSION OF THIS QUESTION

A-T pairs share two hydrogen bonds, G-C pairs share three. Predict which region of DNA a cell can unzip more easily, and where you would expect a replication origin to sit. The AT-rich regions, because fewer hydrogen bonds means less energy to separate. Replication origins are in fact AT-rich, and that is not a coincidence.

WHY IT MATTERS IN PATHOLOGY

PCR, used in everything from infection testing to genetic diagnostics, relies on heating DNA until hydrogen bonds break. GC-rich sequences require higher temperatures, which is why lab protocols specify melting temperatures.

WHAT IT SETS UP LATER

This exact fact underlies DNA replication in genetics and PCR in any molecular lab.

Why is water polar, and what does that cause?

A HARDER VERSION OF THIS QUESTION

Why does ice float, and why does that matter for life? Hydrogen bonds lock water into an open lattice when it freezes, making solid water less dense than liquid. So lakes freeze from the top down, insulating the water below, which lets aquatic life survive winter. If ice sank, bodies of water would freeze solid from the bottom.

WHAT IT SETS UP LATER

Intermolecular forces are a full unit in Chem 1A and explain boiling points, solubility, and phase behavior.

The four emergent properties of water, each with a biological example.

A HARDER VERSION OF THIS QUESTION

Pick sweating and trace it all the way down to electronegativity in one chain, no skipped links. Oxygen is more electronegative, so bonds are polar, so the molecule has partial charges, so molecules hydrogen bond to each other, so a lot of energy is needed to separate them into vapor, so evaporating sweat removes a lot of heat from your skin. Being able to say that without stopping is the depth this course wants.

WHY IT MATTERS IN PATHOLOGY

Fever management, burn patients losing fluid through damaged skin, and dehydration all come back to water's thermal properties.

WHAT IT SETS UP LATER

Thermoregulation is a whole chapter in A&P.

Hydrophilic vs hydrophobic, and why water is called the universal solvent.

A HARDER VERSION OF THIS QUESTION

Soap has a charged head and a long nonpolar tail. Explain how that lets it remove grease with water. The tails bury themselves in the grease while the heads face outward into the water, forming a micelle that carries the grease away. Notice this is exactly the phospholipid principle, just with one tail instead of two.

WHY IT MATTERS IN PATHOLOGY

Drug design depends on this constantly. A drug that is too hydrophilic will not cross a cell membrane; one too hydrophobic will not dissolve in blood. Most drugs are engineered compromises.

WHAT IT SETS UP LATER

Membrane transport in A&P and pharmacology both start here.

Acids, bases, and the pH scale.

A HARDER VERSION OF THIS QUESTION

Stomach acid sits around pH 2 and blood around pH 7.4. That is more than five pH units, which means over 100,000 times the hydrogen ion concentration. How does the stomach avoid digesting itself? Mucus layer, bicarbonate secretion, and rapid cell turnover. When those defenses fail, you get an ulcer.

WHY IT MATTERS IN PATHOLOGY

Arterial blood gas analysis is a core diagnostic test, and it is read almost entirely in terms of pH, bicarbonate, and carbon dioxide.

WHAT IT SETS UP LATER

Acid-base chemistry is a major Chem 1A unit and a heavily tested MCAT topic.

Strong vs weak acids and bases.

A HARDER VERSION OF THIS QUESTION

Why can you drink vinegar, which is an acid, but not hydrochloric acid at the same concentration? Because acetic acid is weak and only partially dissociates, so the actual hydrogen ion concentration is far lower even at the same molarity. Concentration and strength are two different things, and mixing them up is one of the most common chemistry errors.

WHY IT MATTERS IN PATHOLOGY

Acidosis and alkalosis are managed by understanding which buffer systems are still available. A patient's bicarbonate level tells you how much buffering capacity is left.

WHAT IT SETS UP LATER

Titration curves and pKa in Chem 1A are the mathematical version of this card.

What is a buffer, and why is it crucial?

A HARDER VERSION OF THIS QUESTION

Someone hyperventilating blows off carbon dioxide. Carbon dioxide combines with water to form carbonic acid, so losing it removes acid, and blood pH rises. That is respiratory alkalosis, and it causes tingling and dizziness. Breathing into a bag works because it puts carbon dioxide back. Trace that chain out loud, it links three chapters at once.

WHY IT MATTERS IN PATHOLOGY

The bicarbonate buffer system is measured directly in every arterial blood gas, and interpreting it is a daily clinical task.

WHAT IT SETS UP LATER

Buffer calculations and Henderson-Hasselbalch are standard Chem 1A material.

Why is carbon the backbone of life?

A HARDER VERSION OF THIS QUESTION

Silicon also has four valence electrons and sits directly below carbon. Why is life not silicon-based? Silicon-silicon bonds are weaker and longer, silicon does not readily form double bonds, and silicon dioxide is a solid rather than a gas, so a silicon-based organism could not exhale its waste. That last point is usually the one people miss.

WHAT IT SETS UP LATER

Organic chemistry is an entire two-semester course built on carbon's bonding behavior.

What are hydrocarbons, and why do they store energy well?

A HARDER VERSION OF THIS QUESTION

Why does fat store roughly 9 calories per gram while carbohydrate stores about 4? Because carbohydrates are already partly oxidized, with oxygen atoms attached throughout, while hydrocarbon chains are fully reduced. There is more energy to release when you oxidize something that has not been oxidized yet. Same reason gasoline is a hydrocarbon.

WHY IT MATTERS IN PATHOLOGY

This is the metabolic basis for why the body stores long-term energy as fat rather than glycogen. Glycogen also carries water with it; fat does not.

WHAT IT SETS UP LATER

Oxidation states and metabolism connect here in both Chem 1A and biochemistry.

Isomers: structural, cis-trans, and enantiomers.

A HARDER VERSION OF THIS QUESTION

Thalidomide was sold as a mixture of both enantiomers. One treated morning sickness, the other caused severe birth defects. Even worse, the body converts one into the other, so separating them would not have solved it. Explain why a cell can tell them apart at all: receptors and enzymes have three-dimensional binding sites, so only one handedness fits, exactly like a glove.

WHY IT MATTERS IN PATHOLOGY

Modern drug development requires testing single enantiomers separately, and this tragedy is the direct reason that regulation exists.

WHAT IT SETS UP LATER

Stereochemistry is a major organic chemistry topic and a recurring MCAT theme.

Cis-trans isomers: how are they different from structural isomers?

A HARDER VERSION OF THIS QUESTION

Vision starts with a cis-trans isomerization. Light hits retinal in your eye and flips it from the cis form to the trans form, which changes the shape of the attached protein and fires a signal to your brain. A single double bond changing sides is the first physical step in seeing anything.

WHY IT MATTERS IN PATHOLOGY

Trans fats were banned from the US food supply because of their effect on cholesterol, a regulatory decision made entirely on the basis of molecular geometry.

WHAT IT SETS UP LATER

Alkene geometry is early organic chemistry material.

Name all seven functional groups and what each one does.

A HARDER VERSION OF THIS QUESTION

An amino acid has both a carboxyl group, which donates H^+ , and an amino group, which accepts it. At physiological pH, what happens? Both ionize at once: the carboxyl loses its hydrogen and goes negative, the amino gains one and goes positive. The molecule carries both charges simultaneously, which is called a zwitterion, and it is why amino acids are so soluble.

WHY IT MATTERS IN PATHOLOGY

Drug behavior in the body is largely predicted by functional groups, since they determine solubility, charge at blood pH, and which enzymes can process the molecule.

WHAT IT SETS UP LATER

Organic chemistry is organized entirely around functional groups. Learning them now is a genuine head start.

Macromolecules

What are the four classes of biological macromolecules, and what are their monomers?

A HARDER VERSION OF THIS QUESTION

Rank the four by how much energy each stores per gram and explain the ranking. Lipids are highest at about 9 calories per gram because their hydrocarbon chains are fully reduced. Carbohydrates and proteins are both around 4. Nucleic acids are not used for energy at all, because their job is information, and burning your own instructions would be a design flaw.

WHAT IT SETS UP LATER

Biochemistry devotes a unit to each of these four.

Dehydration synthesis vs hydrolysis.

A HARDER VERSION OF THIS QUESTION

Every digestive enzyme you have is a hydrolase, meaning it performs hydrolysis. Amylase hydrolyzes starch, lipase hydrolyzes fat, protease hydrolyzes protein. Explain why digestion has to be hydrolysis and not something else: polymers are too large to cross a membrane, so they must be cut back into monomers, and water is the tool that cuts.

WHY IT MATTERS IN PATHOLOGY

Pancreatic insufficiency is diagnosed partly by undigested fat in stool, which is hydrolysis failing at the organ level.

WHAT IT SETS UP LATER

Enzyme kinetics in biochemistry begins with reactions exactly like these.

What are carbohydrates, and what is the general formula?

A HARDER VERSION OF THIS QUESTION

Deoxyribose is $C_5H_{10}O_4$. Does it fit the 1:2:1 rule? No, it is one oxygen short, which is exactly what deoxy means. So is it still a carbohydrate? Yes. The formula is a generalization, not a law, and modified sugars break it routinely.

WHAT IT SETS UP LATER

You will meet many modified sugars in biochemistry, and almost none follow the ratio exactly.

Monosaccharides: name three and why they matter.

A HARDER VERSION OF THIS QUESTION

Why does the body convert almost everything to glucose specifically, rather than running on fructose or galactose? Because glycolysis, the central energy pathway, takes glucose as its input, so every other sugar gets converted in the liver first. That conversion step is why fructose is metabolized differently and why high fructose intake stresses the liver.

WHY IT MATTERS IN PATHOLOGY

Blood glucose, not blood sugar in general, is what gets measured in diabetes, and this is why.

WHAT IT SETS UP LATER

Glycolysis is one of the first pathways you memorize in biochemistry.

Aldose vs ketose.

A HARDER VERSION OF THIS QUESTION

Glucose and fructose are both $C_6H_{12}O_6$, so why is fructose noticeably sweeter? Sweetness comes from how the molecule fits taste receptors on your tongue, and the ketose arrangement fits better. Same atoms, different shape, different receptor fit, different perception. It is the structure-function principle showing up in something you can taste.

WHAT IT SETS UP LATER

Sugar chemistry and ring formation are early biochemistry topics.

How do disaccharides form? Name three.

A HARDER VERSION OF THIS QUESTION

Lactose intolerance is extremely common worldwide, and the ability to digest lactose as an adult is the unusual trait, not the other way around. It evolved separately in populations with a long history of dairy farming. Explain what that is an example of: natural selection acting on a digestive enzyme within human history.

WHY IT MATTERS IN PATHOLOGY

Lactose intolerance is diagnosed by a hydrogen breath test, because undigested lactose gets fermented by gut bacteria which produce hydrogen gas.

WHAT IT SETS UP LATER

This is a standard example in genetics for recent human evolution.

Starch, glycogen, and cellulose: compare structure and function.

A HARDER VERSION OF THIS QUESTION

Cows cannot digest cellulose either. So how do they live on grass? Symbiotic microbes in their rumen produce cellulase and break it down for them, and the cow absorbs the products. Termites do the same thing. The lesson: the enzyme is what is missing, not the ability to use the energy.

WHY IT MATTERS IN PATHOLOGY

Gut microbiome composition affects how much energy a person extracts from the same food, and it is an active research area in metabolic disease.

WHAT IT SETS UP LATER

Microbiome work is one of the fastest growing areas in medical research.

Chitin and the other structural polysaccharides.

A HARDER VERSION OF THIS QUESTION

Fungi have chitin cell walls, while human cells have no cell wall at all. Why is that useful for treating fungal infections? Because a drug that attacks chitin synthesis has no human target to damage. Selective toxicity, the principle of hitting something the pathogen has and you do not, is the basis of nearly every antimicrobial.

WHY IT MATTERS IN PATHOLOGY

Antifungal drugs are harder to design than antibacterials precisely because fungi are eukaryotes like us, so there are fewer unique targets.

WHAT IT SETS UP LATER

Selective toxicity is a core principle in microbiology and pharmacology.

What defines a lipid, and what do lipids do?

A HARDER VERSION OF THIS QUESTION

Vitamins A, D, E, and K are lipids, which means they are stored in body fat rather than excreted in urine. What is the clinical consequence? Fat-soluble vitamins can accumulate to toxic levels if over-supplemented, while excess water-soluble vitamins mostly get urinated out. Solubility decides the toxicity risk.

WHY IT MATTERS IN PATHOLOGY

Vitamin D deficiency is one of the most commonly measured lab values, and fat malabsorption disorders cause deficiency in all four at once.

WHAT IT SETS UP LATER

Nutrition and pharmacokinetics both organize around this distinction.

Structure of a triglyceride, and saturated vs unsaturated.

A HARDER VERSION OF THIS QUESTION

Trans fats are worse for you than saturated fats, even though both are straight chains. Why? They raise LDL like saturated fat does, but they also lower HDL, which saturated fat does not. The body also has no enzyme that evolved to handle a trans double bond, because they barely exist in nature.

WHY IT MATTERS IN PATHOLOGY

A lipid panel measuring LDL, HDL, and triglycerides is one of the most ordered blood tests in medicine, and everything on it traces back to this card.

WHAT IT SETS UP LATER

Lipid metabolism is a full biochemistry unit.

Structure of a phospholipid, and why it forms membranes.

A HARDER VERSION OF THIS QUESTION

Membranes need to stay fluid, not freeze into a solid sheet. Two things keep them fluid: unsaturated tails with kinks that prevent tight packing, and cholesterol wedged between them. Fish living in cold water have more unsaturated tails in their membranes than tropical fish. Predict why, then check: colder environment, more kinks needed to stay fluid.

WHY IT MATTERS IN PATHOLOGY

Membrane fluidity affects how drugs cross into cells and how cells signal, and it changes measurably in some disease states.

WHAT IT SETS UP LATER

Fluid mosaic model and membrane transport are the opening chapters of cell biology.

What makes steroids structurally different, and why are they still lipids?

A HARDER VERSION OF THIS QUESTION

Steroid hormones pass straight through the membrane and bind receptors inside the cell, which means they change gene expression. Protein hormones like insulin cannot cross, so they bind surface receptors and trigger a relay inside. Predict which type acts faster and which lasts longer: protein hormones act in seconds, steroid hormones take hours but the effects persist.

WHY IT MATTERS IN PATHOLOGY

This is why corticosteroid treatment takes time to work and why stopping it abruptly is dangerous, since the body has adjusted its own production.

WHAT IT SETS UP LATER

Endocrinology in A&P is built on exactly this two-pathway distinction.

List the functions of proteins.

A HARDER VERSION OF THIS QUESTION

One gene mutation, one changed amino acid, one misshapen protein, one disease. Sickle cell anemia is exactly that: a single base change swaps glutamic acid for valine in hemoglobin, the protein clumps, and red blood cells distort into a sickle shape that blocks capillaries. Trace the whole chain from DNA to symptom out loud.

WHY IT MATTERS IN PATHOLOGY

Hemoglobin electrophoresis, a lab technique that separates proteins by charge, is the standard diagnostic for sickle cell. It works because that one amino acid change alters the protein's charge.

WHAT IT SETS UP LATER

This example appears in genetics, biochemistry, hematology, and on the MCAT.

Structure of an amino acid, and how peptide bonds form.

A HARDER VERSION OF THIS QUESTION

R groups sort into nonpolar, polar, acidic, and basic. Predict where each type ends up in a folded protein sitting in water: nonpolar R groups get buried in the interior away from water, polar and charged ones face outward. That single prediction explains most of tertiary structure, and it is the hydrophobic effect again.

WHY IT MATTERS IN PATHOLOGY

Protein misfolding causes Alzheimer's, Parkinson's, and prion disease, all of which are diagnosed by pathologists identifying characteristic protein deposits in tissue.

WHAT IT SETS UP LATER

You will memorize all 20 R groups in biochemistry. Learning the four categories now makes that far easier.

The four levels of protein structure.

A HARDER VERSION OF THIS QUESTION

Which level does a mutation change directly, and which level actually causes the disease? A mutation changes primary structure, the sequence. But the disease comes from the tertiary structure being wrong, because that is what determines shape and therefore function. Primary determines tertiary, so changing one changes the other, but they are not the same level.

WHY IT MATTERS IN PATHOLOGY

Immunohistochemistry, a foundational pathology technique, uses antibodies that recognize the three-dimensional shape of specific proteins in tissue. If the protein is misfolded or absent, the stain shows it.

WHAT IT SETS UP LATER

Protein folding is a major biochemistry unit and a large open research problem.

What is denaturation, and is it reversible?

A HARDER VERSION OF THIS QUESTION

Why is a fever above roughly 104 F dangerous, but a fever of 101 F is useful? Mild fever speeds immune response and slows pathogen growth. High fever begins to denature your own enzymes, and once enzymes lose their shape the reactions they catalyze stop. Above about 106 F, protein damage becomes widespread and irreversible.

WHY IT MATTERS IN PATHOLOGY

Formalin fixation, which is the first step for nearly every tissue sample that reaches a pathology lab, works by cross-linking proteins to lock structure in place before it degrades. It is controlled denaturation used deliberately.

WHAT IT SETS UP LATER

Enzyme activity and optimal conditions are standard biochemistry material.

What are nucleic acids and nucleotides?

A HARDER VERSION OF THIS QUESTION

ATP is a nucleotide. It is adenine, ribose, and three phosphates instead of one. So the energy currency of every cell is structurally an RNA building block with extra phosphates. Explain where the energy is: in the repulsion between those clustered negative phosphates, which is released when one is broken off.

WHY IT MATTERS IN PATHOLOGY

Many chemotherapy drugs are nucleotide analogs. They resemble real nucleotides closely enough to get incorporated into DNA, then stop replication, which kills fast-dividing cells.

WHAT IT SETS UP LATER

ATP appears in literally every metabolism chapter you will ever read.

Purines vs pyrimidines.

A HARDER VERSION OF THIS QUESTION

Gout is caused by uric acid crystals in joints, and uric acid is the breakdown product of purines specifically, not pyrimidines. That is why gout diets restrict organ meats and shellfish, which are purine-rich. A chemistry distinction you are learning for an exam turns out to explain a dietary restriction.

WHY IT MATTERS IN PATHOLOGY

Joint fluid aspiration examined under polarized light for needle-shaped urate crystals is the definitive gout diagnosis, and it is done in the lab.

WHAT IT SETS UP LATER

Nucleotide metabolism is a biochemistry unit, and gout is its standard clinical example.

Describe DNA's double helix and complementary base pairing.

A HARDER VERSION OF THIS QUESTION

Antiparallel means the two strands run in opposite directions, 5 prime to 3 prime on one and the reverse on the other. Why does that complicate replication? Because DNA polymerase can only build in one direction, so one strand is made continuously and the other in backwards fragments that get stitched together. That awkwardness is a direct consequence of antiparallel structure.

WHY IT MATTERS IN PATHOLOGY

Understanding strand directionality is required for reading any sequencing result, and sequencing is now routine in cancer diagnosis.

WHAT IT SETS UP LATER

Replication, transcription, and PCR all depend on this in genetics.

DNA vs RNA, and the three types of RNA.

A HARDER VERSION OF THIS QUESTION

RNA has an extra hydroxyl group on its sugar that DNA lacks, and it makes RNA more chemically reactive and less stable. Why would the cell want its long-term storage molecule to be the stable one and its messenger to be the unstable one? Because messages should expire. If mRNA lasted forever, the cell could never stop making a protein it no longer needs.

WHY IT MATTERS IN PATHOLOGY

mRNA vaccines exploit this instability deliberately. The mRNA delivers instructions, gets translated, and degrades within days, which is why it cannot alter your genome.

WHAT IT SETS UP LATER

Transcription and translation are the central topics of any genetics course.

Viruses

Describe the basic structure of a virus.

A HARDER VERSION OF THIS QUESTION

Naked viruses survive far longer on surfaces than enveloped ones and resist alcohol sanitizer better. Explain why using the lipid card. The envelope is lipid, and both soap and alcohol disrupt lipids. A naked virus has only a protein capsid, so there is no lipid to disrupt. Norovirus is naked, which is why outbreaks spread through surfaces so effectively.

WHY IT MATTERS IN PATHOLOGY

Disinfection protocols in clinical labs are chosen based on whether the target pathogen is enveloped, and this is the reason.

WHAT IT SETS UP LATER

Microbiology covers viral classification and disinfection in detail.

Why is a virus so specific about which cells it infects?

A HARDER VERSION OF THIS QUESTION

HIV infects helper T cells specifically because its surface protein binds the CD4 receptor, which those cells carry. That single binding specificity explains the entire disease: destroy the cells that coordinate immune response, and every other infection becomes dangerous. Specificity is not a detail, it determines what the disease looks like.

WHY IT MATTERS IN PATHOLOGY

CD4 counts are the standard monitoring test in HIV care, measuring exactly the cell population the virus targets.

WHAT IT SETS UP LATER

Receptor-ligand specificity underlies immunology and pharmacology both.

Walk through RNA virus replication inside a host cell.

A HARDER VERSION OF THIS QUESTION

Retroviruses like HIV add a step: reverse transcriptase copies their RNA genome into DNA, which then integrates into the host genome permanently. Explain why that makes the infection so hard to cure. The viral DNA becomes part of the cell's own chromosome, so eliminating it means eliminating the infected cell.

WHY IT MATTERS IN PATHOLOGY

Antiretroviral drugs target specific steps of this cycle, and combination therapy works because hitting several steps at once makes resistance far less likely.

WHAT IT SETS UP LATER

Viral replication strategies are a core microbiology topic.

Why do RNA viruses mutate so fast?

A HARDER VERSION OF THIS QUESTION

Influenza changes in two distinct ways. Antigenic drift is gradual mutation, and it is why the vaccine is updated yearly. Antigenic shift is two different flu strains infecting one cell and swapping whole genome segments, producing something the population has no immunity to at all. Shift is what causes pandemics, drift causes seasons.

WHY IT MATTERS IN PATHOLOGY

Flu strain surveillance and vaccine composition decisions are made annually from global lab data.

WHAT IT SETS UP LATER

Viral evolution is heavily tested on the MCAT and central to epidemiology.

Build a full answer to 'are viruses alive?'

A HARDER VERSION OF THIS QUESTION

Giant viruses like mimivirus are bigger than some bacteria and carry genes for protein synthesis machinery, which was thought to be impossible for a virus. Does that change the answer? It complicates it. They still cannot replicate independently, so most biologists still say no, but they show the boundary is blurrier than a textbook implies. Acknowledging that nuance is what a strong answer does.

WHAT IT SETS UP LATER

The definition of life stays genuinely unsettled, and you will meet it again in microbiology and astrobiology.

How do viruses fit into the characteristics-of-life checklist?

A HARDER VERSION OF THIS QUESTION

Some argue viruses should count as alive while inside a host cell and not alive outside it, treating life as a state rather than a property. Evaluate that. It is coherent, but it breaks how we use the word everywhere else, since nothing else switches between alive and not alive. Whether you accept it or not, being able to weigh the argument is what a full-credit answer looks like.

WHAT IT SETS UP LATER

This framing appears in current origin-of-life research.

Practice questions

68 multiple choice · answer key at the back

1. A researcher tests three fertilizer doses on basil and measures final plant height. What is the dependent variable?
 - a) Fertilizer dose
 - b) Plant height
 - c) Type of soil
 - d) The basil variety
2. Which statement about a scientific theory is correct?
 - a) It is an educated guess awaiting testing
 - b) It becomes a law once enough evidence accumulates
 - c) It is a broad explanation supported by many independent lines of evidence
 - d) It cannot be revised once accepted
3. Which is NOT a requirement of a scientific hypothesis?
 - a) It must be testable
 - b) It must be falsifiable
 - c) It must turn out to be correct
 - d) It should be based on prior observation
4. The control group in an experiment exists in order to:
 - a) Make the sample size larger
 - b) Provide a baseline with no treatment for comparison
 - c) Keep all variables constant
 - d) Prove the hypothesis correct
5. On a standard scientific graph, the independent variable belongs on:
 - a) The y-axis
 - b) The x-axis
 - c) Either, it makes no difference
 - d) The legend
6. Which is the correct order, smallest to largest?
 - a) Cell, organelle, tissue, organ
 - b) Organelle, cell, tissue, organ
 - c) Tissue, cell, organ, organelle
 - d) Molecule, cell, organelle, tissue
7. A single neuron cannot store a memory, but a network of neurons can. This is an example of:
 - a) Reductionism
 - b) Homeostasis
 - c) Emergence
 - d) Natural selection
8. Which is the strongest argument that a virus is NOT alive?
 - a) It is very small
 - b) It cannot reproduce without hijacking a host cell's machinery
 - c) It causes disease
 - d) It contains RNA instead of DNA
9. Sweating when body temperature rises is an example of:
 - a) Emergence
 - b) Homeostasis
 - c) Reductionism
 - d) Metabolism only
0. Which statement about natural selection is accurate?
 - a) Individual organisms evolve during their lifetime to meet needs
 - b) Populations change over generations as heritable traits that aid survival become more common
 - c) Organisms develop new traits because the environment requires them
 - d) Natural selection creates variation from nothing
1. Why is the cell considered the fundamental unit of life?
 - a) It is the smallest structure visible under a microscope
 - b) It is the smallest unit that independently shows all characteristics of life
 - c) It contains DNA
 - d) It is present in every organism except viruses

2. An atom has 6 protons, 8 neutrons, and 6 electrons. It is:
 - a) An ion of carbon
 - b) An isotope of carbon
 - c) An ion of oxygen
 - d) An isotope of nitrogen
3. An ionic bond forms when:
 - a) Electrons are shared equally
 - b) Electrons are shared unequally
 - c) One atom transfers electrons to another and opposite charges attract
 - d) A hydrogen is attracted to a nearby oxygen
4. Water is a polar molecule because:
 - a) It contains hydrogen
 - b) Oxygen is more electronegative and the molecule is bent, creating partial charges
 - c) It dissolves salt
 - d) It forms hydrogen bonds
5. Sweating cools you down because water has:
 - a) High heat capacity
 - b) High heat of vaporization
 - c) Strong cohesion
 - d) Solvent properties
6. A solution at pH 4 compared to a solution at pH 6 is:
 - a) 2 times more acidic
 - b) 10 times more acidic
 - c) 100 times more acidic
 - d) 2 times more basic
7. Which functional group makes a molecule acidic?
 - a) Amino, -NH₂
 - b) Carboxyl, -COOH
 - c) Hydroxyl, -OH
 - d) Methyl, -CH₃
8. Carbon forms the backbone of organic molecules because it:
 - a) Is the most abundant element on Earth
 - b) Has four valence electrons and forms four stable covalent bonds
 - c) Is the heaviest element in living things
 - d) Forms ionic bonds easily
9. Hydrogen bonds are essential in biology despite being weak because:
 - a) They are actually stronger than covalent bonds
 - b) Large numbers of them together stabilize structures, and their weakness allows DNA to unzip
 - c) They never break
 - d) They transfer electrons completely
0. Two molecules share the formula C₆H₁₂O₆ but have different structures. They are:
 - a) Isotopes
 - b) Ions
 - c) Isomers
 - d) Polymers
1. Two monomers join and a water molecule is released. This is:
 - a) Hydrolysis
 - b) Dehydration synthesis
 - c) Denaturation
 - d) Ionization
2. Humans cannot digest cellulose because:
 - a) It is too large
 - b) We lack the enzyme to break its beta glycosidic bonds
 - c) It is not made of glucose
 - d) It is a lipid
3. Glycogen is more heavily branched than starch because:
 - a) Animals need to release glucose rapidly and branches provide more ends for enzymes
 - b) Plants cannot make branches
 - c) Glycogen uses beta bonds
 - d) Branching makes it waterproof

4. Butter is solid at room temperature because its fatty acids:
 - a) Contain many double bonds that kink the chains
 - b) Contain no double bonds, so chains are straight and pack tightly
 - c) Are hydrophilic
 - d) Are shorter than those in oil
5. Phospholipids form a bilayer in water because they are:
 - a) Entirely hydrophobic
 - b) Entirely hydrophilic
 - c) Amphipathic, with a hydrophilic head and hydrophobic tails
 - d) Positively charged
6. Which level of protein structure do alpha helices and beta pleated sheets represent?
 - a) Primary
 - b) Secondary
 - c) Tertiary
 - d) Quaternary
7. When an egg is fried, the protein is denatured. What is preserved?
 - a) Tertiary structure
 - b) Quaternary structure
 - c) Primary structure
 - d) Its biological function
8. Which base pairing is correct in DNA?
 - a) A with G, C with T
 - b) A with T, G with C
 - c) A with U, G with C
 - d) A with C, T with G
9. Which is true of RNA but not DNA?
 - a) Contains phosphate
 - b) Contains uracil and ribose, and is usually single stranded
 - c) Is double stranded
 - d) Contains adenine
0. Which macromolecule class is NOT a true polymer?
 - a) Carbohydrates
 - b) Lipids
 - c) Proteins
 - d) Nucleic acids
1. Which structure do ALL viruses have?
 - a) An envelope
 - b) Glycoprotein spikes
 - c) A capsid surrounding nucleic acid
 - d) Ribosomes
2. A viral envelope is made of:
 - a) Protein subunits made by the virus
 - b) Lipid membrane taken from the host cell
 - c) Carbohydrate
 - d) Nucleic acid
3. Viruses infect only specific host cells because:
 - a) They can only survive at certain temperatures
 - b) Their attachment proteins must physically match a specific host receptor
 - c) They only infect weakened cells
 - d) They are too large for most cells
4. Which step of viral replication is impossible without a host cell?
 - a) Attachment
 - b) Replication and protein synthesis
 - c) Existing as a particle
 - d) Being detected by a microscope
5. RNA viruses mutate rapidly primarily because:
 - a) They replicate slowly
 - b) Their polymerase lacks proofreading ability
 - c) They have larger genomes
 - d) They have envelopes

6. You measured heart rate at rest, 1, 3 and 5 minutes into exercise. Which graph fits best?
- Bar graph, because there are four groups
 - Line graph, because time is continuous
 - Pie chart, to show proportions
 - Scatter plot, because there are two variables
7. Which is the clearest sign a graph is being used to mislead?
- It has a title
 - The y-axis starts partway up instead of at zero
 - It includes error bars
 - It labels units on both axes
8. Why must a buffer be made from a weak acid rather than a strong one?
- Weak acids are safer to handle
 - A weak acid dissociates only partially, so it can both donate and take back H^+
 - Strong acids are too expensive
 - Weak acids have a higher pH
9. Vinegar and hydrochloric acid at the same concentration differ because:
- Vinegar contains no hydrogen ions
 - Acetic acid is weak and only partially dissociates, so fewer H^+ are released
 - Vinegar is a base
 - Hydrochloric acid is more dilute
0. Cis and trans isomers are different molecules because:
- They contain different atoms
 - A carbon-carbon double bond cannot rotate, locking each arrangement in place
 - They have different molecular formulas
 - One is an ion
1. Which ion pair is responsible for generating a nerve signal?
- Calcium and chloride
 - Sodium and potassium
 - Magnesium and iron
 - Hydrogen and hydroxide
2. Which statement about purines and pyrimidines is correct?
- Purines have one ring; adenine and guanine are purines
 - Purines have two rings; adenine and guanine are purines
 - Pyrimidines have two rings; cytosine and thymine are pyrimidines
 - All four DNA bases are purines
3. Chitin differs from cellulose mainly because chitin:
- Uses alpha linkages
 - Contains a nitrogen-bearing group added to glucose
 - Is a storage polysaccharide
 - Is made of fructose
4. Which is the best description of what biology studies?
- Only animals and plants
 - Life at every level, from molecules inside cells up to whole ecosystems
 - Only things visible under a microscope
 - Chemistry that happens in a lab
5. Which statement about the scientific process is most accurate?
- It always runs in the same fixed order and ends with a conclusion
 - It loops, and results often send you back to revise the hypothesis or the method
 - Observation always comes before any hypothesis
 - Peer review only happens at the very end
6. A supplement study reports 8 of 10 people felt more energetic after a week. The biggest problem is:
- The sample is slightly small
 - There is no control group, no blinding, and the outcome is self-reported
 - They should have used a bar graph
 - One week is too long
7. What does peer review NOT reliably catch?
- Flawed reasoning
 - Unclear methods
 - Fabricated data
 - Overstated conclusions

8. Researchers studying how bacteria defend against viruses accidentally enabled CRISPR gene editing. This shows:
- a) Applied science is more valuable than basic science
 - b) Basic research can produce applications nobody could have predicted
 - c) Bacteria research is a waste of money
 - d) Gene editing was the original goal
9. Fire grows, consumes fuel, produces waste and spreads. Why is it not alive?
- a) It moves too fast
 - b) It is not made of cells, has no genetic material, and does not maintain homeostasis
 - c) It produces heat
 - d) It is not found in nature
0. Studying the structure of DNA to explain how traits are inherited is an example of:
- a) Emergence
 - b) Reductionism
 - c) Homeostasis
 - d) Natural selection
1. A cell packed with mitochondria and a heavily folded membrane most likely:
- a) Does very little work
 - b) Performs energy-expensive transport, like a kidney tubule cell
 - c) Is dead
 - d) Stores fat
2. Why do carbon, hydrogen, oxygen and nitrogen make up about 96% of living matter?
- a) They are the most abundant elements on Earth
 - b) They are small and form strong, stable covalent bonds, including multiple bonds
 - c) They are the lightest elements
 - d) They are all metals
3. Which statement about atomic structure is correct?
- a) Protons are negative and electrons are positive
 - b) Nearly all the mass is in the nucleus, and nearly all the volume is electron shells
 - c) Neutrons carry a positive charge
 - d) Electrons sit inside the nucleus
4. Why is carbon's atomic mass listed as 12.011 rather than a whole number?
- a) Rounding error
 - b) It is a weighted average across naturally occurring isotopes
 - c) Electrons add fractional mass
 - d) Carbon has a partial proton
5. Nitrogen has 5 valence electrons. How many covalent bonds will it form?
- a) 1
 - b) 2
 - c) 3
 - d) 5
6. Carbon dioxide has two polar bonds but is a nonpolar molecule. Why?
- a) Its bonds are actually nonpolar
 - b) It is linear, so the two bond dipoles point opposite ways and cancel
 - c) Carbon is not electronegative
 - d) It contains no oxygen
7. Oil separates from water because:
- a) Oil is heavier than water
 - b) Oil is nonpolar and cannot hydrogen bond with water, so water excludes it
 - c) Oil is positively charged
 - d) Oil evaporates
8. A buffer resists pH change because it:
- a) Neutralizes all acids permanently
 - b) Contains a weak acid and its conjugate base, so it can both accept and donate H⁺
 - c) Removes all hydrogen ions from solution
 - d) Raises pH to exactly 7
9. Fat stores about 9 calories per gram while carbohydrate stores about 4 because:
- a) Fat molecules are larger
 - b) Hydrocarbon chains are fully reduced, so oxidizing them releases more energy
 - c) Carbohydrates are not digestible
 - d) Fat contains more oxygen

0. Which formula fits the general carbohydrate ratio?
- a) $C_6H_{12}O_6$
 - b) $C_6H_{14}O_2$
 - c) C_2H_6O
 - d) $C_5H_5N_5$
1. Why does the body convert fructose and galactose into glucose?
- a) They are toxic otherwise
 - b) Glycolysis, the central energy pathway, takes glucose as its input
 - c) They taste too sweet
 - d) They cannot dissolve in blood
2. Fructose is classified as a ketose because its carbonyl group sits:
- a) At the end of the chain
 - b) In the middle of the chain
 - c) On a side branch
 - d) Outside the molecule
3. Lactose intolerance happens because a person lacks:
- a) Lactose
 - b) Lactase, the enzyme that hydrolyzes the bond in lactose
 - c) Glucose
 - d) Stomach acid
4. What do all lipids have in common?
- a) They share the same monomer
 - b) They are hydrophobic and do not dissolve in water
 - c) They all contain glycerol
 - d) They all store energy
5. Steroid hormones can enter a cell directly because they are:
- a) Very small
 - b) Nonpolar, so they pass through the hydrophobic membrane interior
 - c) Positively charged
 - d) Made of protein
6. Which of these is NOT a typical protein function?
- a) Catalyzing reactions as an enzyme
 - b) Providing structure, like collagen
 - c) Storing genetic information long term
 - d) Transporting oxygen, like hemoglobin
7. What makes one amino acid different from another?
- a) The amino group
 - b) The carboxyl group
 - c) The R group, or side chain
 - d) The central carbon
8. A nucleotide is made of:
- a) A phosphate, a sugar and a nitrogenous base
 - b) Two sugars and a base
 - c) An amino group and a carboxyl group
 - d) Glycerol and fatty acids

Short answer practice

Say each one out loud first, then write the version you would put on the exam

1. Explain the difference between a controlled variable and a control group, using an example.

2. A friend says evolution is 'just a theory.' Explain why that misuses the word.

3. Give an example of emergence and an example of reductionism, each using two specific levels of biological organization.

4. Describe one homeostatic feedback loop from start to finish.

5. Explain, step by step, how water's electronegativity difference leads to sweating cooling you down.

6. Name four functional groups and explain how each changes the molecule it is attached to.

7. Explain the four levels of protein structure and what stabilizes each.

8. Explain how the structure of DNA makes accurate inheritance possible.

9. Explain why a phospholipid bilayer forms spontaneously in water.

0. Are viruses alive? Build a full answer using the characteristics of life.

1. Walk through the replication cycle of an RNA virus and name the host component used at each step.

2. Design an experiment testing whether background music affects study performance. Include every required element.

3. State the general rule linking linkage type, chain shape and function in polysaccharides.

4. Walk the characteristics of life and score a virus on each one.

Answer key

Multiple choice

- b) Plant height**
Height is what gets measured, and it depends on the dose. The dose is the independent variable; soil and variety are controlled variables.
- c) It is a broad explanation supported by many independent lines of evidence**
Theories do not get promoted into laws. Laws describe what happens, theories explain why. And theories can always be revised with new evidence.
- c) It must turn out to be correct**
A hypothesis that gets disproven still did its job. Being wrong is a normal, useful outcome in science.
- b) Provide a baseline with no treatment for comparison**
Watch the wording carefully. Keeping variables constant is what CONTROLLED VARIABLES do. The control GROUP is the untreated comparison baseline.
- b) The x-axis**
DRY MIX: Dependent, Responding, Y-axis; Manipulated, Independent, X-axis.
- b) Organelle, cell, tissue, organ**
Organelles are parts inside a cell, so they come before it. Then cells build tissues, tissues build organs.
- c) Emergence**
Emergence is a property that appears at a higher level of organization and does not exist in the individual parts.
- b) It cannot reproduce without hijacking a host cell's machinery**
Size and disease are irrelevant, and plenty of living things use RNA. Lacking independent reproduction and metabolism is the real argument.
- b) Homeostasis**
Homeostasis is maintaining a stable internal state through feedback when the external environment changes.
- b) Populations change over generations as heritable traits that aid survival become more common**
Individuals do not evolve, populations do. Selection acts on variation that already exists; it does not generate traits on demand.
- b) It is the smallest unit that independently shows all characteristics of life**
Organelles contain DNA too and some are visible, but nothing below the cell can independently metabolize, maintain homeostasis, and reproduce.
- b) An isotope of carbon**
6 protons means carbon, always. Protons equal electrons, so no charge and therefore not an ion. Extra neutrons make it an isotope, carbon-14.
- c) One atom transfers electrons to another and opposite charges attract**
Sharing equally is nonpolar covalent, sharing unequally is polar covalent, and the hydrogen option describes a hydrogen bond.
- b) Oxygen is more electronegative and the molecule is bent, creating partial charges**
Watch the direction of causation. Polarity is the CAUSE; dissolving salt and hydrogen bonding are the RESULTS. Reversing cause and effect is a classic lost point.
- b) High heat of vaporization**
Heat of vaporization is the heat needed to turn liquid into gas. That heat leaves your skin with the escaping water. High heat capacity is about resisting temperature change, a different property.
- c) 100 times more acidic**
The pH scale is logarithmic, so each unit is a factor of 10. Two units means $10 \times 10 = 100$.
- b) Carboxyl, -COOH**
Carboxyl donates an H^+ , which is the definition of an acid. Amino accepts H^+ , making it basic.
- b) Has four valence electrons and forms four stable covalent bonds**
Four bonding sites, including bonds to other carbons, is what allows chains, branches, and rings of unlimited variety.
- b) Large numbers of them together stabilize structures, and their weakness allows DNA to unzip**
The weakness is a feature. If hydrogen bonds were strong, DNA could never be separated for replication.
- c) Isomers**
Isotopes involve neutrons, ions involve charge. Same formula with different arrangement is the definition of an isomer, like glucose and fructose.
- b) Dehydration synthesis**
Building removes water; breaking adds water. Hydrolysis is the reverse and is what digestion does.
- b) We lack the enzyme to break its beta glycosidic bonds**
Cellulose is pure glucose, same as starch. The difference is the beta linkage, and we have no cellulase to cut it, so it passes through as fiber.
- a) Animals need to release glucose rapidly and branches provide more ends for enzymes**
More branch ends means more places enzymes can start cutting at once, so energy releases faster. Animals move; plants can afford to wait.
- b) Contain no double bonds, so chains are straight and pack tightly**
Saturated means no double bonds, so straight chains, tight packing, solid. Double bonds cause kinks and keep oils liquid.

5. **c) Amphipathic, with a hydrophilic head and hydrophobic tails**
Two opposite personalities in one molecule. Heads face the water on both sides, tails hide in the middle, and it assembles itself.
6. **b) Secondary**
Secondary, and it is stabilized by hydrogen bonds along the backbone, not by R-group interactions. R groups drive tertiary structure.
7. **c) Primary structure**
Denaturation destroys folding but does not break peptide bonds, so the amino acid sequence survives. Function is lost because function depends on shape.
8. **b) A with T, G with C**
A-T and G-C in DNA. A-U appears in RNA, where uracil replaces thymine.
9. **b) Contains uracil and ribose, and is usually single stranded**
Both contain phosphate and adenine. The RNA-only features are uracil, ribose, and single-stranded structure.
0. **b) Lipids**
Lipids are not built from repeating identical monomers strung into a chain. They are grouped together by being hydrophobic, not by a shared building block.
1. **c) A capsid surrounding nucleic acid**
Envelopes and spikes are optional, and no virus has ribosomes. Genome plus capsid is the universal minimum.
2. **b) Lipid membrane taken from the host cell**
It is stolen host membrane, which is lipid. That is exactly why soap, which disrupts lipids, destroys enveloped viruses.
3. **b) Their attachment proteins must physically match a specific host receptor**
Key and lock. No matching receptor means no attachment and no entry, which is why host range is narrow.
4. **b) Replication and protein synthesis**
Viruses have no ribosomes, no metabolic enzymes, and no ATP. Every bit of the manufacturing is borrowed, and that is the core argument against them being alive.
5. **b) Their polymerase lacks proofreading ability**
No proofreading means errors stay in. Combined with huge numbers of copies, variants appear fast, which is why flu shots are reformulated yearly.
6. **b) Line graph, because time is continuous**
Time is continuous, so a line graph shows the change across it. Bar graphs are for separate categories, not points along a continuum.
7. **b) The y-axis starts partway up instead of at zero**
Truncating the y-axis exaggerates small differences. Error bars, units and titles are all signs of a graph being made honestly.
8. **b) A weak acid dissociates only partially, so it can both donate and take back H⁺**
Buffering requires a reversible reaction. A strong acid dissociates completely and has nothing left to take back, so it cannot absorb added base.
9. **b) Acetic acid is weak and only partially dissociates, so fewer H⁺ are released**
Concentration and strength are different things. Same molarity, very different actual hydrogen ion concentration, because a weak acid only partly lets go.
0. **b) A carbon-carbon double bond cannot rotate, locking each arrangement in place**
Identical atoms, identical connections. The double bond is a locked hinge rather than a swivel, so the two spatial arrangements cannot interconvert.
1. **b) Sodium and potassium**
Sodium and potassium gradients across the membrane store the charge difference that becomes an action potential. Calcium triggers muscle contraction, which is a different job.
2. **b) Purines have two rings; adenine and guanine are purines**
Pure As Gold: purines are adenine and guanine, and they carry two rings. A purine always pairs with a pyrimidine so the helix stays a constant width.
3. **b) Contains a nitrogen-bearing group added to glucose**
Both are straight beta-linked structural polysaccharides. Chitin is built from a nitrogen-containing glucose derivative, and it forms shells and fungal cell walls.
4. **b) Life at every level, from molecules inside cells up to whole ecosystems**
Biology claims anything alive, at any level. That range is why it overlaps chemistry below it and ecology above it.
5. **b) It loops, and results often send you back to revise the hypothesis or the method**
The textbook draws a clean loop, but the important part is that it is iterative. Results feed back into the question.
6. **b) There is no control group, no blinding, and the outcome is self-reported**
Several fatal flaws at once. Self-reported feelings with no control group and no blinding cannot separate the product from the placebo effect.
7. **c) Fabricated data**
Reviewers evaluate the reasoning and methods described, not the raw data. Independent replication is what catches fabrication.
8. **b) Basic research can produce applications nobody could have predicted**
You cannot know in advance which basic question becomes a technology, which is the core argument for funding basic research.
9. **b) It is not made of cells, has no genetic material, and does not maintain homeostasis**
Life requires the whole checklist, not a majority of it. Fire fails cells, heredity, homeostasis and evolution.

0. **b) Reductionism**
Reductionism explains a system by breaking it into parts. Emergence is the opposite direction, looking at what appears when parts combine.
1. **b) Performs energy-expensive transport, like a kidney tubule cell**
Predicting function from structure is the core skill here. Lots of mitochondria means high energy demand; folded membrane means lots of surface for transport.
2. **b) They are small and form strong, stable covalent bonds, including multiple bonds**
Small atoms form short, strong bonds. Bigger atoms form weaker, longer ones that would break too easily to hold a genome together.
3. **b) Nearly all the mass is in the nucleus, and nearly all the volume is electron shells**
Protons positive, neutrons neutral, electrons negative. The nucleus holds the mass; the shells take up essentially all the space.
4. **b) It is a weighted average across naturally occurring isotopes**
Carbon is mostly carbon-12 with a small fraction of carbon-13, and averaging the two lands just above 12.
5. **c) 3**
It needs 3 more electrons to reach 8, so it forms 3 bonds. You can predict bonding for any of these atoms this way instead of memorizing.
6. **b) It is linear, so the two bond dipoles point opposite ways and cancel**
Shape decides molecular polarity, not just bond polarity. Water has polar bonds too, but it is bent so they do not cancel.
7. **b) Oil is nonpolar and cannot hydrogen bond with water, so water excludes it**
The real answer is about what water does. Water molecules stick to each other and exclude anything nonpolar, which forces the oil together.
8. **b) Contains a weak acid and its conjugate base, so it can both accept and donate H+**
Buffering needs a reversible pair. The bicarbonate system holds blood near 7.4 by absorbing or releasing H+ as needed.
9. **b) Hydrocarbon chains are fully reduced, so oxidizing them releases more energy**
Carbohydrates are already partly oxidized. There is more energy left to release in a chain of C-H bonds that has not been oxidized yet.
0. **a) C₆H₁₂O₆**
The ratio is roughly one carbon to two hydrogens to one oxygen, which glucose fits exactly. Modified sugars like deoxyribose break the rule.
1. **b) Glycolysis, the central energy pathway, takes glucose as its input**
Every other sugar gets converted in the liver first, which is why blood glucose specifically is what gets measured.
2. **b) In the middle of the chain**
Aldose means at the end, forming an aldehyde. Ketose means internal, forming a ketone. Glucose is an aldose, fructose a ketose.
3. **b) Lactase, the enzyme that hydrolyzes the bond in lactose**
The disaccharide is fine; the enzyme that cuts the glycosidic bond is missing, so undigested lactose reaches gut bacteria.
4. **b) They are hydrophobic and do not dissolve in water**
Lipids are grouped by a physical property rather than a building block, which is why steroids count despite looking nothing like a fat.
5. **b) Nonpolar, so they pass through the hydrophobic membrane interior**
Being nonpolar lets them cross the membrane and bind receptors inside the cell, which is why they act slowly but last longer than protein hormones.
6. **c) Storing genetic information long term**
Long-term genetic storage is DNA's job. DNA is the manual, proteins are the workers.
7. **c) The R group, or side chain**
All 20 share the same central carbon, amino group, carboxyl group and hydrogen. Only the R group changes, and it decides how that residue behaves.
8. **a) A phosphate, a sugar and a nitrogenous base**
Three parts, every time. The sugar is deoxyribose in DNA and ribose in RNA, and ATP is a nucleotide with two extra phosphates.

Short answer — what a full-credit answer sounds like

1. **Explain the difference between a controlled variable and a control group, using an example.**
A controlled variable is a factor deliberately kept identical across all groups so it cannot explain the results, such as giving every basil plant the same soil, light, and water. A control group is an entire experimental group that receives no treatment, such as the basil plants given zero fertilizer, and it exists to provide a baseline to compare the treated groups against. The first is about keeping conditions constant; the second is about having something to compare to.
2. **A friend says evolution is 'just a theory.' Explain why that misuses the word.**
In everyday speech a theory means a guess, but in science a theory is the opposite: it is a broad explanation that has been tested repeatedly and is supported by many independent lines of evidence. A guess in science is called a hypothesis. Evolution by natural selection is supported by fossil evidence, genetics, comparative anatomy, and direct observation, so calling it 'just a theory' confuses the everyday meaning with the scientific one.
3. **Give an example of emergence and an example of reductionism, each using two specific levels of biological organization.**
Emergence: individual heart muscle cells can only contract, but organized together at the organ level the heart pumps blood through an entire body, a function no single cell possesses. Reductionism: studying the structure of DNA at the molecule level to explain how traits are inherited at the organism level. Emergence looks upward at what appears when parts combine; reductionism looks downward by breaking a system into parts to explain it.

4. **Describe one homeostatic feedback loop from start to finish.**
Body temperature rises above the set point. Temperature sensors in the skin and hypothalamus detect the increase. The hypothalamus acts as the control center and signals effectors: sweat glands release sweat and blood vessels in the skin dilate. Evaporation of sweat removes heat and dilated vessels release heat at the surface, so body temperature falls back toward the set point. Because the response reverses the original change, this is negative feedback.
5. **Explain, step by step, how water's electronegativity difference leads to sweating cooling you down.**
Oxygen is more electronegative than hydrogen, so it pulls the shared electrons closer and the bent shape of the molecule gives oxygen a partial negative charge and each hydrogen a partial positive charge. These partial charges let water molecules form hydrogen bonds with each other. Breaking those many hydrogen bonds to convert liquid water into vapor takes a large amount of energy, which is water's high heat of vaporization. When sweat evaporates from skin it takes that heat energy with it, lowering body temperature and helping maintain homeostasis.
6. **Name four functional groups and explain how each changes the molecule it is attached to.**
Hydroxyl (-OH) is polar, so it makes the molecule more soluble in water; it is found in alcohols and sugars. Carboxyl (-COOH) is acidic because it donates a hydrogen ion; it is found in amino acids and fatty acids. Amino (-NH₂) is basic because it accepts a hydrogen ion; it is the other reactive end of every amino acid. Phosphate (-PO₄) carries a negative charge and is central to energy transfer in ATP and to the backbone of DNA. (Sulfhydryl forming disulfide bridges, carbonyl defining aldehydes and ketones, and nonpolar methyl acting as a tag are also acceptable.)
7. **Explain the four levels of protein structure and what stabilizes each.**
Primary structure is the linear sequence of amino acids joined by peptide bonds, determined by the gene. Secondary structure is local folding into alpha helices and beta pleated sheets, stabilized by hydrogen bonds between atoms of the backbone rather than the R groups. Tertiary structure is the overall three-dimensional shape of the whole polypeptide, stabilized by interactions between R groups including hydrophobic clustering, hydrogen bonds, ionic bonds, and covalent disulfide bridges. Quaternary structure is the assembly of two or more separate polypeptide chains into one functional protein, as in hemoglobin's four subunits, and not every protein has it.
8. **Explain how the structure of DNA makes accurate inheritance possible.**
DNA is a double helix with a sugar-phosphate backbone on the outside and bases pointing inward. Because adenine always pairs with thymine and guanine always pairs with cytosine, the sequence of one strand exactly specifies the sequence of the other. That means when the two strands separate, each one serves as a template for building a new complementary partner, producing two identical double helices. The complementary base pairing is what makes the copying accurate, so the structure of the molecule is the direct reason inheritance works.
9. **Explain why a phospholipid bilayer forms spontaneously in water.**
A phospholipid is amphipathic: its phosphate head is polar and hydrophilic while its two fatty acid tails are nonpolar and hydrophobic. In water, the hydrophilic heads form favorable interactions with the surrounding water while the hydrophobic tails cannot, so the tails cluster together to minimize contact with water. The most stable arrangement is a double layer with heads facing outward into the water on both sides and tails sequestered in the interior. Nothing assembles it deliberately; the hydrophobic effect makes this the lowest-energy arrangement, and the resulting bilayer is what allows a cell to exist as a separate compartment.
0. **Are viruses alive? Build a full answer using the characteristics of life.**
Viruses fail most of the standard criteria for life. They are acellular, so they violate cell theory. They have no metabolism, no ribosomes, and cannot produce ATP, so they cannot process energy. They do not maintain homeostasis or grow and develop. They can only reproduce by entering a host cell and hijacking its machinery, so reproduction is not independent. However, they do possess genetic material in the form of DNA or RNA, and they clearly mutate and undergo natural selection, as seen with new influenza and coronavirus variants. Because they meet only the heredity and evolution criteria, most biologists describe viruses as existing at the border of life, or as alive only while inside a host cell. The deciding property is the inability to reproduce or metabolize independently.
1. **Walk through the replication cycle of an RNA virus and name the host component used at each step.**
First, attachment: a glycoprotein spike on the virus binds a specific receptor protein on the host cell surface. Second, entry and uncoating: the virus enters the cell and the capsid comes off, releasing the RNA genome into the cytoplasm. Third, replication and biosynthesis: the host cell's ribosomes, amino acids, and ATP are used to translate viral proteins, while viral RNA polymerase copies the genome. Fourth, assembly: newly made capsid proteins package copies of the genome into new virions. Fifth, release: new viruses either bud out through the host membrane, taking an envelope with them, or burst the cell open by lysis. The virus supplies only the instructions; the ribosomes, energy, and raw materials are all the host's.
2. **Design an experiment testing whether background music affects study performance. Include every required element.**
Hypothesis: students who study with background music will score lower on a recall test than students who study in silence. Independent variable: presence of background music. Dependent variable: score on the recall test. Controlled variables: same study material, same study duration, same room, same time of day, same test. Control group: the silence group. Design: recruit enough participants for a meaningful result, randomly assign them to the two groups so that existing study habits are distributed evenly, and run the test the same way for both. Analysis method decided in advance: compare the mean scores of the two groups. A limitation worth naming is that people already used to studying with music may respond differently, which random assignment helps but does not fully remove.
3. **State the general rule linking linkage type, chain shape and function in polysaccharides.**
Alpha glycosidic linkages produce chains that coil and branch, which suits energy storage because branching creates many ends for enzymes to attack, as in starch and glycogen. Beta glycosidic linkages produce straight chains that stack and hydrogen bond to each other into rigid fibers, which suits structure, as in cellulose and chitin. Most animals lack the enzymes to break beta linkages, which is why cellulose passes through as fiber and why chitin is indigestible. Alpha coils for storage, beta is a beam for building.
4. **Walk the characteristics of life and score a virus on each one.**
Made of cells: no, viruses are acellular. Own metabolism and energy: no, they have no ribosomes and cannot make ATP. Homeostasis: no, they do not regulate an internal environment. Grows and develops: no, they are assembled rather than grown. Reproduces independently: no, they can only replicate inside a host cell using its machinery. Has genetic material: yes, DNA or RNA. Evolves by natural selection: yes, clearly and quickly, as new influenza and coronavirus variants show. That is two out of seven, which is why most biologists describe viruses as sitting at the border of life rather than answering yes or no.